Dangerous Talk[*]

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Abstract

I explore continuous-time partnerships in an organization. Each partner has a productivity-irrelevant hidden type, either normative or rogue. An exogenous public signal process gradually reveals partners' types to each other. The organization incentivizes calling out rogue types, but punishes false accusations. Opportunities to make an accusation or abandon the partnership arrive randomly, and are privately observed — so the partnership's very survival is an endogenous source of information.

I fully solve the one-sided incomplete information game (say, a stranger entering a purely normative organization), and show how false accusations and partnership dissolutions respond to organizational incentives. In the general case with two-sided incomplete information, the equilibrium is unique when exit opportunities are sufficiently rare. Here, I characterize the equilibrium in terms of fear—exiting because of one's reputation—and trust—collaborating regardless of one's reputation. Lastly, I show that partnerships in this environment are assortative in the long run.

This analysis offers a lens to understand the dynamics of past organizational witch hunts, such as during the Red Scare or the Lavender Scare of the 1950s.

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1. Introduction

Partnerships are formed around shared objectives, but their members may differ in beliefs and other characteristics unrelated to those objectives. When these differences concern conformity to a prevailing social norm, they can create tension and lead otherwise productive relationships to dissolve. There are plenty of examples of such cases throughout history, particularly in the United States. The Red Scare campaign that ran through the late 1940s and 1950s is one such example, where the fear of communist influence turned the suspicion of being a communist into grounds for professional exclusion across many fields, regardless of one's professional record. A prominent example is the Hollywood blacklist (Ceplair and Englund, 2003), through which studios denied work to screenwriters and actors suspected of communist sympathies. The consequences extended beyond those formally blacklisted: Pontikes et al. (2010) find that artists who were not themselves blacklisted were less likely to obtain film work when a former coworker was subsequently blacklisted.

The employment consequences of differences unrelated to job performance are difficult to study because they rarely leave a record. The federal loyalty program of the Red Scare is an exception: its scrutiny of employees' political beliefs and associations generated administrative records of investigations and their outcomes (Handler, 2026). Figure 1 summarizes these outcomes, distinguishing the decisions of loyalty boards from resignations tendered while cases were pending. Resignations, unlike verdicts, were the employees' own decisions, so the records also capture employees' strategic responses to scrutiny.

These tensions remain relevant today and have gained renewed prominence in debates over "cancel culture," where expressions deemed objectionable can prompt social or economic sanctions (Dias et al., 2025). The prospect of such sanctions can influence behavior even among people who are never directly targeted. Gibson and Sutherland (2023) document that the share of Americans reporting that they felt less free to speak their minds than they used to rose from approximately 13 percent in 1954 to 46 percent in 2020. They suggest that concerns about alienating friends, family, and neighbors may help explain self-censorship. These concerns give contemporary relevance to the question of how people navigate productive relationships when revealing their beliefs can put those relationships at risk.

I study how learning about a partner's conformity to a prevailing social norm shapes accusations, disclosure, and exit from an otherwise productive relationship. To do so, I develop a continuous-time partnership model. Each partner privately knows whether

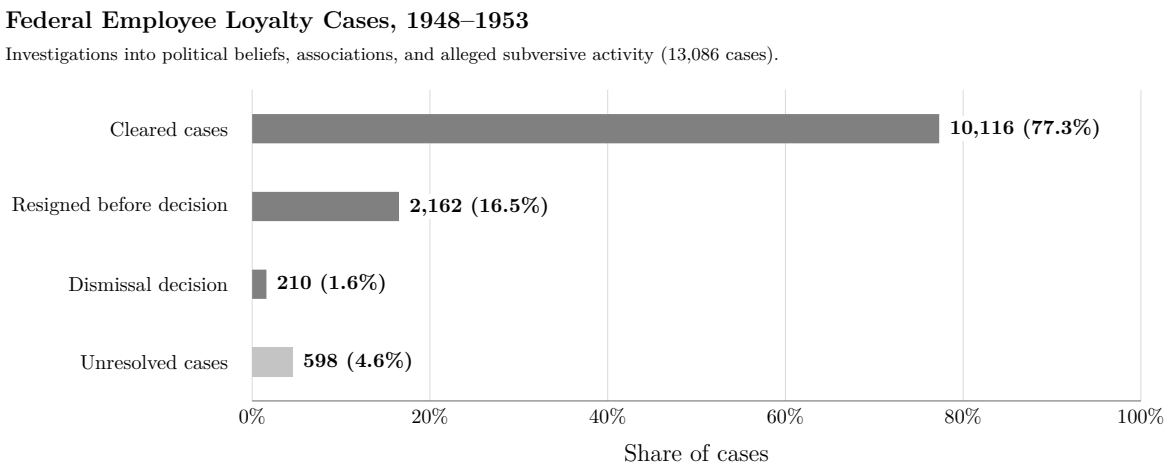


Figure 1: **Outcomes of Federal Loyalty Investigations.** Data from Handler (2026, Table 1, pp. 1240–1241). Cleared cases are decisions favorable to the employee. Remaining cases are the difference between the reported total and the three displayed outcomes.

she conforms to the prevailing social norm (normative) or departs from it (rogue). Types do not affect the partnership's output, and both partners earn a fixed flow payoff while it lasts, which gives them a reason to stay. Partners learn about each other's types through a pair of public signals that capture their daily interactions. Opportunities to act on this information arrive at random, and each partner observes only her own. When opportunity knocks, a normative type may call out her partner (whistleblow): exposing a rogue earns a reward, while a false accusation is costly to both. A rogue may reveal herself to her partner (scheme): if both are rogue, they carry on as accomplices, in a collaboration worth more than before; otherwise, the scheme exposes the rogue and implicates her partner, imposing a loss on both. Either type may also simply leave (quit) to avoid trouble. A partner may instead pass on the opportunity and leave the partnership undisturbed (collaborate).

Each partner's *reputation* represents the public belief about her type, expressed below in log-odds form. As public signals change reputations, the two types may have different incentives to continue collaborating, so survival itself can carry information. This information accumulates over time because opportunities to act arrive at random, and continuously because they are private, contributing an additional drift to the reputation dynamics driven by the public signals. Since both the signals and survival are publicly observed, reputations are common knowledge at all times. Yet the two partners assign different laws to the evolution of the same reputation, since only its owner knows her type (Proposition 1).

As in Aumann and Maschler (1995), the analysis begins with one-sided incomplete information and then turns to the two-sided case. In the one-sided case, a *stranger*, whose type is private, works with an *insider* known to be normative, as when someone new joins a purely normative organization. As long as the partnership's output is small relative to both the reward for exposing a rogue and the losses from being called out, the normative-insider game admits a unique cutoff stationary Markov equilibrium (Proposition 2). The order of these cutoffs depends on what an accusation costs each type (Proposition 3). In the unique equilibrium, the partnership's value falls as the stranger's reputation deteriorates, whatever her type; only partnerships with a normative stranger can last forever, and even these end in a false accusation with positive probability (Proposition 4). In the two-sided case, where each partner's type is private information, there is a unique symmetric equilibrium when opportunities to act are rare (Proposition 7). Survival is also assortative: mixed partnerships dissolve, while like types stay together longer (Proposition 9).

Related literature. One perspective on my paper is that it explores a game with the possibility of arbitrarily long cheap (Aumann and Hart, 2003), but where the stopping set is not exogenously determined, but is an equilibrium choice. So this paper imports optimal stopping into communication games.

The model studies how partnerships dissolve as beliefs about the partners evolve. In matching models, these beliefs concern match quality (Jovanovic, 1979) or individual productivity (Anderson and Smith, 2010); here, they concern a trait that is irrelevant to joint output. While the partnership lasts, these beliefs respond to two sources of information: the public signals and the partnership's survival. The signals follow the standard Brownian learning framework (Bolton and Harris, 1999; Moscarini and Smith, 2001), in which beliefs evolve as a diffusion. In these models, and in reputation models such as Faingold and Sannikov (2011) and Ekmekci et al. (2022), actions affect learning directly through the signal process. In this model, actions affect learning only indirectly: exit decisions make the partnership's survival informative in equilibrium, and survival adds an endogenous drift on top of the signal-driven diffusion, much as the absence of news does in Keller et al. (2005). There, no news is always bad news. By contrast, survival can move a partner's reputation in either direction, depending on which type wants to exit, and whether that is good or bad news for the other partner depends on her own type. Survival also conveys information in Furusawa and Kawakami (2008), where types differ in their exogenous probabilities of receiving an outside option. Here, types face the same opportunity process, and survival becomes informative through differences in their equilibrium choices about whether to continue collaborating. Exit

decisions therefore reveal private information. In such settings, an informed player often randomizes to control how fast she reveals it (Anderson and Smith, 2013; De Angelis et al., 2022). Here, the random arrival of exit opportunities does this work, so the analysis can focus on pure strategies.

Substantively, the model concerns how suspicion erodes the value of productive partnerships when partners care about how others perceive their type. Such concerns shape behavior (Bénabou and Tirole, 2006) and can sustain conformity to a norm (Bernheim, 1994), so people may misrepresent their private views in public (Kuran, 1995). Reputational concerns can also distort advice and actions, destroying information (Morris, 2001; Ely and Välimäki, 2003). In these models, actions affect learning through the signals they generate. Here, the public signal process is exogenous, allowing me to isolate the additional learning generated by endogenous partnership survival.

The paper proceeds as follows. Section 2 presents the model, and Section 3 derives the reputation dynamics and defines equilibrium. Section 4 solves the one-sided case, and Section 5 turns to the two-sided game. The appendices contain all omitted proofs.

2. Model

Players and Types. There are two players $i \in \{1,2\}$[1] with privately known types $\theta_i \in \{n, r\}$ drawn independently from a common prior $\mathbb{P}(\theta_i = r) = 1 - \mathbb{P}(\theta_i = n) = p_0$, where $p_0 \in (0,1)$. Here, n is a *normative* type and r a *rogue* type, with the distinction reflecting whether the player conforms to or departs from a prevailing social norm.

Signals. Time is continuous and indexed by $t \in [0, \infty)$. At $t = 0$, the players enter a partnership, and the game continues until one of them takes a terminal action, described below. Until that time, information about the partners' types is gradually and unintentionally revealed through dialogue and interactions. This information is represented by a pair of public signal processes (Z_t^1, Z_t^2), observed by both players. Specifically, for each $i \in \{1,2\}$, the process $(Z_t^i)_{t\geq 0}$ evolves according to

$$dZ_t^i = \mu(\theta_i)dt + \sigma dB_t^i, \qquad Z_0^i = 0,$$

where B^1 and B^2 are independent Brownian motions, independent of the types. The common diffusion coefficient is $\sigma > 0$, and the type-dependent drifts satisfy $\mu(r) > \mu(n)$. The *signal-to-noise ratio* of the process is denoted by $\psi \equiv (\mu(r) - \mu(n))/\sigma$.

Opportunities. Players can end the game only at exogenous opportunities, which

[1] Throughout, player i denotes the player under consideration and player $j \neq i$ her partner.

arrive for player i according to a Poisson process $(N_t^i)_{t\geq 0}$ of intensity $\lambda > 0$, her *opportunity process*. These processes are independent of each other, of the types, and of the Brownian motions (B^1, B^2), so opportunity arrivals by themselves carry no information. Player i's opportunity times are the jump times of N^i,

$$T_k^i := \inf\{t \geq 0 : N_t^i \geq k\}, \qquad k \in \mathbb{N},$$

and $T^i := \{T_k^i : k \in \mathbb{N}\}$ collects them. Since the opportunity processes are independent, players almost surely never receive simultaneous opportunities: $\mathbb{P}(T^1 \cap T^2 = \varnothing) = 1$.

Actions. When opportunity knocks for player i, at time T_k^i, she chooses an action from a type-dependent set,

$$\mathcal{A}_n := \{c, q, w\}, \qquad \mathcal{A}_r := \{c, q, s\}.$$

She may collaborate (c) by passing on the opportunity. This choice leaves no trace: since opportunities are private, her partner cannot tell whether she passed on an opportunity or had none. She may instead take a *terminal action*, which both players observe and which ends the game. She may quit (q), and end the game. There are two *blunt* actions, impacting the game in a way dependent on the other's type: a normative type may *whistleblow* (w), that is, call out her partner, and a rogue type may offer to *scheme* (s), that is, come out to her partner. A player who quits or whistleblows dissolves the partnership. A rogue who schemes dissolves the partnership if her partner is normative; if her partner is rogue, the two carry on as accomplices. Since the model concerns only collaboration under uncertainty about types, the payoff from a terminal action summarizes, in present value, everything that follows.

Payoffs. Each player accrues a flow payoff $\pi > 0$ while the game continues and receives a terminal payoff when a terminal action is taken. The terminal payoff depends on this action, the partner's type, and which player takes the action. Table 1 reports the terminal payoffs when player i takes the type-specific action $a \in \{w, s\}$ (whistleblow or scheme): her own payoff $u(a; \theta_j)$ first and her partner's $u^p(a; \theta_j)$ second. The common terminal action, quit, pays both types the same amount, normalized to zero, irrespective of who takes it: $u(q; \cdot) = u^p(q; \cdot) = 0$.

Thus u denotes the actor's payoff and u^p the nonactor's payoff, each evaluated at the nonactor's type.

Both players discount at rate $\rho > 0$. Let τ denote the time of the first terminal action and a^τ that action on $\{\tau < \infty\}$. The discounted terminal payoff is defined to be

Type-specific Action	Partner's type: Normative	Partner's type: Rogue
whistleblow w	$-F_n,\ -L_n$	$G_n,\ -L_r$
scheme s	$-L_r,\ -L_n$	$G_r,\ G_r$

Table 1: **Terminal Payoffs with Blunt Action.** Rows indicate the type-specific action $a \in \{w, s\}$: normative players whistleblow (w) and rogues scheme (s). Columns indicate the partner's type, θ_j. Each cell lists player i's payoff first, $u(a;\theta_j)$, and her partner's payoff second, $u^p(a;\theta_j)$. All parameters are positive: G_θ and L_θ denote a type-θ player's gain and loss, respectively, and F_n is the cost of making a false accusation.

zero on $\{\tau = \infty\}$. Player i's discounted payoff is

$$\int_0^\tau e^{-\rho t}\pi\, dt + e^{-\rho\tau}\left[u(a^\tau;\theta_j)\,\mathbb{1}\{\tau \in T^i\} + u^p(a^\tau;\theta_i)\,\mathbb{1}\{\tau \in T^j\}\right].$$

The payoff structure reflects the central tension of the game. Calling out a player (whistleblow) triggers an investigation, which reveals her type and imposes on her a loss L_θ. For a rogue type, the accusation brings a severe backlash, while for a normative type it is a nuisance until the investigation clears her name. For the accuser, calling someone out earns a reward G_n when it correctly targets a rogue, but a false accusation against a normative type carries a cost F_n. Coming out as a rogue (scheme) is beneficial between rogues, who each earn G_r, but detrimental when the partner is normative: the rogue loses L_r and her partner L_n. The stakes of accusations and schemes are large relative to the value of the partnership, so each player must be wary of what others think about her type. Together, these considerations motivate the parameter restrictions $L_r > L_n \geq F_n$ and, for each type $\theta \in \{n, r\}$, $\rho G_\theta > \pi$ and $\lambda L_\theta > \pi$. The last two give the tension its bite: a player sure that her partner is a rogue acts on it, and one who expects her partner to act against her would rather leave than wait.[2]

3. Analysis

3.1. Information and Beliefs

Players learn from two public sources of information. First, they observe the signals (Z^1, Z^2), which generate the filtration $\{\mathcal{Z}_t\}_{t\geq 0}$.[3] Second, they observe whether the game

[2] The analysis also excludes knife-edge parameter values — specifically, $\pi + \rho F_n \neq \lambda(L_n - F_n)$.

[3] Let $\mathcal{Z}_t^0 := \sigma(Z_s^1, Z_s^2 : 0 \leq s \leq t)$ be the raw filtration generated by the public signals. I use the usual augmentation, defined by $\mathcal{Z}_t := \bigcap_{u>t}(\mathcal{Z}_u^0 \vee \mathcal{N})$, where $\mathcal{N}$ consists of all subsets of $\mathbb{P}$-null events in the underlying probability space. The resulting filtration is complete and right-continuous.

continues. That is, at time t, they observe the *survival event* $\{\tau > t\}$. Since a player's decision to take a terminal action may depend on her type, the absence of such actions can convey information beyond what the public signals contain. Accordingly, I define player i's public posterior as[4]

$$P_t^i := \mathbb{P}(\theta_i = r \mid \mathcal{Z}_t, \{\tau > t\}),$$

the probability that she is rogue conditional on the signal history and continued play. For the analysis, it is convenient to express beliefs in log-odds form. I define player i's *reputation* as

$$X_t^i := \log\left(\frac{P_t^i}{1 - P_t^i}\right), \quad \text{with inverse} \quad P_t^i = p(X_t^i), \quad \text{where} \quad p(x) := \frac{1}{1 + e^{-x}}.$$

Since $p(\cdot)$ is a strictly increasing bijection from $\mathbb{R}$ onto $(0, 1)$, reputation X_t^i and posterior belief P_t^i carry the same information. The reputation profile is a measurable function of public information and is therefore common knowledge while the game continues. Moreover, the players' types remain conditionally independent given the public signal history and continued play: the signal likelihoods factor across players, as does the survival likelihood conditional on the signals. Consequently, (X_t^1, X_t^2) determines the joint posterior distribution of (θ_1, θ_2) and summarizes the public information about types. Motivated by this representation, I focus on stationary Markov strategies, under which each player's action at an opportunity depends only on her type and the current reputation profile.

NOTATION. The variables x and y denote realizations of X_t^i and X_t^j, respectively, and θ' denotes the type opposite to θ. A higher reputation means greater suspicion of being rogue. The analysis is conducted in log-odds space, while figures are reported in belief space to ease exposition.

3.2. Strategies and State Dynamics

Player i acts only when opportunity knocks, selecting an available action based on her privately known type θ_i and the current reputation profile (x, y). Formally, a pure stationary Markov strategy is a pair of measurable mappings $\alpha = (\alpha_n, \alpha_r)$, where $\alpha_\theta : \mathbb{R}^2 \to \mathcal{A}_\theta$ is the *decision rule* for type θ, specifying the action taken at state (x, y). Each decision rule partitions the state space according to the prescribed action. For

[4]For a sigma-algebra $\mathcal{F}$ and events E_1, E_2, I use $\mathbb{P}(E_1 \mid \mathcal{F}, E_2)$ as shorthand for $\mathbb{P}(E_1 \cap E_2 \mid \mathcal{F})/\mathbb{P}(E_2 \mid \mathcal{F})$, wherever $\mathbb{P}(E_2 \mid \mathcal{F}) > 0$.

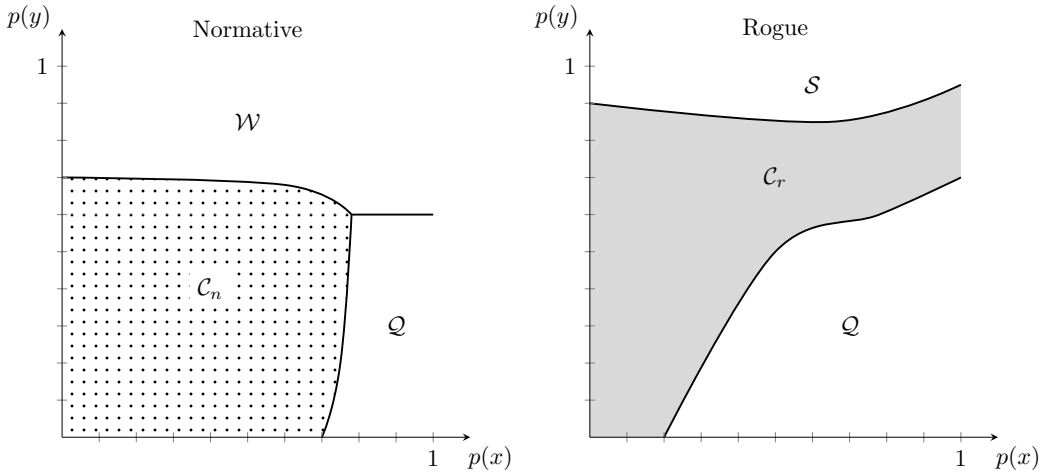


Figure 2: **Decision Rules**. The figure illustrates a strategy $\alpha = (\alpha_n, \alpha_r)$. Each type's decision rule partitions the state space into a collaboration region and stopping regions.

each type θ, the *collaboration region*

$$\mathcal{C}_\theta(\alpha) := \{(x, y) \in \mathbb{R}^2 : \alpha_\theta(x, y) = c\}$$

consists of the states at which that type passes on an opportunity, allowing the game to continue. Analogously, $\mathcal{Q}_\theta(\alpha)$ denotes the quitting region for type θ, while $\mathcal{W}(\alpha)$ and $\mathcal{S}(\alpha)$ denote the normative type's whistleblowing region and the rogue type's scheming region, respectively. An example of the resulting partitions is shown in Figure 2.

Collaboration regions need not coincide across types: at a given state, a strategy may prescribe collaboration for one type and a terminal action for the other. These differences make continued play informative about a player's type. To identify the relevant states in which the survival event $\{\tau > t\}$ is informative, define the *informational region* of type θ as

$$\mathcal{I}_\theta(\alpha) := \mathcal{C}_\theta(\alpha) \setminus \mathcal{C}_{\theta'}(\alpha).$$

This region consists of the states at which type θ collaborates but type θ' would take a terminal action. While the state remains in this region, the game continues under type θ' only if no opportunity arrives. Since opportunities arrive at rate λ, the probability of no arrival falls as time passes, so the survival of the game increasingly favors type θ. Because opportunities are private, this evidence accumulates continuously until the game ends or the state leaves the informational regions. Measured in log-odds, the evidence grows at the constant rate λ: continued play moves the reputation toward rogue in $\mathcal{I}_r(\alpha)$, toward normative in $\mathcal{I}_n(\alpha)$, and is uninformative elsewhere. The *survival drift* measures this

effect at each state:

$$\Lambda(x, y; \alpha) := \lambda \left(\mathbb{1}\{(x, y) \in \mathcal{I}_r(\alpha)\} - \mathbb{1}\{(x, y) \in \mathcal{I}_n(\alpha)\} \right).$$

The next proposition gives the laws of motion of reputations, taking into account both the information conveyed by the public signals and what continued play reveals given the strategies.

Proposition 1 (Reputation Processes)**.** *Fix a strategy profile* (α, β)*. From player* i*'s perspective, the reputations* (X_t^i, X_t^j) *evolve according to*

$$dX_t^i = \left[\Lambda(X_t^i, X_t^j; \alpha) + \frac{\psi^2}{2} \left(\mathbb{1}\{\theta_i = r\} - \mathbb{1}\{\theta_i = n\} \right) \right] dt + \psi dB_t^i, \tag{1}$$

$$dX_t^j = \left[\Lambda(X_t^j, X_t^i; \beta) + \frac{\psi^2}{2} \tanh\left(X_t^j / 2 \right) \right] dt + \psi \, d\widetilde{B}_t^j, \tag{2}$$

for some Brownian motion $\widetilde{B}^j$ *with respect to the public information.*

The asymmetry between the two laws of motion reflects private knowledge rather than disagreement about the public state. Player i knows θ_i and therefore conditions the signal component of X_t^i on her true type, which gives rise to the indicator functions. By contrast, she does not know θ_j, so she averages the signal drift of X_t^j over the current posterior, which yields the term $\tanh(X_t^j/2)$. The noise term changes accordingly: $\widetilde{B}^j$ is the innovation in player j's signal, its surprise relative to that average. Both players nonetheless observe the same reputation profile at every instant; private information only gives them different expectations about how it will evolve.

The survival drift, however, enters both laws of motion in the same way. Since the strategy profile is common knowledge in equilibrium, both players draw the same inference from continued play. Through the survival drift, players' actions shape the evolution of reputations even though they do not affect the signal process. Figure 3 illustrates the survival drift induced by an example strategy.

3.3. Equilibrium Concept

I focus on symmetric stationary Markov perfect equilibria (MPE). In such an equilibrium, the common strategy α plays two roles: it prescribes behavior through the decision rules α_n and α_r, and it determines how continued play updates reputations through the survival drift $\Lambda(\cdot; \alpha)$. Equilibrium requires consistency between these roles: each type's decision rule must be optimal given the partner's behavior and the reputation updating induced by α.

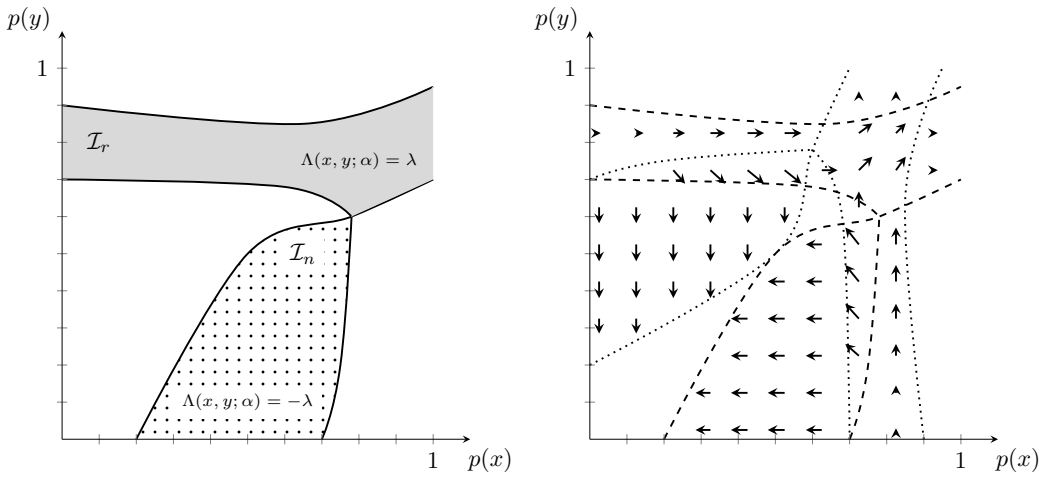


Figure 3: **Survival Drift.** Left: informational regions and values of Λ for the strategy α shown in Figure 2. Right: survival-induced drift field in belief space under the symmetric strategy profile (α, α). Arrows lengths are proportional to the drift magnitude in probability coordinates, accounting for the conversion from log odds to probabilities.

To make the equilibrium condition explicit, fix a conjectured strategy α, common to both players, and allow player i to deviate to an arbitrary measurable decision rule $\varphi : \mathbb{R}^2 \to \mathcal{A}_{\theta_i}$. Public reputations continue to be computed from observed histories using the conjectured profile (α, α), while φ determines player i's actions at opportunity times. The deviation induces the stopping time and terminal action:

$$\tau_i(\varphi;\alpha) := \inf\{T_k^i : \varphi(X^i_{T_k^i}, X^j_{T_k^i}) \in \mathcal{A}_{\theta_i} \setminus \{c\}\}, \quad a_i(\varphi;\alpha) := \varphi(X^i_{\tau_i(\varphi;\alpha)}, X^j_{\tau_i(\varphi;\alpha)}).$$

The action $a_i(\varphi;\alpha)$ is defined on $\{\tau_i(\varphi;\alpha) < \infty\}$; the analogous convention applies to player j. The game ends at $\tau = \min\{\tau_i(\varphi;\alpha), \tau_j(\alpha_{\theta_j};\alpha)\}$, and, on $\{\tau < \infty\}$, the terminal action a^τ is $a_i(\varphi;\alpha)$ if player i stops first and $a_j(\alpha_{\theta_j};\alpha)$ if player j does. The expected payoff of player i's deviation is

$$J_{\theta_i}(x,y;\varphi,\alpha) := \mathbb{E}^{\theta_i;\alpha,\varphi}_{x,y}\left[\int_0^\tau e^{-\rho t}\pi\, dt + e^{-\rho\tau}\left(u(a^\tau;\theta_j)\,\mathbb{1}\{\tau_i=\tau\} + u^p(a^\tau;\theta_i)\,\mathbb{1}\{\tau_j=\tau\}\right)\right],$$

where reputations evolve as in Proposition 1 for the conjectured profile (α, α), while φ determines player i's actions.[5]

A symmetric MPE is a strategy $\alpha^* = (\alpha_n^*, \alpha_r^*)$ such that, for every type $\theta \in \{n, r\}$

[5]The operators $\mathbb{E}$ and $\mathbb{P}$ (used later) denote expectation and probability over the path of play. Subscripts give initial reputations, superscripts give types, and labels after the semicolon give strategies, absent once fixed.

and every state $(x, y) \in \mathbb{R}^2$,

$$\alpha_\theta^* \in \operatorname*{argmax}_{\varphi} J_\theta(x, y; \varphi, \alpha^*).$$

Write $\tau_\theta^* := \tau_i(\alpha_\theta^*; \alpha^*)$ and $V_\theta^*(x, y) := J_\theta(x, y; \alpha_\theta^*, \alpha^*)$ for its induced stopping times and continuation values. Each type's equilibrium rule is therefore optimal given the partner's behavior and the belief updating induced by α^*.

4. One-Sided Learning

To fix ideas, consider the boundary case of one-sided learning. A newcomer, the stranger (she), enters a partnership with an insider (he) who is publicly known to be normative. The insider's reputation never moves and drops out of the state; the stranger's reputation x is the sole state variable. The collaboration regions, informational regions, and survival drift keep their definitions, now as sets and functions of x alone.

4.1. Equilibrium Strategies

In this one-sided environment, Proposition 1 describes the dynamics of the stranger's reputation x. Equation (1) gives its law of motion from her perspective. Equation (2), with the players' roles reversed, gives the insider's view of the same process, so its innovation process is the stranger's.

I focus on games where the insider is normative, since it is the only one of economic interest. If the insider were rogue, either type of stranger would have a profitable terminal action available, making the equilibrium trivial. With a normative insider, by contrast, the stranger's gains come entirely from the flow payoff π. A rogue stranger weighs this payoff against the risk that the signals reveal her type, while a normative stranger weighs it against the risk that they misrepresent it. Either way, the stranger collaborates until the risk of being called out grows too high and then quits.

A stationary Markov strategy is a *cutoff strategy* if, at each decision opportunity, it calls for collaboration when the stranger's reputation lies below a threshold and for stopping otherwise; the stranger stops by quitting and the insider by whistleblowing. Three cutoffs then describe an equilibrium in cutoff strategies: x_n and x_r for the two stranger types and x_I for the insider. The following proposition characterizes equilibrium in terms of these cutoffs.

Proposition 2. *The normative-insider game admits a unique cutoff stationary Markov equilibrium, with cutoffs* (x_n^*, x_r^*, x_I^*) *satisfying* $x_r^* < x_n^*$.

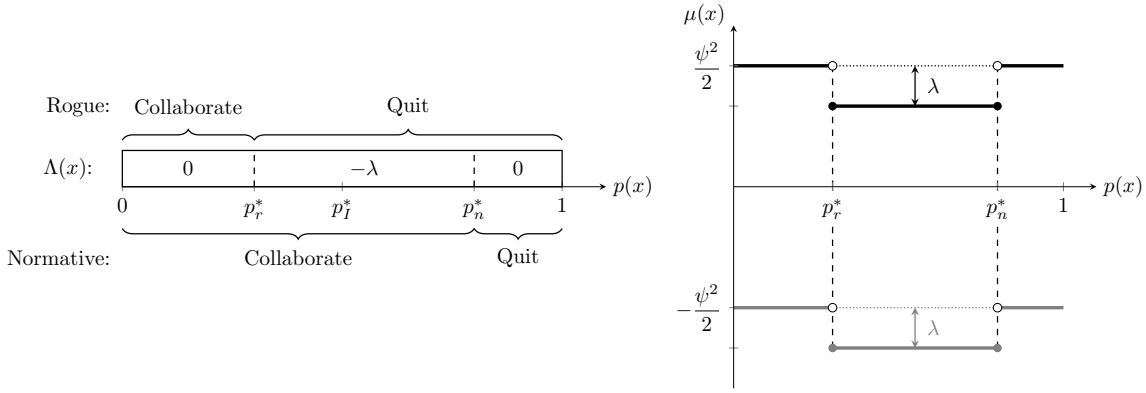


Figure 4: **One-Sided Equilibrium.** Left panel: braces indicate each stranger type's action at a stopping opportunity, and the strip reports the survival drift $\Lambda(x)$. All cutoffs are stated in belief space: $p_\theta^* \equiv p(x_\theta^*)$ for $\theta \in \{r, n\}$ and $p_I^* \equiv p(x_I^*)$. Right panel: reputation drift $\mu(x)$ of a rogue (black) and a normative (gray) stranger in equilibrium. Parameters: $\pi = 0.03$, $\rho = 0.1$, $\lambda = 0.1$, $\psi = 0.7$, $F_n = L_n = 0.73$, $L_r = 0.97$, $G_n = 1.37$.

Because the rogue's reputation drifts up faster and $L_n \leq L_r$, the rogue stranger's cutoff always lies below the normative stranger's, $x_r^* < x_n^*$, so the equilibrium characterization reduces to locating the insider's cutoff relative to this pair. Each location yields a different pattern of exit and accusation as the stranger's reputation deteriorates. The equilibrium exhibits a *crackdown* if the insider's cutoff lies below both types' cutoffs, so that his threat starts before either type would quit; *rogue flight* if it lies between them, so that the rogue quits before the threat starts while the normative stays on despite it; and *general flight* if it lies above both, so that both types quit before the threat starts. Figure 4 illustrates the rogue-flight equilibrium.

The primitives of the game determine which pattern arises. To see how, let $\bar{L}(\mu)$ denote the critical loss at which a stranger whose reputation has constant drift μ is indifferent between quitting and collaborating at the insider's cutoff:

$$\bar{L}(\mu) = \frac{\pi}{\lambda}\left[1 + \frac{\sqrt{\mu^2 + 2\psi^2(\rho + 2\lambda)} - \mu}{\sqrt{\mu^2 + 2\psi^2\rho} + \mu}\right]. \tag{3}$$

A stranger with loss L_θ and constant drift μ prefers to collaborate at the insider's cutoff if $L_\theta < \bar{L}(\mu)$, so its cutoff lies above x_I^*, and prefers to quit there if $L_\theta > \bar{L}(\mu)$, so its cutoff lies below x_I^*. The threshold is decreasing in μ, since the numerator of the fraction falls and its denominator rises with μ. A stranger whose reputation drifts up faster spends more time in the insider's stop region, where the insider's call out arrives at rate λ, so a smaller loss suffices to make quitting optimal.

In equilibrium, however, the drift is piecewise constant. A rogue stranger's reputation

drifts at $\Lambda + \psi^2/2$, while a normative stranger's drifts at $\Lambda - \psi^2/2$, where $\Lambda \in \{-\lambda, 0\}$. Between the two types' cutoffs, only the rogue quits when an opportunity arrives, so continued collaboration is evidence that the stranger is normative and $\Lambda = -\lambda$; elsewhere, both types act alike and $\Lambda = 0$. Each stranger's drift therefore switches between two values, and her actual critical loss lies between the values of $\bar{L}$ at those two drifts. Applying these bounds to both types yields sufficient conditions for each pattern.

Proposition 3. *The unique cutoff equilibrium exhibits a crackdown if the rogue's loss from being called out is small, $L_r < \bar{L}(\psi^2/2)$; rogue flight if the rogue's loss is large, $L_r > \bar{L}(\psi^2/2 - \lambda)$, and the normative's is small, $L_n < \bar{L}(-\psi^2/2)$; and general flight if even the normative's loss is large, $L_n > \bar{L}(-\psi^2/2 - \lambda)$.*

4.2. Partnership Dissolution

In any equilibrium, the game ends in finite time with positive probability. Let $\mathbb{P}^x_\theta$ denote the law of equilibrium starting from initial reputation x, conditional on the stranger being of type θ, and write $\mathbb{E}^x_\theta$ for the corresponding expectation. Denote the opportunity times of the insider and the stranger by $(T^I_k)_{k\geq 1}$ and $(T^S_k)_{k\geq 1}$, respectively, and define the stopping times

$$\begin{aligned} \tau_I &:= \inf\{T^I_k : X_{T^I_k} \geq x^*_I\}, \\ \tau_\theta &:= \inf\{T^S_k : X_{T^S_k} \geq x^*_\theta\}. \end{aligned} \tag{4}$$

with the convention $\inf \varnothing = \infty$. The partnership ends at $\tau = \tau_I \wedge \tau_\theta$. Since the two opportunity processes are independent, $\tau_I \neq \tau_\theta$ almost surely on $\{\tau < \infty\}$, so every dissolution is either a call-out or a quit. Let

$$\begin{aligned} \omega_\theta(x) &:= \mathbb{P}^\theta_x(\tau_I < \tau_\theta), \\ q_\theta(x) &:= \mathbb{P}^\theta_x(\tau_\theta < \tau_I), \end{aligned}$$

denote the probabilities that the insider ends the partnership by whistleblowing and that the stranger ends it by quitting, respectively. Their sum, $\omega_\theta(x) + q_\theta(x) = \mathbb{P}^\theta_x(\tau < \infty)$, is the probability that the partnership dissolves.

How much collaboration the equilibrium sustains depends on how long the partnership lasts. Expected duration, however, is uninformative: when the stranger is normative, since her reputation drifts down the partnership survives forever with positive probability, $\omega_n(x) + q_n(x) < 1$, and $\mathbb{E}^n_x[\tau] = \infty$. To obtain a useful measure of the duration of the partnership I consider the discounted value of the flow payoff from the collaboration. Define the *value of the partnership* as the expected discounted flow payoff

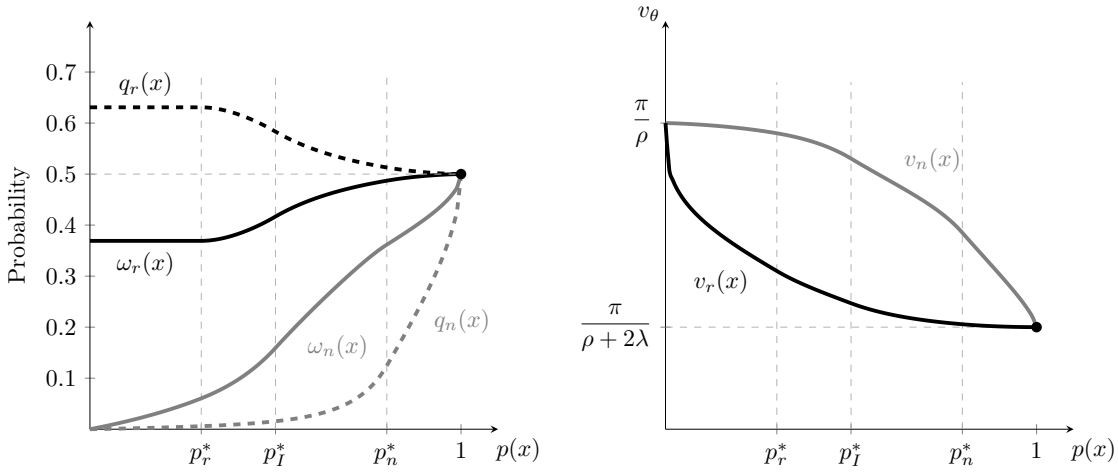


Figure 5: **Partnership Dissolution & Value.** For the equilibrium of Figure 4, the left panel plots the probabilities that the stranger is called out, $\omega_\theta(x)$, and that she quits, $q_\theta(x)$. The right panel plots the value of the partnership, $v_\theta(x)$. Rogue strangers are in black and normative strangers in gray.

that each partner accrues until termination,

$$v_\theta(x) := \mathbb{E}_x^\theta\left[\int_0^\tau e^{-\rho t}\pi\, dt\right] = \frac{\pi}{\rho}\left(1 - \mathbb{E}_x^\theta\left[e^{-\rho\tau}\right]\right),$$

where $e^{-\rho\tau} = 0$ on $\{\tau = \infty\}$. Unlike expected duration, it is always finite, and it increases as dissolution is delayed. An everlasting partnership has discounted duration $1/\rho$, so its value is π/ρ. At the other extreme, when both partners stop as soon as they get an opportunity, its value is $\pi/(\rho + 2\lambda)$. Hence

$$\frac{\pi}{\rho + 2\lambda} \leq v_\theta(x) \leq \frac{\pi}{\rho}. \tag{5}$$

From left to right, the panels of Figure 5 show how the stranger's initial reputation affects the probabilities of a call-out and of quitting, and the value of the partnership in a sorting equilibrium. The intuition behind this figure follows from how reputation evolves and the logic of cutoff strategies. First, observe that for $x < \min\{x_r^*, x_n^*, x_I^*\}$, the only driving force behind the dissolution of the partnership is the reputation dynamics, as no player actively acts to terminate it.

Consider how this applies to a rogue stranger, whose reputation drifts up. Since no player takes a terminal action before reputation reaches a cutoff, the rogue's reputation reaches the minimum cutoff almost surely, and hence ω_r is constant below that cutoff. Note that although the probability of reaching the minimum cutoff does not change

with the initial reputation, the closer the initial reputation is to the cutoff, the sooner the rogue's reputation reaches it, and so v_r falls. Once reputation reaches the minimum cutoff, a race begins. The rogue stranger wants to quit but must wait for an opportunity, and before one arrives, her reputation may cross the insider's cutoff, exposing her to being called out. Hence ω_r increases with the stranger's reputation. Because a rogue's partnership ends almost surely, she quits whenever she is not called out, so $q_r = 1 - \omega_r$. Her cutoff lies below the insider's, so she is more likely to quit than to be called out. As $x \to \infty$, the race becomes a coin flip and $\omega_r(x)$ and $q_r(x)$ converge to $1/2$: both players end the partnership at the first opportunity, and the chance that her reputation falls below a cutoff before then vanishes.

Now, consider a normative stranger, whose reputation drifts down. Starting below a cutoff, her reputation therefore has a positive probability of never reaching it. Nonetheless, unfavorable signals may still push her reputation above any cutoff, and since the probability of this happening increases with the initial reputation, ω_n increases everywhere. The same holds for q_n, but since the order of the cutoffs is the reverse of the rogue's (the insider's cutoff comes first), $\omega_n > q_n$. As a result, v_n falls with the initial reputation, but it always lies above v_r: the downward drift tends to offset any unfavorable signal realization, so with positive probability her partnership never ends, whereas a rogue's ends almost surely. The two forces at work in this example, the drift of reputation and the race between opportunities, operate in every equilibrium. The next proposition shows how they shape dissolution in general.

Proposition 4. *In equilibrium,*

(i) only partnerships with normative strangers may last forever: $\mathbb{P}^n_x(\tau < \infty) < 1$, *while* $\mathbb{P}^r_x(\tau < \infty) = 1$;

(ii) the partner with the lower cutoff is more likely to end the partnership: $q_\theta(x) > \omega_\theta(x)$ *if and only if* $x^*_\theta < x^*_I$;

(iii) the accusation probability against a rogue stranger satisfies $\omega_r(x) > 0$ *for all* $x \in \mathbb{R}$. *Furthermore,* $\omega_r'(x) \geq 0$ *if* $x^*_r < x^*_I$ *and* $\omega_r'(x) \leq 0$ *if* $x^*_I < x^*_r$, *with equality in either case if and only if* $x \leq \min\{x^*_I, x^*_r\}$;

(iv) the accusation probability against a normative stranger satisfies $\omega_n(x) > 0$ *for all* $x \in \mathbb{R}$. *Furthermore,* $\omega_n'(x) > 0$ *for every* $x \in \mathbb{R}$ *if* $x^*_n \leq x^*_I$. *If* $x^*_I < x^*_n$, *then* $\omega_n'(x) > 0$ *for* $x \leq x^*_I$, *and* ω_n *is either nondecreasing everywhere or has a unique maximum in* (x^*_I, x^*_n).

Proposition 4 distinguishes whether a partnership ends from how it ends. Partnerships with rogue strangers eventually dissolve, whereas those with normative strangers

may survive indefinitely. Nevertheless, false accusations remain a possible outcome at every reputation. The ordering of the cutoffs determines which partner is more likely to end the relationship, but the effect of initial reputation on accusation probabilities also depends on the opportunity to quit. When the insider's cutoff is lower, a higher initial reputation can bring the stranger closer to quitting and reduce the insider's opportunity to accuse first. Consequently, higher initial reputation need not imply a higher probability of accusation.

The next proposition examines how these outcomes respond to changes in economic incentives. A higher collaboration payoff increases what each player gives up by ending the partnership and raises all three equilibrium cutoffs. A higher reward for exposing a rogue instead lowers the insider's accusation cutoff and induces both stranger types to quit at lower reputations; a higher cost of false accusation reverses these movements. For the latter two changes, all cutoffs move by the same amount, preserving their ordering and the distances between them.

Proposition 5. *The following comparative statics hold in equilibrium for the insider and each stranger type $\theta \in \{n, r\}$:*

(i) An increase in π strictly raises each equilibrium cutoff, $\partial x_I^*/\partial \pi > 0$, $\partial x_\theta^*/\partial \pi > 0$.

(ii) An increase in G_n strictly lowers all three equilibrium cutoffs by the same amount, $\partial(x_I^* - x_\theta^*)/\partial G_n = 0$.

(iii) An increase in F_n strictly raises all three equilibrium cutoffs by the same amount, $\partial(x_I^* - x_\theta^*)/\partial F_n = 0$.

(iv) For every fixed initial reputation $x \in \mathbb{R}$,

$$\frac{\partial \omega_\theta(x)}{\partial G_n} = -\omega_\theta'(x)\frac{\partial x_I^*}{\partial G_n}, \qquad \frac{\partial q_\theta(x)}{\partial G_n} = -q_\theta'(x)\frac{\partial x_I^*}{\partial G_n}.$$

The same identities hold with F_n replacing G_n.

At a fixed initial reputation, lowering all cutoffs has the same effect on termination probabilities as increasing reputation while holding the cutoffs fixed. Consequently, an increase in G_n or a decrease in F_n leaves $\omega_r(x)$ unchanged whenever the initial reputation remains below the lowest cutoff: the rogue adjusts her quitting threshold along with the insider's accusation threshold. Stronger incentives to accuse then shorten the partnership and reduce its collaboration value without making the rogue more likely to be identified. A normative stranger also loses collaboration value, but additionally faces a higher probability of false accusation. Beyond this region, the effect on false accusations can reverse, as earlier quitting may reduce the insider's opportunity to accuse.

5. Equilibrium

5.1. Best Response and Optimal Stopping

The best response is characterized indirectly through the value function, from which the stopping regions are recovered. For a player of type θ facing a partner using strategy α, the value function

$$V_\theta(x, y; \alpha) := \sup_{\varphi} J_\theta(x, y; \varphi, \alpha). \tag{6}$$

A key observation is that the optimal terminal action depends only on the partner's current reputation y. The optimal stopping action therefore solves a static decision problem. Denote the *myopic action* and *myopic payoff* of type θ by

$$m_\theta(y) \in \operatorname*{argmax}_{a \in \mathcal{A}_\theta \setminus \{c\}} \overline{u}(a; y), \quad M_\theta(y) = \max_{a \in \mathcal{A}_\theta \setminus \{c\}} \overline{u}(a; y),$$

where the expected terminal payoff is

$$\overline{u}(a; y) = p(y)u(a; r) + (1 - p(y))u(a; n).$$

Because each type has only two terminal actions available, the myopic action admits a threshold characterization:

$$m_n(y) = \begin{cases} w & y \geq y_n \\ q & y < y_n \end{cases}; \qquad m_r(y) = \begin{cases} s & y \geq y_r \\ q & y < y_r \end{cases}; \qquad \begin{aligned} y_n &= \log(F_n/G_n) \\ y_r &= \log(L_r/G_r) \end{aligned}$$

A strategy α is called *myopically optimal* if, for every type θ, its decision rule prescribes the myopic action whenever the player stops:

$$\alpha_\theta(x, y) = m_\theta(y) \quad \text{for all } (x, y) \notin \mathcal{C}_\theta(\alpha).$$

At a terminal-action tie, m_θ selects the type-specific action, as above. For a bounded continuous pair $f = (f_n, f_r)$, define the selected rule

$$\alpha_\theta[f](x, y) := \begin{cases} c, & f_\theta(x, y) > M_\theta(y), \\ m_\theta(y), & f_\theta(x, y) \leq M_\theta(y). \end{cases} \tag{7}$$

This convention selects stopping at collaboration–stopping indifference. Given a conjecture α, write $\varphi^*_\theta(\cdot; \alpha) := \alpha_\theta[V(\cdot; \alpha)]$ for the selected response.

Proposition 6. *For any strategy α, there exists a unique bounded continuous value function $V_\theta(\cdot;\alpha)$, attained by a myopically optimal decision rule φ^*_θ with collaboration region*

$$\mathcal{C}^{BR}_\theta(\alpha) := \{(x,y) : V_\theta(x,y;\alpha) > M_\theta(y)\}. \tag{8}$$

The associated indifference set is

$$\mathcal{U}_\theta(\alpha) := \{(x,y) : V_\theta(x,y;\alpha) = M_\theta(y)\}. \tag{9}$$

When the conjecture is fixed, its argument is suppressed. Players continue collaborating, gathering information and receiving the flow payoff, until they stop, at which point they select their terminal action myopically. Collaboration, however, is risky: normative players may be mistaken for rogues, while rogue players may be exposed. At each stopping opportunity, a player compares the forward-looking value of collaboration, V_θ, with the myopic payoff from stopping, M_θ. Figure 6 illustrates this comparison by plotting their difference, $V_\theta(x,y) - M_\theta(y)$, at two levels of own reputation, $x_1 < x_2$.

At the lower own-reputation state x_1, normative players collaborate unless the partner's reputation is sufficiently high, in which case whistleblowing becomes optimal. Rogue players, by contrast, collaborate only when their partner's reputation lies in an intermediate range. When it is low, the expected evolution of reputations depresses their continuation value: by Proposition 1, a rogue player anticipates their own reputation drifting upward and y downward. Because the partner may receive the next opportunity first, this unfavorable drift lowers the value of waiting, and quitting is optimal. Higher values of y make a normative type less likely, weakening this concern until collaboration is optimal. Once y is sufficiently high, the rogue player switches to scheming.

At the high own-reputation state x_2, the prospect that the partner ends the relationship matters more. For normative players, this unambiguously lowers the collaboration value. The signal component of their reputation drift works in their favor, but the survival drift works against them at high reputations, where only rogue types collaborate. A normative player's reputation is therefore unlikely to improve before the partner receives an opportunity to call her out or scheme. For rogue players, the effect depends on y. At low y, where the odds favor a normative type, moving from x_1 to x_2 makes calling out more likely: the collaboration value falls, and the collaboration region contracts from the left. At high y, where the odds favor a rogue type, the same move makes scheming more likely: the collaboration value rises, and the region expands to the right.

Figure 6 illustrates the distinction between stopping at Poisson opportunities and standard continuous-time stopping. Here, V_θ is the continuation value between oppor-

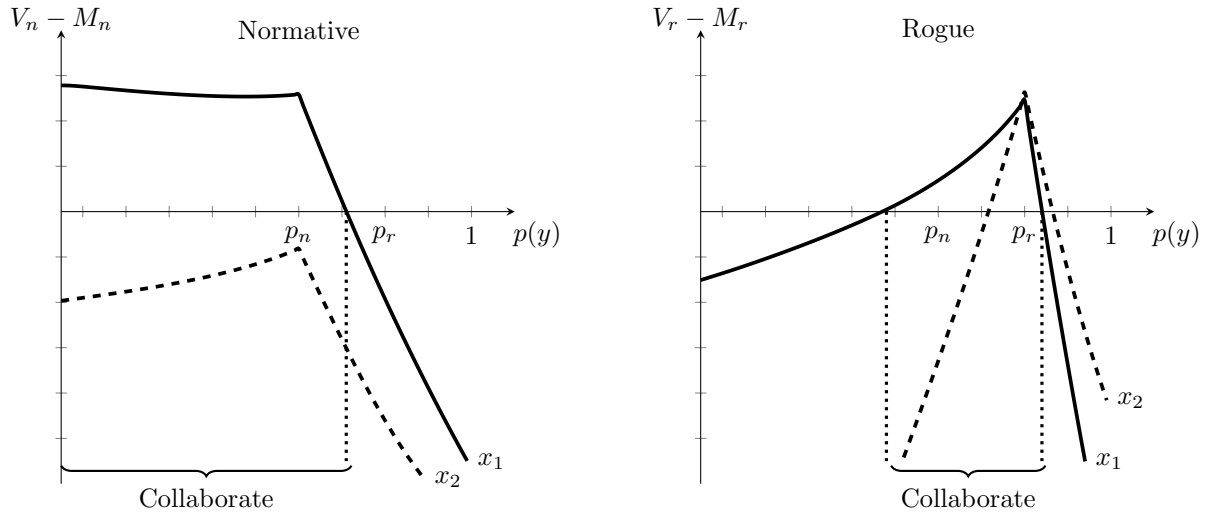


Figure 6: **Collaboration Decision**. For each type $\theta \in \{n, r\}$, the figure plots $V_\theta(x, y) - M_\theta(y)$ at two fixed levels of own reputation, $x_1 < x_2$ (solid and dashed, respectively). The positive portion of each curve identifies the cross-section of the collaboration region. The thresholds $p_n \equiv p(y_n)$ and $p_r \equiv p(y_r)$ denote the myopic stopping thresholds.

tunities, whereas the terminal payoff M_θ becomes available at an opportunity. Consequently, V_θ can lie above or below M_θ, and the usual continuous-exercise argument for smooth fit does not apply. Lemma A.3 establishes that the indifference set and the boundary of the selected optimal-response collaboration region have measure zero. It does not require a nonzero gradient at every boundary point

5.2. Equilibrium Characterization

Given any conjecture about the partner's stopping behavior, Proposition 6 determines a unique continuation value and specifies an optimal decision rule through the stated selection convention. The reputation dynamics governing that response, however, depend on the conjectured behavior itself through the survival drift. An equilibrium must therefore be self-confirming: stopping behavior must be optimal under the reputational consequences that it generates.

The comparison below makes opportunities rarer while holding π/λ fixed. This scaling preserves the balance between collaboration profits and exposure to adverse terminal actions. Specifically, setting $\pi = \kappa\lambda$ with $\kappa \in (0, L_n)$ maintains the restriction $\lambda L_\theta > \pi$ for both types. At a fixed positive flow payoff, sufficiently infrequent opportunities would violate this restriction, because collaboration profits would exceed the instantaneous expected loss from an adverse action. The following proposition establishes existence and uniqueness along this proportionally scaled family.

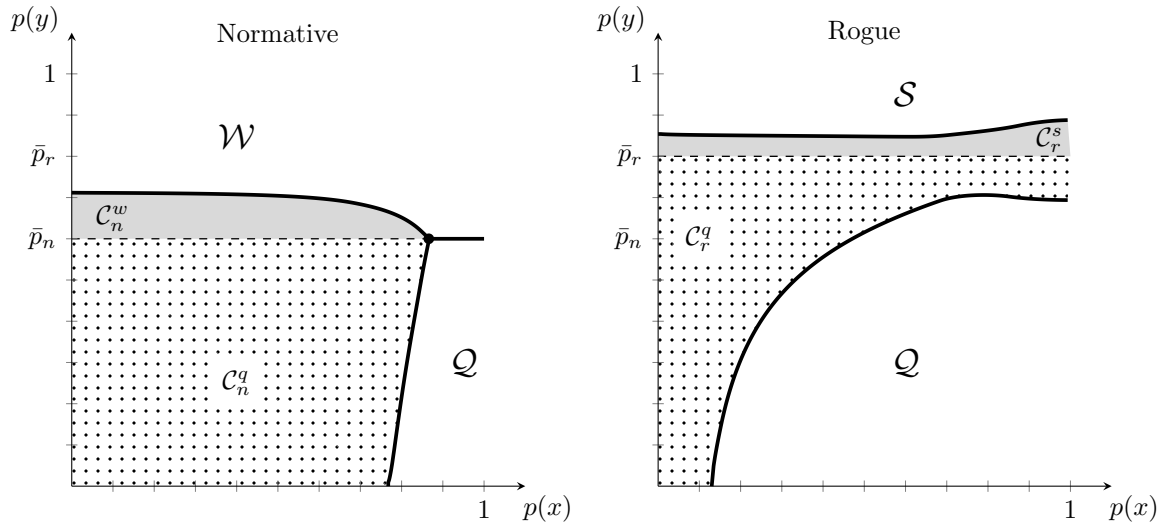


Figure 7: **Equilibrium.** Shaded areas denote collaboration regions, partitioned according to the myopically optimal terminal action that replaces: $\mathcal{C}_\theta^a = \{(x, y) \in \mathcal{C}_\theta : m_\theta(y) = a\}$. Parameters: $\pi = 0.03$, $\rho = 0.1$, $\lambda = 0.1$, $\psi = 0.7$, $F_n = L_n = 3/2$, $L_r = 4$, $G_n = G_r = 1$.

Proposition 7. *Fix $\kappa \in (0, L_n)$ and hold all terminal-payoff parameters, ρ, and ψ fixed. Consider the family of games indexed by $\lambda > 0$, with collaboration flow payoff $\pi = \kappa\lambda$. There exists $\bar{\lambda} > 0$, depending only on these fixed parameters, such that, for every $\lambda \in (0, \bar{\lambda})$, the game admits a unique symmetric stationary Markov perfect equilibrium.*

Within this family, small λ limits two channels through which strategic feedback may generate multiple self-consistent stopping patterns. First, frequent opportunities strengthen preemption: when players can react almost immediately to changes in reputation, a player approaching a region in which her partner would act against her may quit before reaching it. This anticipated stopping region thus shapes exit before players reach it, and earlier exit changes what survival reveals.[6] Second, frequent opportunities make survival a stronger signal of type, due to the survival drift being proportional to λ. Thus differences in stopping behavior can become the main force shaping reputation evolution when λ is large relative to $\psi^2/2$.

Two notions describe the shape of the collaboration regions. Type θ *exhibits fear* on an interval (x_1, x_2), possibly unbounded, if she ends the game at the first opportunity at which her own reputation lies in that interval, whatever her partner's reputation is. Type θ *exhibits trust* on (x_1, x_2) if, for some partner reputation y, she collaborates at

[6]Continuous-time timing games may lack pure-strategy equilibria or admit multiple equilibria; see Fudenberg and Tirole (1985); Riedel and Steg (2017) for preemption and Hendricks et al. (1988); Myatt (2025) for attrition games.

every own reputation in that interval,

$$\begin{aligned} \textit{Fear} \text{ on } (x_1, x_2): &\quad \mathcal{C}_\theta \cap \big((x_1, x_2) \times \mathbb{R}\big) = \varnothing, \\ \textit{Trust} \text{ on } (x_1, x_2): &\quad (x_1, x_2) \times \{y\} \subseteq \mathcal{C}_\theta \text{ for some } y \in \mathbb{R}. \end{aligned}$$

Fear captures terminations that are triggered by her own reputation, within the interval, no partner reputation is reassuring enough to sustain collaboration. Trust, by contrast, means that with the right partner, no reputation in the interval prompts her to end the game. The two notions are mutually exclusive: a type that exhibits fear on an interval cannot exhibit trust on any interval that overlaps it.

Proposition 8. *There exist $\bar{L}_n^\infty, \bar{L}_r^\infty > 0$ and $\bar{\pi} > 0$ such that the unique equilibrium has the following property:*

(i) Normative players exhibit fear on (x_{L_n}, ∞) for some finite x_{L_n} if $L_n > \bar{L}_n^\infty$, and exhibit trust otherwise.

(ii) Rogue players exhibit trust if $\pi > \bar{\pi}$, and exhibit fear at some interior interval (x_1, x_2), if $\pi < \bar{\pi}$ and $L_r > \bar{L}_r^\infty$.

Figure 7 plots an example of the unique equilibrium when the normative type exhibit fear. Distinguish the end of collaboration under uncertainty from dissolution of the economic partnership. The stopping time τ continues to denote the first terminal action and hence the end of the game under uncertainty. A successful scheme between two rogues leads to an absorbing accomplice relationship, whose continuation payoff is summarized by G_r. No subsequent dissolution is modeled in that state. Using the terminal action a^τ defined in the Model, define the partnership dissolution time by

$$\tau^P := \begin{cases} \infty, & \text{if } \tau = \infty \text{ or } (\tau < \infty,\ a^\tau = s,\ \theta_i = \theta_j = r), \\ \tau, & \text{otherwise.} \end{cases}$$

Thus, $\tau < \infty$ need not imply $\tau^P < \infty$. The probabilities below use $\mathbb{P}_{x,y}^{\theta_i,\theta_j}$, with both actual types fixed.

Proposition 9. *In the equilibrium of Proposition 7, the following statements hold for every finite initial reputation profile $(x, y) \in \mathbb{R}^2$:*

1. Mixed partnerships dissolve almost surely:

$$\mathbb{P}_{x,y}^{n,r}(\tau^P < \infty) = \mathbb{P}_{x,y}^{r,n}(\tau^P < \infty) = 1.$$

2. *Two normative players remain together forever with positive probability:*

$$\mathbb{P}^{n,n}_{x,y}(\tau^P = \infty) = \mathbb{P}^{n,n}_{x,y}(\tau = \infty) > 0.$$

3. *For two rogues, the game under uncertainty ends almost surely, but the economic partnership survives forever with positive probability:*

$$\mathbb{P}^{r,r}_{x,y}(\tau < \infty) = 1, \qquad \mathbb{P}^{r,r}_{x,y}(\tau^P = \infty) = \mathbb{P}^{r,r}_{x,y}(\tau < \infty,\ a^\tau = s) > 0.$$

6. Discussion

In this model, reputational concerns destroy partnerships rather than information. These concerns arise because the organization in which partnerships develop enforces a social norm, and players differ in whether they adhere to it. Yet adherence to the norm is irrelevant to joint output. Partnerships therefore end for reasons that have nothing to do with their productivity. The model suggests that this enforcement not only purges those who defy the norm but also generates false accusations in equilibrium. A lower cost of false accusations makes normative players act on weaker evidence and their partners more fearful, lowering the value of every partnership. Beyond these direct effects, partnership survival sorts the organization's members, so small cliques of rogues emerge.

Three simplifications qualify this model. First, talk is exogenous: the signals capture what players reveal unintentionally, not what they choose to say, so the model shuts down the channel by which reputational concerns destroy information (Morris, 2001; Ely and Välimäki, 2003). Second, the uniqueness result for the two-sided game covers only rare opportunities. When opportunities are frequent, preemption and learning from survival may sustain several equilibria, which might help explain how abruptly reputation concerns can escalate into a witch hunts. Finally, the analysis follows a single pair. In an organization, workers whose collaboration ends would seek new matches, and assortative survival suggests that reputation concern would gradually segregate the organization by type.

A. Technical Foundations

All stochastic processes are defined on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The main text treats the reputation profile (X^i, X^j) as a well-posed Markov diffusion and each value function V_θ as an almost-everywhere differentiable solution of a dynamic-programming equation. This appendix establishes both. The starting point is the well-posedness of the system introduced in Proposition 1.

Lemma A.1. *The system of SDEs in Proposition 1 admits a unique strong solution. Moreover, the resulting process is strong Markov and strong Feller.*

The strong Markov property supports the dynamic-programming representation below, while the strong Feller property is used to establish continuity of the value functions. The transition-density property also implies zero occupation time on Lebesgue-null sets, which will make boundary selections immaterial for realized play. The proof of Lemma A.1 gives the diffusion results used here.

A.1. HJB Equation and Regularity

Proposition 6 characterizes the best-response value. I collect the generator and opportunity maps used in its dynamic-programming representation here. For a type-θ player facing the conjectured profile (α, β), define

$$\mathcal{L}_\theta^{\alpha,\beta} = \frac{\psi^2}{2}(\partial_{xx} + \partial_{yy}) + \mu_i(x, y; \theta, \alpha)\partial_x + \mu_j(x, y; \beta)\partial_y, \tag{A.1}$$

where

$$\mu_i(x, y; \theta, \alpha) := \Lambda(x, y; \alpha) + \frac{\psi^2}{2}\big(\mathbb{1}\{\theta = r\} - \mathbb{1}\{\theta = n\}\big), \tag{A.2}$$

$$\mu_j(x, y; \beta) := \Lambda(y, x; \beta) + \frac{\psi^2}{2}\tanh(y/2). \tag{A.3}$$

For a symmetric conjecture, write $\mathcal{L}_\theta^\alpha := \mathcal{L}_\theta^{\alpha,\alpha}$. I suppress fixed strategy arguments in the generator, its drifts, and the opportunity maps below. For a bounded measurable candidate value f, the own-opportunity map is

$$\Upsilon_\theta(x, y; f) := \max\{M_\theta(y), f(x, y)\}. \tag{A.4}$$

The partner-opportunity map under α is[7]

$$\Gamma^{\alpha}_{\theta}(x,y;f) := p(y)\left[\mathbb{1}\{(y,x)\in\mathcal{C}_r(\alpha)\}\, f(x,y) + \mathbb{1}\{(y,x)\notin\mathcal{C}_r(\alpha)\}\, u^p(\alpha_r(y,x);\theta)\right] \\ + (1-p(y))\left[\mathbb{1}\{(y,x)\in\mathcal{C}_n(\alpha)\}\, f(x,y) + \mathbb{1}\{(y,x)\notin\mathcal{C}_n(\alpha)\}\, u^p(\alpha_n(y,x);\theta)\right].$$

The first bracket conditions on a rogue partner. If she collaborates at state (y,x), player i's value remains $f(x,y)$; otherwise, she ends the game at her first opportunity with action $\alpha_r(y,x)$, and player i receives $u^p(\alpha_r(y,x);\theta)$. The second bracket does the same for a normative partner.

Write $\delta := \rho + 2\lambda$. The value function $V_\theta(\cdot;\alpha)$ from Proposition 6 is characterized as the solution of the Hamilton–Jacobi–Bellman equation

$$\rho V_\theta = \pi + \mathcal{L}_\theta V_\theta + \lambda\big(M_\theta(y) - V_\theta\big)^+ + \lambda\big(\Gamma_\theta(x,y;V_\theta) - V_\theta\big). \tag{A.5}$$

Equivalently,

$$\mathcal{L}_\theta V_\theta - \delta V_\theta = -\pi - \lambda(\Upsilon_\theta + \Gamma_\theta)(x,y;V_\theta). \tag{A.6}$$

The corresponding resolvent representation is

$$V_\theta(x,y) = \mathbb{E}^{\theta}_{x,y}\left[\int_0^\infty e^{-\delta s}\big[\pi + \lambda(\Upsilon_\theta + \Gamma_\theta)(X^i_s, X^j_s; V_\theta)\big]\, ds\right]. \tag{A.7}$$

The proof of Lemma A.2 establishes this representation from the Bellman fixed point. In expressions evaluated along the diffusion, the state arguments of the opportunity maps are suppressed when unambiguous. Equation (A.5) admits a natural asset-pricing interpretation. The value V_θ can be viewed as the value of an asset that pays the dividend π while the partnership remains active through collaboration. Changes in the reputation profile generate the expected capital gain $\mathcal{L}_\theta V_\theta$. In addition, liquidation opportunities arrive at rate λ. At the player's own opportunity, the asset is liquidated for M_θ whenever liquidation is optimal, producing the term $\lambda(M_\theta - V_\theta)^+$. Similarly, at the partner's opportunity, control over liquidation passes to them, with the resulting value summarized by $\lambda(\Gamma_\theta - V_\theta)$. Thus, the discounted value ρV_θ equals the dividend plus expected capital gains from reputation dynamics and from the possibility that either player ends the partnership.

Because stopping is constrained to the opportunity process, a player cannot exercise

[7]For notational convenience, terminal-payoff functions are extended by zero at the continuation action: $u^p(c;\theta) = \bar{u}(c;y) = 0$. Continuation values enter separately through $f(x,y)$.

the stopping option immediately at an arbitrary state. As a result, M_θ does not impose the pointwise constraint $V_\theta \geq M_\theta$ familiar from standard continuous-time optimal stopping, and the usual smooth-pasting condition need not hold at the stopping boundary. In addition, the indicator functions entering Γ_θ generate discontinuities across the partner's stopping boundaries, so classical C^2 solutions need not exist. This motivates the next result.

Lemma A.2 (Regularity)**.** *Let α be a myopically optimal strategy and fix $\theta \in \{n, r\}$. The value function $V_\theta(\cdot; \alpha)$ defined in* (6) *satisfies:*

(i) $V_\theta(\cdot; \alpha) \in W^{2,p}_{\text{loc}}(\mathbb{R}^2)$ for every $p \in (1, \infty)$;

(ii) $V_\theta(\cdot; \alpha) \in C^{1,\gamma}_{\text{loc}}(\mathbb{R}^2)$ for every $\gamma \in (0, 1)$;

(iii) for every $p > 2$, $V_\theta(\cdot; \alpha)$ is the unique bounded $W^{2,p}_{\text{loc}}$ strong solution of (A.5).

Lemma A.2 upgrades continuity to continuous differentiability despite the stopping-rule discontinuities, and identifies the probabilistic value with the unique bounded strong solution of the HJB equation. Its proof supplies the local Sobolev estimates and embedding argument.

Lemma A.3 (Null indifference sets)**.** *Let α be a myopically optimal conjectured strategy. For each $\theta \in \{n, r\}$, the indifference set $\mathcal{U}_\theta(\alpha)$ in* (9) *has two-dimensional Lebesgue measure zero. Consequently, the selected-response collaboration region $\mathcal{C}^{BR}_\theta(\alpha)$ in* (8) *has boundary of two-dimensional Lebesgue measure zero.*

The conclusion concerns the optimal response to α. It imposes no restriction on the boundary of the collaboration region $\mathcal{C}_\theta$ prescribed by an arbitrary conjecture. By the transition-density property in Lemma A.1, the reputation process spends zero time in $\mathcal{U}_\theta(\alpha)$. The compensation formula for Poisson opportunities also implies that, almost surely, no positive opportunity time occurs in this set. Thus, alternative choices at collaboration-stopping indifference do not affect realized play.

Lemma A.4 (Stability)**.** *Consider two possible strategies α and $\widetilde{\alpha}$ employed by the partner. Suppose $\|V_\theta(\cdot; \alpha) - V_\theta(\cdot; \widetilde{\alpha})\|_\infty \leq \varepsilon$ for each $\theta \in \{n, r\}$. Then the optimal decision rules $\varphi^*_\theta(\cdot; \alpha)$ and $\varphi^*_\theta(\cdot; \widetilde{\alpha})$ can differ only on the ε-indifference region:*

$$\{(x, y) \in \mathbb{R}^2 : \varphi^*_\theta(x, y; \alpha) \neq \varphi^*_\theta(x, y; \widetilde{\alpha})\} \subseteq \{(x, y) : |V_\theta(x, y; \alpha) - M_\theta(y)| \leq \varepsilon\}.$$

Away from indifference, a sufficiently small change in the continuation value preserves the selected action. For a fixed myopically optimal conjecture α, Lemma A.3 implies that, for every compact set $K \subset \mathbb{R}^2$,

$$\mathrm{Leb}_2(K \cap \{(x,y) : |V_\theta(x,y;\alpha) - M_\theta(y)| \le \varepsilon\}) \longrightarrow 0 \quad \text{as } \varepsilon \downarrow 0.$$

Consequently, uniform convergence of continuation values to $V_\theta(\cdot;\alpha)$ implies convergence of the selected optimal decision rules pointwise outside the limiting indifference set and in Lebesgue measure on every compact set, using the same selection convention throughout.

Uniform value bound and stopping region. Fix a selected symmetric equilibrium. Every terminal payoff to a type-θ player is at most G_θ, and $\pi/\rho < G_\theta$. No terminal action can occur before the first opportunity of either player, so paying G_θ at that opportunity gives

$$V_\theta(x,y) \le \overline{V}_\theta := \frac{\pi + 2\lambda G_\theta}{\rho + 2\lambda} < G_\theta. \tag{A.8}$$

Since $M_\theta(y) \to G_\theta$ as $y \to \infty$, choose finite b_n, b_r such that

$$M_\theta(y) > \overline{V}_\theta \quad \text{for } y > b_\theta, \qquad b := \max\{b_n, b_r\}. \tag{A.9}$$

By Proposition 6, a normative player whistleblows and a rogue player schemes whenever her partner's reputation exceeds b, regardless of her own reputation. These bounds and thresholds will be used in the geometry and survival proofs.

B. Solution of the normative insider game

This appendix derives the results of a stranger facing a normative insider.

B.1. Equilibrium

Let $\alpha = (\alpha_n, \alpha_r)$ describe the stranger's conjectured decision rules and β the insider's. Since the insider is known to be normative, the stranger's decision rule is restricted to collaborate or quit, as scheme and whistleblow are strictly dominated. Similarly, the insider's decision rule is restricted to collaborate or whistleblow, since quit is dominated by collaboration. The reputation process follows from Proposition 1, which specializes

to

$$dX_t = \left(\Lambda(X_t;\alpha) + \frac{\psi^2}{2}\big(\mathbb{1}\{\theta = r\} - \mathbb{1}\{\theta = n\}\big)\right) dt + \psi dB_t^i, \tag{B.10}$$

$$dX_t = \left(\Lambda(X_t;\alpha) + \frac{\psi^2}{2}\tanh\left(\frac{X_t}{2}\right)\right) dt + \psi\, d\widetilde{B}_t^i, \tag{B.11}$$

where $\theta \in \{n, r\}$ is the stranger's type. Equation (B.10) describes X under the stranger's information, while equation (B.11) describes it under the insider's information; $\widetilde{B}^i$ is the innovation Brownian motion associated with the stranger's signal in the insider's filtration. For each type $\theta \in \{n, r\}$, the stranger's Bellman equation is

$$\rho V_\theta = \pi + \frac{\psi^2}{2}V_\theta'' + \left(\Lambda + \frac{\psi^2}{2}\big(\mathbb{1}\{\theta = r\} - \mathbb{1}\{\theta = n\}\big)\right) V_\theta' + \lambda(-V_\theta)^+ - \lambda(V_\theta + L_\theta)\mathbb{1}\{\beta = w\}, \tag{B.12}$$

while the insider's Bellman equation is

$$\rho W = \pi + \frac{\psi^2}{2}W'' + \left[\Lambda + \frac{\psi^2}{2}\tanh(x/2)\right] W' + \lambda[(\bar{u}(w;x) - W)^+ - HW],$$

Here $H(x) := p(x)\mathbb{1}\{\alpha_r(x) = q\} + (1 - p(x))\mathbb{1}\{\alpha_n(x) = q\} \in [0, 1]$ is the probability that the stranger ends the game at an opportunity.

Fix a finite insider cutoff x_I and conjectured stranger cutoff decision rules α. The conjectured cutoffs determine Λ, while V_θ is the optimized value against that drift and insider cutoff. Its zero need not coincide with the type's conjectured cutoff until consistency is imposed. To characterize the stranger's optimal response, the first step is to show that her continuation value $V_\theta(x)$ is non-increasing in reputation x. Since quitting yields zero, this implies that the region in which collaboration is strictly optimal is a lower interval.

Construct successive continuation values, starting from zero. Each update computes the expected discounted flow until the next opportunity and the discounted value at that opportunity. At the stranger's own opportunity, the value is the maximum of the preceding continuation value and zero. At an insider opportunity, the value to the stranger is continuation below x_I and $-L_n$ if normative, or $-L_r$ if rogue, above it. If the preceding continuation value is non-increasing and bounded below by the corresponding terminal payoff, both functions are non-increasing: the switch at x_I cannot create an upward jump. The lower bound is preserved by every update because the flow payoff is positive and terminal payoffs are bounded below by $-L_\theta$.

Holding the conjectured strategy fixed, the one-dimensional reputation dynamics

preserve the ordering of initial reputations when paths are compared under the same shocks. Consequently, taking discounted expectations of the non-increasing opportunity payoffs preserves monotonicity in the initial reputation x. The discounted Bellman operator resulting from this process is a contraction, as shown later in the proof of Proposition 6. Its iterates therefore converge to V_θ, which inherits their monotonicity and is thus non-increasing in x. The tail limits of V_θ are

$$\lim_{x\to-\infty} V_\theta(x) = \frac{\pi}{\rho} > 0, \qquad \lim_{x\to+\infty} V_\theta(x) = \frac{\pi - \lambda L_\theta}{\rho + 2\lambda}. \tag{B.13}$$

Hence, continuity implies that V_θ has a zero. Since V_θ is non-increasing, its zero set is an interval. Furthermore, that interval cannot have a positive length, as the Bellman equation (B.12) would reduce, on a subinterval not containing x_I, to

$$0 = \pi - \lambda L_\theta \mathbb{1}\{x \geq x_I\}.$$

This identity cannot hold: below x_I it requires $\pi = 0$, and above x_I it requires $\pi = \lambda L_\theta$, which contradicts the assumption $\pi < \lambda L_\theta$. Thus, each type satisfying its condition has a unique zero, that is, there is a unique reputation x_θ such that $V_\theta(x_\theta) = 0$. Thus, a stranger of type θ optimally chooses to collaborate for all reputations $x < x_\theta$ and quits at all reputations $x \geq x_\theta$, given the opportunity. The strict comparison $V_n(x) > V_r(x)$, established below, implies $x_n > x_r$.

A equilibrium must satisfy the following Bellman equations and matching conditions: the strangers' Bellman equations

$$\begin{aligned}
\rho V_n &= \pi + \frac{\psi^2}{2} V_n'' + \left(\Lambda - \frac{\psi^2}{2}\right) V_n' - \lambda \left[\mathbb{1}\{x \geq x_n^*\} V_n + \mathbb{1}\{x \geq x_I^*\}(V_n + L_n)\right], \\
\rho V_r &= \pi + \frac{\psi^2}{2} V_r'' + \left(\Lambda + \frac{\psi^2}{2}\right) V_r' - \lambda \left[\mathbb{1}\{x \geq x_r^*\} V_r + \mathbb{1}\{x \geq x_I^*\}(V_r + L_r)\right],
\end{aligned}$$

the insider's Bellman equation

$$\rho W = \pi + \frac{\psi^2}{2} W'' + \left[\Lambda + \frac{\psi^2}{2}\tanh(x/2)\right] W' + \lambda[\mathbb{1}\{x \geq x_I^*\}(\bar{u}(w;x) - W) - HW],$$

and the survival drift

$$\Lambda(x;\alpha^*) = \lambda[\mathbb{1}\{x \geq x_n^*\} - \mathbb{1}\{x \geq x_r^*\}],$$

with value matching conditions $V_\theta(x_\theta^*) = 0$ and $W(x_I^*) = \bar{u}(w; x_I^*)$, and the values and

first derivatives are continuous across the cutoffs.

Next, I show that, for each fixed finite insider cutoff x_I, there is exactly one consistent stranger pair. Fix a finite insider cutoff x_I. Since $|\Lambda| \leq \lambda$, the reputation drift is uniformly bounded over conjectured cutoffs. Over any fixed time interval, the probability of crossing x_I vanishes uniformly as the initial reputation approaches either extreme. Bounded payoffs and discounting therefore make the tail limits in (B.13) uniform over conjectures. Moving a conjectured cutoff changes the drift only on a shrinking interval. The corresponding discounted occupation tends to zero for this nondegenerate diffusion, so the optimized values depend continuously on the conjectured cutoffs.

For each conjectured x_n, there is at most one consistent rogue cutoff. At a zero of the rogue consistency condition, the rogue's value under that conjecture is strictly negative at every higher reputation. A higher conjectured rogue cutoff raises Λ, weakly lowering her value at each fixed reputation. Its consistency value is therefore strictly negative, ruling out two consistent rogue cutoffs. Continuity and the positive and negative tail limits guarantee existence. Denote the resulting cutoff by $x_r(x_n)$. Uniqueness, continuity of the consistency condition, and the uniform tail limits imply that this response is continuous.

Along this consistent rogue response, the normative value evaluated at the conjectured x_n is continuous. The same uniform tail limits make it positive for sufficiently low conjectured cutoffs and negative for sufficiently high conjectured cutoffs. It therefore has at least one zero. Consider any such zero, so both strangers are consistent. The regularity argument in Appendix A gives continuously differentiable values, smooth between the cutoffs. Differentiating their Bellman equations with respect to reputation, away from the cutoffs, gives

$$\frac{\psi^2}{2}V_r''' + \left(\Lambda + \frac{\psi^2}{2}\right)V_r'' - [\rho + \lambda(\mathbb{1}\{x \geq x_r\} + \mathbb{1}\{x \geq x_I\})]V_r' = 0,$$

and

$$\frac{\psi^2}{2}V_n''' + \left(\Lambda - \frac{\psi^2}{2}\right)V_n'' - [\rho + \lambda(\mathbb{1}\{x \geq x_n\} + \mathbb{1}\{x \geq x_I\})]V_n' = 0,$$

Now substitute $e^{-x}V_n'$ into the differential equation satisfied by V_r', using

$$\Lambda = \lambda[\mathbb{1}\{x \geq x_n\} - \mathbb{1}\{x \geq x_r\}]$$

at the consistent pair,

$$\frac{\psi^2}{2}\left(e^{-x}V_n'\right)'' + \left(\Lambda + \frac{\psi^2}{2}\right)(e^{-x}V_n')' - [\rho + \lambda(\mathbb{1}\{x \geq x_r\} + \mathbb{1}\{x \geq x_I\})]e^{-x}V_n' = 0,$$

$$e^{-x}\left\{\frac{\psi^2}{2}V_n''' + \left(\Lambda - \frac{\psi^2}{2}\right)V_n'' - [\rho + \lambda(\mathbb{1}\{x \geq x_n\} + \mathbb{1}\{x \geq x_I\})]V_n'\right\} = 0,$$

Thus, V_r' and $e^{-x}V_n'$ satisfy the same second-order differential equation away from the cutoffs. Below all cutoffs, the common differential equation has the following characteristic equation and roots,

$$\xi^2 + \xi - \frac{2\rho}{\psi^2} = 0, \qquad \xi = \frac{-1 \pm \sqrt{1 + 8\rho/\psi^2}}{2}.$$

Boundedness of the original value function as $x \to -\infty$ excludes the negative root. Consequently, both V_r' and $e^{-x}V_n'$ are constant multiples of

$$\exp\left[\frac{-1 + \sqrt{1 + 8\rho/\psi^2}}{2}x\right].$$

To extend this proportionality across the stranger cutoffs, suppose first that the cutoffs are distinct. Subtract the left-hand limit of each stranger's Bellman equation from its right-hand limit. The values and first derivatives are continuous. Moreover, the change in a stranger's own quitting term contributes zero at her cutoff because $V_\theta(x_\theta) = 0$. Since Λ decreases by λ at x_r, and increases by λ at x_n,

$$\frac{\psi^2}{2}\left(\lim_{x \downarrow x_r} V_\theta''(x) - \lim_{x \uparrow x_r} V_\theta''(x)\right) - \lambda V_\theta'(x_r) = 0.$$
$$\frac{\psi^2}{2}\left(\lim_{x \downarrow x_n} V_\theta''(x) - \lim_{x \uparrow x_n} V_\theta''(x)\right) + \lambda V_\theta'(x_n) = 0.$$

Because $(e^{-x}V_n')' = e^{-x}(V_n'' - V_n')$, and V_n' is continuous, the jump in this derivative is e^{-x} times the jump in V_n''. Thus, the two functions obey the same matching rules: each is continuous, and its derivative jump at each stranger cutoff is proportional to its own value. If their values and derivatives are proportional immediately before a stranger cutoff, they remain proportional immediately after it, with the same factor. Uniqueness for their common second-order differential equation preserves proportionality throughout the next interval. Repeating this argument establishes proportionality throughout $x < x_I$.

On the right side, start from the right tail $x \to +\infty$. Above all cutoffs, the common differential equation has the following characteristic equation and roots,

$$\xi^2 + \xi - \frac{2\rho + 4\lambda}{\psi^2} = 0, \qquad \xi = \frac{-1 \pm \sqrt{1 + 8(\rho + 2\lambda)/\psi^2}}{2}.$$

Boundedness selects the common decaying root. Then, applying the same argument backward through the stranger cutoffs establishes proportionality through $x > x_I$.

Finally, both functions are continuous at x_I and are strictly negative there, as follows. Monotonicity already gives $V'_\theta \leq 0$, so suppose $V'_\theta(x_I) = 0$. Then, V'_θ attains its maximum at x_I, so V''_θ can only jump downward here. But the Bellman equation gives

$$\lim_{x \downarrow x_I} V''_\theta(x) - \lim_{x \uparrow x_I} V''_\theta(x) = \frac{2\lambda}{\psi^2}[V_\theta(x_I) + L_\theta] > 0,$$

where positivity follows from

$$V_\theta(x) \geq \frac{\pi - \lambda L_\theta}{\rho + \lambda} > -L_\theta.$$

Indeed, always collaborating does at least as well as facing an insider who whistleblows at every opportunity; the payoff in that latter case is the displayed lower bound. The derivative cannot vanish at another finite reputation either. A zero of the non-positive function V'_θ is a maximum. Away from the cutoffs, its derivative must therefore be zero. At a stranger cutoff, the homogeneous jump condition also makes the two one-sided derivatives agree, so both must be zero. Uniqueness for the homogeneous differential equation and its matching conditions would then propagate this zero to x_I, contradicting the argument above. Hence $V'_\theta(x) < 0$ at every finite reputation. Therefore continuity at x_I makes the proportionality factors on its two sides coincide, yielding

$$\frac{V'_n(x)}{V'_n(x_I)} = e^{x - x_I} \frac{V'_r(x)}{V'_r(x_I)}. \tag{B.14}$$

If cutoffs coincide, their jump contributions are added. Proportionality is still propagated separately up to x_I from each tail.

The strict comparison between the stranger types follows from the Bellman equations. The normative stranger has a lower reputation drift, and $V'_r < 0$ makes the gain from that lower drift strictly positive: substituting V_r into the normative Bellman equation contributes $-\psi^2 V'_r(x) > 0$. The weakly smaller whistleblowing loss contributes the additional nonnegative term $\lambda \mathbb{1}\{x \geq x_I\}(L_r - L_n)$. Thus V_r is a strict subsolution of the normative Bellman equation, and strong comparison gives $V_n(x) > V_r(x)$. At x_r, therefore, $V_n(x_r) > V_r(x_r) = 0$, so the unique zeros satisfy $x_r < x_n$.

Before differentiating with respect to conjectured cutoffs, local smooth dependence also needs justification. For prescribed conjectured and own quitting cutoffs, the values solve linear equations with continuous value and first-derivative matching. Their

exponential solutions and matching conditions depend smoothly on the cutoff positions within each strict ordering. The matching system is nonsingular because the bounded discounted value for a prescribed policy is unique. At an optimal own quitting cutoff, moving that cutoff produces no first-order source because its value is zero. Since $V_\theta'(x_\theta) < 0$, the implicit function theorem gives local smooth dependence of the optimal cutoff and value. At rogue consistency, the derivative of its indifference condition with respect to the conjectured rogue cutoff is

$$V_r'(x_r) + \left.\frac{\partial V_r(x;\alpha)}{\partial x_r}\right|_{x=x_r} < 0 :$$

the first term is strictly negative, and the second is nonpositive because raising the conjectured rogue cutoff raises the drift. Hence $x_r(x_n)$ is locally differentiable at the consistent pair. When x_I coincides with a stranger cutoff, the same argument uses one-sided variations, with the coincident jump contributions added.

Equipped with the proportionality relation (B.14), I now bound the first-order change in the rogue stranger's value induced by increasing the conjectured normative cutoff x_n, while readjusting $x_r(x_n)$ to preserve the rogue's indifference at her cutoff. Throughout this comparison, the current reputation x and insider cutoff x_I remain fixed.

To bound the rogue's response to the changing cutoffs, I compare it with $-V_r'(x)$. At each reputation, consider the difference between $-V_r'(x)$ and the change in the rogue's value per unit increase in x_n, with $x_r(x_n)$ preserving the indifference,

$$-V_r'(x) - \frac{d}{dx_n}V_r(x;\alpha) = -V_r'(x) - \left[\frac{\partial V_r}{\partial x_n} + \frac{\partial V_r}{\partial x_r}\frac{dx_r(x_n)}{dx_n}\right]. \tag{B.15}$$

Between cutoffs, the indicators defining Λ are locally unchanged when the cutoff positions move. Differentiating the rogue's Bellman equation with respect to x_n therefore gives the same homogeneous equation satisfied by V_r'. By linearity, the difference (B.15) also satisfies this equation.

Suppose first that the cutoffs are distinct. Differentiating rogue indifference gives,

$$\left.\frac{dV_r(x;\alpha)}{dx_n}\right|_{x=x_r} + V_r'(x_r)\frac{dx_r}{dx_n} = 0.$$

Combining this equality with the differentiated continuity conditions shows that the difference is continuous and that its reputation derivative jumps at x_r by

$$\frac{2\lambda}{\psi^2}\left[-V_r'(x_r) - \left.\frac{dV_r(x;\alpha)}{dx_n}\right|_{x=x_r}\right]$$

At x_n, its derivative is continuous: the jump in the derivative of $-V_r'$ cancels the jump induced by moving x_n. At the fixed x_I, its derivative jumps by

$$-\frac{2\lambda}{\psi^2}\left[V_r(x_I) + L_r\right] < 0.$$

The difference also approaches zero at both extremes ($x \to +\infty$ and $x \to -\infty$). On each tail, V_r equals its corresponding limiting value in (B.13) plus a multiple of a decaying exponential. Changing the cutoffs changes the multiplier, while the exponent remains unchanged. Thus both V_r' and dV_r/dx_n approach zero.

These conditions imply that the difference is strictly positive. If it were negative somewhere, its zero tail limits would imply a negative minimum. Between cutoffs, such a minimum contradicts the homogeneous equation because the discount coefficient is positive. At x_r, a negative minimum would require a nonnegative derivative jump, whereas the prescribed jump is negative. The strictly negative jump at x_I likewise excludes a minimum there. At x_n, the derivative is continuous, so the same one-sided differential-equation argument applies.

A zero minimum is also impossible: uniqueness for the homogeneous equation would propagate zero value and zero derivative to x_I, contradicting the strictly negative jump there. Consequently, the difference in (B.15) is strictly positive at every finite reputation. If x_I coincides with a stranger cutoff, the jump contributions are added: at a negative or zero minimum, their combined jump has the same contradictory sign. One-sided variations therefore give the same conclusion.

Now consider any zero of the normative consistency condition, so that both stranger cutoffs are consistent. Apply the exponential substitution to the linearized Bellman equations for the value changes. Their moving-cutoff derivative jumps involve

$$-\frac{2\lambda}{\psi^2}V_\theta'(x_r)\frac{dx_r}{dx_n} \quad \text{and} \quad \frac{2\lambda}{\psi^2}V_\theta'(x_n).$$

Equation (B.14) makes these jumps proportional after multiplication of the normative value change by e^{-x}. The transformed change $e^{-x}dV_n/dx_n$ tends to zero at both extremes: on each tail, changing the cutoffs changes only the multiplier of the exponential, whose exponent was identified above. The rogue value change also tends to zero at both extremes. Uniqueness for their common linear equation with these tail and matching conditions gives

$$\frac{dV_n(x;\alpha)}{dx_n} = e^{x-x_I}\frac{V_n'(x_I)}{V_r'(x_I)}\frac{dV_r(x;\alpha)}{dx_n}. \tag{B.16}$$

Movement of either stranger's own optimal quitting cutoff contributes no additional first-order boundary term, because her value there is zero.

Lastly, evaluate the normative value at the moving conjectured cutoff. The chain rule and equations (B.14) and (B.16) give

$$\begin{aligned}\frac{dV_n(x_n;\alpha)}{dx_n} &= V_n'(x_n) + \left.\frac{dV_n(x;\alpha)}{dx_n}\right|_{x=x_n} \\ &= e^{x_n - x_I}\frac{V_n'(x_I)}{V_r'(x_I)}\left[V_r'(x_n) + \left.\frac{dV_r(x;\alpha)}{dx_n}\right|_{x=x_n}\right] \\ &< 0.\end{aligned}$$

The multiplier is positive, and the bracket is negative by the bound just proved. This inequality holds at every zero of the normative consistency condition. Since every zero is crossed strictly downward, it cannot have two zeros. Hence there is exactly one consistent stranger pair for the fixed insider cutoff.

Before varying x_I, it remains to show that an insider facing any ordered stranger cutoffs $x_r < x_n$ optimally uses a cutoff. This is needed because indifference at a candidate x_I alone does not verify the insider's behavior at other reputations. Subtracting $\bar{u}(w;x)$ from the insider's Bellman equation gives

$$\begin{aligned}0 = \frac{\psi^2}{2}(W - \bar{u}(w;x))'' + \left(\Lambda + \frac{\psi^2}{2}\tanh(x/2)\right)(W - \bar{u}(w;x))' \\ - (\rho + \lambda H)(W - \bar{u}(w;x)) + \lambda(\bar{u}(w;x) - W)^+ \\ + \pi + \big(\rho + \lambda \mathbb{1}\{\alpha_n = q\}\big)F_n(1 - p(x)) - \big(\rho + \lambda \mathbb{1}\{\alpha_r = q\}\big)G_n p(x).\end{aligned}$$

The value bounds make $W - \bar{u}(w;x)$ positive at sufficiently low reputations and negative at sufficiently high reputations. Suppose it becomes positive again after being negative. Consider the resulting bounded positive interval. At its left endpoint, $\bar{u}(w;x) = W \geq \pi/(\rho + \lambda)$, so the same lower bound on $\bar{u}(w;x)$ holds throughout the interval. If its positive maximum occurred at or above x_r, the rogue would quit there, and the last line of the preceding equation would be at most

$$\pi - (\rho + \lambda)\bar{u}(w;x) \leq 0.$$

The Bellman equation would then make $(W - \bar{u}(w;x))'' > 0$ at the maximum, a contradiction. At x_r, the same conclusion follows from the right-hand equation. Hence the positive maximum lies below x_r.

The preceding negative interval has an earlier negative minimum, also below x_r. Both strangers collaborate at these two reputations, so the last line reduces to $\pi - \rho\bar{u}(w;x)$, a strictly decreasing function. The equation requires this function to be negative at the earlier negative minimum and positive at the later positive maximum, which is impossible. Thus the insider cannot return to collaboration after entering a whistleblowing region.

There can be no additional zero contacts. At a zero minimum followed by positive values, the later positive maximum must lie below x_r, as above. The zero minimum also lies below x_r and requires $\pi - \rho\bar{u}(w;x) \leq 0$, whereas the later positive maximum requires $\pi - \rho\bar{u}(w;x) > 0$, contradicting strict decrease.

A zero maximum preceded by negative values has an earlier negative minimum. If both reputations lie below x_r, the minimum requires $\pi - \rho\bar{u}(w;x) < 0$ and the later zero maximum requires $\pi - \rho\bar{u}(w;x) \geq 0$, again a contradiction. If the zero maximum is at or above x_r, the source is strictly negative there, also contradicting a maximum. This uses $W(x) > \pi/(\rho+\lambda)$ at every finite x: always collaborating delays termination beyond the stranger's first opportunity with positive probability, because both stranger types collaborate in a sufficiently low tail. At x_r, use the right-hand equation. Finally, an interval of zeros is impossible because the source is strictly decreasing between stranger cutoffs. Therefore $W - \bar{u}(w;x)$ has exactly one zero and the insider's best response is a cutoff.

I now let x_I vary. Shifting reputation and all three cutoffs by the same amount leaves the strangers' Bellman equations unchanged: the type-specific drift is constant, and the cutoff indicators are unchanged by this common shift. By uniqueness of the consistent stranger pair, $x_r - x_I$ and $x_n - x_I$ therefore remain constant as x_I varies,

$$x_\theta(x_I + \Delta) = x_\theta(x_I) + \Delta, \qquad \theta \in \{n, r\}. \tag{B.17}$$

Evaluate $W(x_I)$ against the consistent stranger pair associated with that same x_I. Here W is the insider's optimized value against that pair; the candidate cutoff x_I is not imposed on his own decision rule. Holding an insider strategy fixed in the translated coordinates gives conditional payoffs independent of x_I, so its expected payoff is affine in $p(x_I)$. Taking the supremum over insider strategies makes $W(x_I)$ convex in $p(x_I)$. Since the conditional payoffs are uniformly bounded, this supremum is also continuous. Since $\bar{u}(w;x_I)$ is affine in this probability, $W(x_I) - \bar{u}(w;x_I)$ is also convex and continuous.

The value bounds imply

$$\lim_{x_I \to -\infty} \big[W(x_I) - \bar{u}(w; x_I)\big] \geq \frac{\pi}{\rho + \lambda} + F_n > 0,$$
$$\lim_{x_I \to +\infty} \big[W(x_I) - \bar{u}(w; x_I)\big] \leq \frac{\pi - \rho G_n}{\rho + \lambda} < 0.$$

Consequently, this difference has exactly one zero: two zeros would, by convexity, force its right-endpoint value to be nonnegative. Because $p(x_I)$ is strictly increasing, the indifference condition $W(x_I) = \bar{u}(w; x_I)$ uniquely determines x_I. Every cutoff equilibrium must satisfy this condition. Together with uniqueness of the consistent stranger pair for fixed x_I, this proves that there is at most one cutoff equilibrium. Existence also follows. At the unique zero, the associated consistent stranger pair is optimal against the candidate insider cutoff. The insider's optimized value is indifferent at that same reputation, and his best response to the ordered stranger pair has exactly one zero, as proved above. His optimal cutoff therefore equals the candidate x_I. All three decision rules are consequently optimal and consistent, so there is exactly one cutoff equilibrium.

I now derive the critical loss $\bar{L}(\mu)$ defined in (3) and establish the sufficient cutoff-order conditions in Proposition 3. To see this, fix an equilibrium (x_n^*, x_r^*, x_I^*) and translate the reputation so that the insider's cutoff is zero. First consider a stranger facing a constant reputation drift:

$$dX_t = \mu\, dt + \psi dB_t.$$

Let $U_\theta(x; \mu)$ denote the stranger's optimized value for this constant-drift process when the insider's cutoff is zero and the stranger's own opportunities still arrive at rate λ. Suppose the stranger's optimal cutoff also equals zero. This supposition identifies a critical loss, rather than imposing the stranger's cutoff for every loss. Below zero, both players collaborate. Above zero, each terminates at the next opportunity. The value function of the stranger must therefore take the form

$$U_\theta(x; \mu) = \begin{cases} \dfrac{\pi}{\rho}(1 - e^{\xi_1(\mu)x}) & \text{if } x \leq 0, \\ \dfrac{\pi - \lambda L_\theta}{\rho + 2\lambda}(1 - e^{-\xi_2(\mu)x}) & \text{if } x > 0, \end{cases}$$

where

$$\xi_1(\mu) = \frac{\sqrt{\mu^2 + 2\psi^2\rho} - \mu}{\psi^2} > 0, \quad \xi_2(\mu) = \frac{\sqrt{\mu^2 + 2\psi^2(\rho + 2\lambda)} + \mu}{\psi^2} > 0.$$

Matching first derivatives at zero gives

$$\frac{\pi}{\rho}\xi_1(\mu) = \frac{\lambda L_\theta - \pi}{\rho + 2\lambda}\xi_2(\mu).$$

Solving for the loss yields,

$$L_\theta = \bar{L}(\mu).$$

The displayed piecewise formula applies at this critical loss; $U_\theta(x;\mu)$ continues to denote the optimized value at other losses. The value decreases strictly with the loss from the insider's whistleblow: at every finite reputation, there is positive probability of an insider accusation before the stranger's first opportunity, independently of her decision rule. Since the stranger's value decreases strictly with this loss, the critical loss determines which side of the insider's cutoff she quits on:

$$L_\theta < \bar{L}(\mu) \implies U_\theta(0;\mu) > 0 \implies x_\theta > x_I, \qquad L_\theta > \bar{L}(\mu) \implies U_\theta(0;\mu) < 0 \implies x_\theta < x_I,$$

Now return to the equilibrium. Because the rogue stranger quits weakly earlier than the normative stranger, the survival drift satisfies $-\lambda \leq \Lambda(x) \leq 0$. The two types' signals reputation drifts are $\mu_r^0 = \psi^2/2$ and $\mu_n^0 = -\psi^2/2$. Thus, coupling the processes with the same Brownian motion gives

$$x + (\mu_\theta^0 - \lambda)t + \psi B_t \leq X_t \leq x + \mu_\theta^0 t + \psi B_t.$$

A lower reputation path delays the insider's whistleblowing and raises the stranger's continuation value. Comparing these optimal stopping problems yields

$$U_\theta(x;\mu_\theta^0) \leq V_\theta(x) \leq U_\theta(x;\mu_\theta^0 - \lambda).$$

Thus,

$$L_\theta < \bar{L}(\mu_\theta^0) \implies x_\theta^* > x_I^*$$

and

$$L_\theta > \bar{L}(\mu_\theta^0 - \lambda) \implies x_\theta^* < x_I^*$$

Applying these implications to each type establishes the three stated sufficient conditions for the cutoff orderings.

B.2. Partnership Dissolution

This subsection establishes each of the four claims of Proposition 4 in turn. Fix an equilibrium with finite cutoffs. By Proposition 2, $x_r^* < x_n^*$. Conditional on the stranger's type

$$dX_t = \mu_\theta(X_t)dt + \psi dB_t, \tag{B.18}$$

where

$$\mu_\theta(x) = \mu_\theta^0 - \lambda \mathbb{1}\{x_r^* \leq x < x_n^*\}.$$

Throughout this proof, X denotes the auxiliary extension of (B.18) to all $t \geq 0$, driven by the Brownian motion independently of the opportunity processes. It agrees with equilibrium reputation until τ. I use the stopping times in (4) and $\tau = \tau_I \wedge \tau_\theta$.

Proof of Proposition 4. **Claim (i): Only partnerships with normative strangers may last forever.** Conditional on the entire reputation path, the partnership survives forever exactly when neither player receives an opportunity while reputation is above her stopping cutoff. Independence of the opportunity processes therefore gives

$$\mathbb{P}_x^\theta(\tau = \infty) = \mathbb{E}_x^\theta\left[\exp\left(-\lambda \int_0^\infty \left(\mathbb{1}\{X_t \geq x_I^*\} + \mathbb{1}\{X_t \geq x_\theta^*\}\right) dt\right)\right]. \tag{B.19}$$

For a normative stranger,

$$X_t \leq x - \frac{\psi^2}{2}t + \psi B_t \longrightarrow -\infty \qquad \mathbb{P}_x^n\text{-almost surely.}$$

Both cutoffs are finite, so the process eventually remains below them. The occupation integral in (B.19) is consequently finite almost surely. Its exponential is strictly positive, and hence $\mathbb{P}_x^n(\tau = \infty) > 0$.

For a rogue stranger, I first show that $X_t \to +\infty$ almost surely. For $x_1 < x < x_2$, solving the exit-probability equation associated with (B.18), with boundary values zero at x_1 and one at x_2, gives

$$\mathbb{P}_x^r(\text{reach } x_2 \text{ before } x_1) = \frac{\displaystyle\int_{x_1}^{x} \exp\left[-\int_0^y \frac{2\mu_r(z)}{\psi^2}\,dz\right] dy}{\displaystyle\int_{x_1}^{x_2} \exp\left[-\int_0^y \frac{2\mu_r(z)}{\psi^2}\,dz\right] dy}. \tag{B.20}$$

The diffusion exits every bounded interval almost surely because its drift is bounded and $\psi > 0$. Moreover,

$$\exp\left[-\int_0^y \frac{2\mu_r(z)}{\psi^2}\,dz\right] = \exp\left[-y + \frac{2\lambda}{\psi^2}\int_0^y \mathbb{1}\{x_r^* \le z < x_n^*\}\,dz\right].$$

The integral in the last exponent has absolute value at most $x_n^* - x_r^*$. Thus the integrand in (B.20) is bounded above and below by positive multiples of e^{-y}. Sending $x_1 \to -\infty$ in (B.20) shows that every upper level is reached almost surely. Sending the upper boundary to infinity instead gives, for $x_2 > x_1$,

$$\mathbb{P}^r_{x_2}(\text{ever reach } x_1) = \frac{\displaystyle\int_{x_2}^{\infty} \exp\left[-\int_0^y \frac{2\mu_r(z)}{\psi^2}\,dz\right] dy}{\displaystyle\int_{x_1}^{\infty} \exp\left[-\int_0^y \frac{2\mu_r(z)}{\psi^2}\,dz\right] dy} \longrightarrow 0 \qquad (x_2 \to \infty). \tag{B.21}$$

Fix x_1. First visits to successively higher integer levels occur almost surely. By the strong Markov property and (B.21), the probabilities of returning to x_1 after these visits tend to zero. The corresponding return events form a decreasing sequence, so almost surely one such visit is followed by no return to x_1. Continuity then keeps the process above x_1 forever. Applying this argument to every integer lower level proves $X_t \to +\infty$ almost surely. The occupation integral in (B.19) is therefore infinite for the rogue, which implies $\mathbb{P}^r_x(\tau < \infty) = 1$.

Claim (ii): the lower cutoff is more likely to end the partnership. An insider opportunity arrives at rate λ and causes an accusation precisely when the partnership is still active and reputation is at least x_I^*. Integrating this intensity gives

$$\omega_\theta(x) = \lambda\,\mathbb{E}^\theta_x\left[\int_0^\tau \mathbb{1}\{X_t \ge x_I^*\}\,dt\right]. \tag{B.22}$$

Likewise,

$$q_\theta(x) = \lambda\,\mathbb{E}^\theta_x\left[\int_0^\tau \mathbb{1}\{X_t \ge x_\theta^*\}\,dt\right]. \tag{B.23}$$

These identities follow first on finite horizons and then on the infinite horizon by monotone convergence.

Every nonempty open interval has strictly positive expected occupation before τ, from every finite initial reputation. To see this, fix $T > 0$. Bounded drift and $\psi > 0$ imply, by Girsanov's theorem on $[0, T]$, that with positive probability the reputation path enters the interval before $T/2$ and remains there for a positive length of time. Independently, with positive probability neither opportunity process arrives before T, so the partnership survives throughout this event. Their intersection has positive probability.

If $x_\theta^* < x_I^*$, subtracting (B.22) from (B.23) yields

$$q_\theta(x) - \omega_\theta(x) = \lambda\, \mathbb{E}_x^\theta \left[\int_0^\tau \mathbb{1}\{x_\theta^* \le X_t < x_I^*\}\, dt \right] > 0. \tag{B.24}$$

If $x_I^* < x_\theta^*$, reversing the interval gives $\omega_\theta(x) > q_\theta(x)$. If the cutoffs coincide, the two indicators in (B.22) and (B.23) coincide, so the probabilities are equal. This proves the equivalence in claim (ii). The same positive-occupation argument, applied to an interval above x_I^* or above x_θ^*, also gives

$$0 < \omega_\theta(x) < 1, \qquad 0 < q_\theta(x) < 1 \qquad \text{for every finite } x.$$

Here the upper bounds follow from positivity of both probabilities and $\omega_\theta + q_\theta \le 1$. In particular, this establishes the positivity assertions in claims (iii) and (iv).

For the derivative assertions, the backward equation associated with (B.22) is

$$\begin{aligned} 0 = {} & \frac{\psi^2}{2}\omega_\theta''(x) + \mu_\theta(x)\omega_\theta'(x) \\ & - \lambda\big(\mathbb{1}\{x \ge x_I^*\} + \mathbb{1}\{x \ge x_\theta^*\}\big)\omega_\theta(x) + \lambda \mathbb{1}\{x \ge x_I^*\}. \end{aligned} \tag{B.25}$$

The bounded solution is continuously differentiable, smooth between the cutoffs, and has continuous value and first derivative across them, by the same nondegenerate-diffusion regularity used for the value functions. Its second derivative may jump.

Below $\min\{x_I^*, x_r^*\}$, neither type can end the partnership and the survival drift is zero. The rogue reaches this level almost surely. For the normative type, solving the constant-drift first-passage equation gives hitting probability $\exp[x - \min\{x_I^*, x_r^*\}]$. The strong Markov property and memorylessness of the opportunity processes therefore imply

$$\begin{aligned} \omega_r(x) &= \omega_r(\min\{x_I^*, x_r^*\})\,, \\ \omega_n(x) &= e^{x - \min\{x_I^*, x_r^*\}} \omega_n(\min\{x_I^*, x_r^*\})\,, \qquad x \le \min\{x_I^*, x_r^*\}. \end{aligned} \tag{B.26}$$

In particular, the rogue derivative is zero and the normative derivative is strictly positive

in this region, including its upper boundary by continuity. Above all cutoffs, both terminal actions are active and $\mu_\theta = \mu_\theta^0$. Boundedness selects the decaying solution of (B.25): for $x \geq z \geq \max\{x_I^*, x_n^*\}$,

$$\omega_\theta(x) = \frac{1}{2} + \left[\omega_\theta(z) - \frac{1}{2}\right] \exp\left[-\frac{\mu_\theta^0 + \sqrt{(\mu_\theta^0)^2 + 4\lambda\psi^2}}{\psi^2}(x - z)\right]. \tag{B.27}$$

The coefficient multiplying $x - z$ inside the negative exponential is strictly positive. Thus $\omega_\theta(x) \to 1/2$, and its upper-tail derivative has the opposite sign to $\omega_\theta(z) - 1/2$.

Claim (iii): Accusations against rogue strangers. Claim (i) gives $\omega_r + q_r = 1$. Claim (ii) then implies $\omega_r > 1/2$ everywhere if $x_I^* < x_r^*$, and $\omega_r < 1/2$ everywhere if $x_r^* < x_I^*$.

Suppose first that $x_I^* < x_r^*$. On the interval between these cutoffs, (B.25) becomes

$$\frac{\psi^2}{2}\omega_r''(x) + \mu_r(x)\omega_r'(x) = \lambda[\omega_r(x) - 1] < 0.$$

Multiplying by the positive integrating factor $\exp[\int_0^x 2\mu_r(z)/\psi^2\, dz]$, and using $\omega_r'(x_I^*) = 0$ from (B.26), shows that $\omega_r'(x) < 0$ for $x_I^* < x \leq x_r^*$. Above both stopping cutoffs, the right-hand side becomes $2\lambda[\omega_r(x) - 1/2] > 0$. The upper-tail formula (B.27) gives a strictly negative derivative at and above $\max\{x_I^*, x_n^*\}$. Integrating the same integrating-factor equation backward from that level keeps the weighted derivative strictly negative throughout the intervening region. Hence $\omega_r'(x) < 0$ for every $x > x_I^*$.

If $x_r^* < x_I^*$, the equation between these cutoffs instead reads

$$\frac{\psi^2}{2}\omega_r''(x) + \mu_r(x)\omega_r'(x) = \lambda\omega_r(x) > 0.$$

Starting from $\omega_r'(x_r^*) = 0$, the integrating-factor argument gives $\omega_r'(x) > 0$ for $x_r^* < x \leq x_I^*$. Above both stopping cutoffs the right-hand side is now $2\lambda[\omega_r(x) - 1/2] < 0$, while (B.27) gives a strictly positive derivative in the upper tail. Backward integration therefore preserves strict positivity down to the larger stopping cutoff.

These integrating factors and the first derivative are continuous across the remaining drift cutoff, so its position does not alter either conclusion. Together with (B.26), this proves both weak inequalities and their exact equality region. If $x_I^* = x_r^*$, claim (ii) and $\omega_r + q_r = 1$ give $\omega_r = q_r = 1/2$ everywhere.

Claim (iv): Accusations against normative strangers. Below x_I^*, equation (B.25)

gives

$$\frac{\psi^2}{2}\omega_n''(x) + \mu_n(x)\omega_n'(x) = \lambda \mathbb{1}\{x \geq x_n^*\}\omega_n(x) \geq 0.$$

The derivative is strictly positive in the lower region by (B.26). Multiplication by the positive integrating factor $\exp[\int_0^x 2\mu_n(z)/\psi^2\, dz]$ shows that its weighted derivative cannot decrease before x_I^*. Consequently, $\omega_n'(x) > 0$ for all $x \leq x_I^*$.

If $x_n^* \leq x_I^*$, claim (ii) gives $q_n(x_I^*) \geq \omega_n(x_I^*)$, while claim (i) gives $q_n(x_I^*) + \omega_n(x_I^*) < 1$. Thus $\omega_n(x_I^*) < 1/2$. In this ordering, all cutoffs are at or below x_I^*, so (B.27), with $z = x_I^*$, gives $\omega_n'(x) > 0$ above x_I^* as well.

It remains to consider $x_I^* < x_n^*$. Between these cutoffs,

$$\frac{\psi^2}{2}\omega_n''(x) + \mu_n(x)\omega_n'(x) = \lambda[\omega_n(x) - 1] < 0.$$

The weighted derivative is therefore strictly decreasing throughout (x_I^*, x_n^*), including across x_r^* if that cutoff lies inside the interval. Since $\omega_n'(x_I^*) > 0$, the derivative can vanish there at most once, and any such zero changes its sign from positive to negative.

For $x \geq x_n^*$, (B.27) specializes to

$$\omega_n(x) = \frac{1}{2} + \left[\omega_n(x_n^*) - \frac{1}{2}\right] \exp\left[-\frac{\sqrt{1 + 16\lambda/\psi^2} - 1}{2}(x - x_n^*)\right].$$

If $\omega_n(x_n^*) < 1/2$, its derivative at x_n^* and throughout the upper tail is strictly positive. Strict decrease of the weighted derivative on (x_I^*, x_n^*) then implies that the derivative is strictly positive there as well. If $\omega_n(x_n^*) = 1/2$, the derivative is zero at and above x_n^* and strictly positive below it. In both cases ω_n is nondecreasing everywhere.

If $\omega_n(x_n^*) > 1/2$, its derivative at x_n^* is strictly negative. Continuity and strict decrease of the weighted derivative therefore give exactly one zero in (x_I^*, x_n^*). The derivative is strictly positive before that zero and strictly negative after it, including throughout the upper tail. This zero is the unique maximum asserted in claim (iv). □

Partnership value. The value of the partnership is the expected discounted collaboration flow,

$$v_\theta(x) = \mathbb{E}_x^\theta\left[\int_0^\tau e^{-\rho t}\pi\, dt\right].$$

At fixed equilibrium cutoffs, I show separately that

$$v_\theta'(x) < 0 \quad \text{for } \theta \in \{n, r\}, \qquad v_n(x) > v_r(x) \quad \text{for every finite } x.$$

Thus a higher initial reputation lowers the value of collaboration, and collaboration is more valuable conditional on a normative stranger.

First, termination cannot occur before the first opportunity of either player, whereas collaboration can never yield more than the perpetual flow value. This gives the bounds in (5),

$$0 < \frac{\pi}{\rho + 2\lambda} \leq v_\theta(x) \leq \frac{\pi}{\rho}.$$

To establish monotonicity, fix the stranger's type and compare reputation processes X^1 and X^2 started at $x_1 < x_2$, driven by the same Brownian motion and opportunity processes. Pathwise uniqueness for (B.18) prevents their continuous paths from crossing: after they meet, they coincide. Thus $X^1_t \leq X^2_t$ at all times. Every opportunity that would terminate the lower path also terminates the higher path if its partnership is still active. Consequently $\tau^1 \geq \tau^2$ almost surely. Since the flow payoff is positive, v_θ is nonincreasing.

The partnership value satisfies

$$\begin{aligned} 0 = {} & \frac{\psi^2}{2} v''_\theta(x) + \mu_\theta(x) v'_\theta(x) \\ & - \Big(\rho + \lambda \mathbb{1}\{x \geq x^*_I\} + \lambda \mathbb{1}\{x \geq x^*_\theta\}\Big) v_\theta(x) + \pi. \end{aligned} \tag{B.28}$$

Here discounting and either terminal action remove the continuation value, while collaboration supplies the flow π. The value is continuously differentiable, smooth between cutoffs, and has continuous value and first derivative across them.

To establish a strictly negative derivative, monotonicity first gives $v'_\theta \leq 0$. Differentiating (B.28) between cutoffs yields

$$\frac{\psi^2}{2} v'''_\theta + \mu_\theta v''_\theta - \Big(\rho + \lambda \mathbb{1}\{x \geq x^*_I\} + \lambda \mathbb{1}\{x \geq x^*_\theta\}\Big) v'_\theta = 0. \tag{B.29}$$

If v'_θ vanishes at an interior point of such an interval, it attains its maximum there, so v''_θ also vanishes. Uniqueness for the homogeneous equation (B.29) then makes v'_θ identically zero on that interval, and zero at its endpoints by continuity.

At any cutoff z where $v'_\theta(z) = 0$, subtracting the one-sided limits of (B.28) gives

$$\frac{\psi^2}{2}\big[v''_\theta(z+) - v''_\theta(z-)\big] = \lambda v_\theta(z)\big(\mathbb{1}\{z = x^*_I\} + \mathbb{1}\{z = x^*_\theta\}\big). \tag{B.30}$$

The drift contribution vanishes because $v'_\theta(z) = 0$. Since $v'_\theta \leq 0$ and equals zero at z, its one-sided derivatives satisfy $v''_\theta(z-) \geq 0 \geq v''_\theta(z+)$. The left-hand side of (B.30)

is therefore nonpositive. At a stopping cutoff the right-hand side is strictly positive, a contradiction, including when both stopping cutoffs coincide. If only the drift changes at z, the jump is zero, so both one-sided second derivatives vanish. Uniqueness then propagates the zero derivative through the adjacent interval. Repeating this argument would eventually reach a stopping cutoff, where a zero is impossible. Hence $v_\theta'(x) < 0$ at every finite reputation.

Finally, compare the two types. In the remainder of this discussion, X follows the normative conditional law and $\tau = \tau_I \wedge \tau_n$. Applying Dynkin's formula to $e^{-\rho t} v_r(X_t)$, first at $\tau \wedge T$ and then letting $T \to \infty$, gives

$$\begin{aligned} &\mathbb{E}_x^n\big[e^{-\rho\tau} v_r(X_\tau)\mathbb{1}\{\tau < \infty\}\big] - v_r(x) \\ &\quad = \mathbb{E}_x^n\left[\int_0^\tau e^{-\rho t}\left(\frac{\psi^2}{2} v_r''(X_t) + \mu_n(X_t) v_r'(X_t) - \rho v_r(X_t)\right) dt\right]. \end{aligned} \tag{B.31}$$

The formula applies because v_r is continuously differentiable with piecewise continuous second derivative; continuity of its first derivative eliminates local-time terms at the cutoffs. Boundedness and $\rho > 0$ make the terminal term vanish on $\{\tau = \infty\}$, where it is defined to be zero. The constant-coefficient tail equations also give bounded derivatives, so discounting permits passage to the limit in the integral.

Integrating the two termination intensities with terminal amount $v_r(X_t)$ also gives

$$\begin{aligned} &\mathbb{E}_x^n\big[e^{-\rho\tau} v_r(X_\tau)\mathbb{1}\{\tau < \infty\}\big] \\ &\quad = \lambda\, \mathbb{E}_x^n\left[\int_0^\tau e^{-\rho t}\big(\mathbb{1}\{X_t \geq x_I^*\} + \mathbb{1}\{X_t \geq x_n^*\}\big) v_r(X_t)\, dt\right]. \end{aligned} \tag{B.32}$$

Substitute (B.32) into (B.31) and use the rogue version of (B.28). Since $\mu_r - \mu_n = \psi^2$ and $x_r^* < x_n^*$, the result is

$$v_n(x) - v_r(x) = \mathbb{E}_x^n\left[\int_0^\tau e^{-\rho t}\left(\lambda \mathbb{1}\{x_r^* \leq X_t < x_n^*\} v_r(X_t) - \psi^2 v_r'(X_t)\right) dt\right].$$

The integrand is strictly positive: $v_r > 0$ and $v_r' < 0$. Since $\tau > 0$ almost surely, $v_n(x) > v_r(x)$ for every finite initial reputation.

Proof of Proposition 5. The proof uses the common-translation identity (B.17) to link changes in the insider's cutoff to movements in the consistent stranger cutoff and outcome probabilities. The derivative proportionality relation (B.14), is then use to compare the

two types' marginal continuation values. The strict positivity of the difference in (B.15) bounds the rogue cutoff's response to a change in the normative cutoff, while the value-change identity (B.16) relates the resulting changes in the two stranger's continuation values. Together, these bounds and proportionality relations make the reduced normative consistency condition cross zero strictly downward, allowing value comparisons to determine the direction of cutoff adjustment.

Claim (i): Collaboration flow payoff π. The main step is to show that, holding the candidate insider cutoff fixed, increasing π raises both consistent stranger cutoffs and raises the insider's continuation advantage.

Under the stopping rules (4), the stranger receives the collaboration flow until termination, zero if she quits, and $-L_\theta$ if the insider accuses. Hence

$$\frac{v_\theta(x) - V_\theta(x)}{L_\theta} = \mathbb{E}_x^\theta\big[e^{-\rho\tau_I}\mathbb{1}\{\tau_I < \tau_\theta\}\big]. \tag{B.33}$$

The values $V_\theta(x)$ and $v_\theta(x)$ are evaluated conditional on the stranger's type, whereas the insider's value $W(x)$ averages over both possible types. To use these conditional values in the insider's comparative statics, let $W_\theta(x)$ denote his continuation payoff conditional on the stranger's true type, under the same equilibrium strategies. This puts both players' payoffs under the same reputation law and stopping rules, allowing us to express $W_\theta(x)$ in terms of $V_\theta(x)$ and $v_\theta(x)$. Averaging over types using the insider's posterior beliefs then recovers $W(x)$ and allows us to study its response to cutoff changes.

Conditional on a given type, the two players' values average over the same reputation paths, opportunity clocks, and termination time. Both players receive the flow π until termination and zero if the stranger quits. Their payoffs differ only upon accusation: relative to the stranger, the insider receives $L_n - F_n$ more against a normative type and $L_r + G_n$ more against a rogue. Multiplying these payoff differences by the discounted accusation probability in (B.33) therefore gives

$$\begin{aligned} W_n(x) &= V_n(x) + \frac{L_n - F_n}{L_n}\big[v_n(x) - V_n(x)\big], \\ W_r(x) &= V_r(x) + \frac{L_r + G_n}{L_r}\big[v_r(x) - V_r(x)\big], \\ W(x) &= [1 - p(x)]W_n(x) + p(x)W_r(x). \end{aligned} \tag{B.34}$$

These are payoff identities under a common strategy profile. They also apply in the local prescribed-cutoff calculations when all values are evaluated under that same profile and termination rule.

To determine how changes in the stranger cutoffs affect the insider's continuation value, we need comparisons across types. Direct payoff comparisons are complicated by the fact that accusation imposes a loss on the insider against a normative stranger but gives him a gain against a rogue. I therefore introduce an auxiliary value consisting entirely of collaboration flows, whose monotonicity can be established by comparing paths and then related back to the original payoffs

Let $\widetilde{v}_\theta(x)$ denote the expected discounted collaboration flow under the same type-conditional reputation dynamics and opportunity clocks, with reputation reset to x_I whenever the insider would accuse. After each reset, collaboration continues; the process ends only when the stranger quits. The reset map at an insider opportunity is $x \mapsto \min\{x, x_I\}$, which preserves the ordering of reputations. Choosing x_I as the reset state also connects the resulting value identities to the insider's indifference condition, used below.

Until the first termination of the original partnership, the two processes coincide and generate collaboration value $v_\theta(x)$. If the stranger quits, there is no further payoff. If the insider would accuse, the auxiliary process restarts at x_I with continuation value $\widetilde{v}_\theta(x_I)$. Weighting that continuation value by the discounted accusation probability gives the first identity below; evaluating it at x_I and rearranging gives the second:

$$\begin{aligned} \widetilde{v}_\theta(x) &= v_\theta(x) + \frac{v_\theta(x) - V_\theta(x)}{L_\theta}\widetilde{v}_\theta(x_I), \\ \widetilde{v}_\theta(x_I) &= \frac{L_\theta v_\theta(x_I)}{L_\theta - v_\theta(x_I) + V_\theta(x_I)}. \end{aligned} \tag{B.35}$$

The denominator is positive because it equals L_θ times one minus the discounted accusation probability, which is at most $\lambda/(\rho+\lambda) < 1$. The choice of reset state makes this transformation useful: combining these identities with insider indifference at x_I will cancel the terms with opposing signs in the weighted derivatives of W_n and W_r below. The remaining terms can then be signed using the monotonicity of the reset values.

The order-preserving coupling from Appendix B.1, now applying the same insider and stranger opportunity clocks to both paths. The reset map is nondecreasing, so it preserves their order; the higher path reaches a quitting opportunity no later than the lower one. Thus $\widetilde{v}_\theta$ is nonincreasing in initial reputation.

To obtain a strict derivative, its backward equation is (B.28) with v_θ replaced by $\widetilde{v}_\theta$ and the additional source $\lambda \mathbb{1}\{x \geq x_I\}\widetilde{v}_\theta(x_I)$. Between cutoffs, its spatial derivative satisfies the associated homogeneous equation. A zero of the nonpositive derivative is a maximum, so its own derivative must vanish there. At a drift cutoff, the matching condition propagates these two zero values. The reset term causes no second-derivative

jump at x_I because the value there equals the reset value. A zero would therefore propagate to the stranger's quitting cutoff. But if $\widetilde{v}'_\theta(x_\theta) = 0$, the one-sided equations imply

$$\lim_{x\downarrow x_\theta} \widetilde{v}''_\theta(x) - \lim_{x\uparrow x_\theta} \widetilde{v}''_\theta(x) = \frac{2\lambda}{\psi^2}\widetilde{v}_\theta(x_\theta) > 0,$$

contradicting a maximum of $\widetilde{v}'_\theta$ there. This also covers a quitting cutoff coinciding with x_I. Consequently,

$$\widetilde{v}'_\theta(x) < 0 \quad \text{for every finite } x. \tag{B.36}$$

Coupling the two types with the same Brownian motion and opportunity clocks gives a weakly lower reputation for the normative type, whose quitting cutoff is strictly higher. With positive probability the rogue quits while the normative type continues. Therefore

$$\widetilde{v}_n(x_I) > \widetilde{v}_r(x_I) > 0. \tag{B.37}$$

The derivative jump at x_I, together with (B.14), gives

$$\frac{V'_n(x_I)}{V'_r(x_I)} = \frac{V_n(x_I) + L_n}{V_r(x_I) + L_r} > 0. \tag{B.38}$$

By (B.35),

$$\frac{v_\theta(x_I)}{V_\theta(x_I) + L_\theta} = \frac{\widetilde{v}_\theta(x_I)}{L_\theta + \widetilde{v}_\theta(x_I)}.$$

Since $L_r > L_n$, (B.37) and (B.38) imply

$$v_n(x_I) > \frac{V'_n(x_I)}{V'_r(x_I)} v_r(x_I). \tag{B.39}$$

This comparison extends to every $x \le x_I$:

$$v_n(x) > e^{x-x_I}\frac{V'_n(x_I)}{V'_r(x_I)} v_r(x), \qquad x \le x_I. \tag{B.40}$$

To verify the extension, apply the exponential substitution used in (B.14) to the two linear equations (B.28). The normative Green kernel equals the rogue Green kernel multiplied by e^{x-y}, where y is the destination reputation. Consequently, $e^{x_I - x} v_n(x)/v_r(x)$ is the average of $e^{x_I - y}$ under the rogue Green kernel normalized by its total mass. For $x_1 < x_2$, the ratio of the rogue Green kernels at initial states x_2 and x_1 is nondecreasing in y. Indeed, the one-dimensional kernel factors into an increasing positive homogeneous solution at the smaller state and a decreasing positive homogeneous solution at

the larger state; the ratio is constant outside (x_1, x_2) and increasing inside it. The normalized weights therefore shift toward larger y as x increases. Since $e^{x_I - y}$ is decreasing, $e^{x_I - x} v_n(x)/v_r(x)$ is nonincreasing in x. Equation (B.39) now proves (B.40).

Two effects of the stranger cutoffs on the insider. At the equilibrium under consideration, insider indifference, (B.35), (B.38), and (B.14) give the identity

$$\begin{aligned} &[1 - p(x)]W_n'(x) + p(x)W_r'(x) \\ &\quad = [1 - p(x)]\frac{L_n - F_n}{L_n + \widetilde{v}_n(x_I)}\widetilde{v}_n'(x) + p(x)\frac{L_r + G_n}{L_r + \widetilde{v}_r(x_I)}\widetilde{v}_r'(x) < 0. \end{aligned} \tag{B.41}$$

For completeness, substituting (B.35) into (B.34) leaves, besides the two displayed terms, the expression

$$[1 - p(x)]\frac{F_n + \widetilde{v}_n(x_I)}{L_n + \widetilde{v}_n(x_I)}V_n'(x) + p(x)\frac{\widetilde{v}_r(x_I) - G_n}{L_r + \widetilde{v}_r(x_I)}V_r'(x).$$

It is zero by insider indifference at x_I, (B.14), and $p(x)/(1 - p(x)) = e^{x - x_I}p(x_I)/(1 - p(x_I))$. The strict sign in (B.41) follows from $L_n \geq F_n$, $L_r + G_n > 0$, and (B.36).

At the normative quitting cutoff, $W_n(x_n) = (1 - F_n/L_n)v_n(x_n) \geq 0$. Differentiating the insider's value equation with respect to the conjectured normative cutoff generates a source at x_n with coefficient

$$\lambda\big([1 - p(x_n)]W_n(x_n) - [1 - p(x_n)]W_n'(x_n) - p(x_n)W_r'(x_n)\big) > 0.$$

This expression includes both the change in the quitting hazard and the change in the survival drift. Positivity of the discounted Green kernel therefore gives

$$\frac{\partial W(x_I)}{\partial x_n} > 0. \tag{B.42}$$

Next translate all three prescribed cutoffs by the same amount, keeping the current reputation fixed. The conditional insider values are translated horizontally, while the first-order contribution from moving the insider's own rule is zero at indifference. Hence

$$\frac{\partial W(x_I)}{\partial x_r} + \frac{\partial W(x_I)}{\partial x_n} = -[1 - p(x_I)]W_n'(x_I) - p(x_I)W_r'(x_I) > 0. \tag{B.43}$$

These two signs do not require a sign for $\partial W(x_I)/\partial x_r$ by itself.

Rogue consistency and profit derivatives. As in Appendix B.1, first fix x_I and a conjectured x_n, and adjust $x_r(x_n)$ to preserve rogue indifference. Until stated otherwise,

$dV_\theta(x)/d\pi$ refers to this adjustment: x, x_n, and x_I are held fixed. Differentiating (B.12) between cutoffs gives

$$\begin{aligned} 0 = \frac{\psi^2}{2}\left(\frac{dV_r}{d\pi}\right)'' + \mu_r\left(\frac{dV_r}{d\pi}\right)' \\ - \left[\rho + \lambda\mathbb{1}\{x \geq x_I\} + \lambda\mathbb{1}\{x \geq x_r\}\right]\frac{dV_r}{d\pi} + 1. \end{aligned} \tag{B.44}$$

Differentiated indifference gives

$$\left.\frac{dV_r(x)}{d\pi}\right|_{x=x_r} = -V_r'(x_r)\left.\frac{\partial x_r}{\partial \pi}\right|_{x_n,x_I}.$$

The spatial derivative of $dV_r/d\pi$ jumps at x_r by

$$\left.\frac{2\lambda}{\psi^2}\,\frac{dV_r(x)}{d\pi}\right|_{x=x_r},$$

and is continuous at the other fixed cutoffs. Its tail limits are $1/\rho$ and $1/(\rho+2\lambda)$. A negative minimum is impossible between cutoffs by (B.44); at x_r it would contradict the displayed jump condition. The strictly positive source also excludes a zero minimum. Thus

$$\frac{dV_r(x)}{d\pi} > 0, \qquad \left.\frac{\partial x_r}{\partial \pi}\right|_{x_n,x_I} > 0. \tag{B.45}$$

The analogous comparison for an increase in x_n, together with the strict positivity established for (B.15), gives

$$0 < \frac{dx_r(x_n)}{dx_n} < 1. \tag{B.46}$$

The strict lower bound follows because increasing x_n creates a positive source in the rogue's value equation, while rogue consistency gives the same additional killing condition at x_r as above.

At a consistent stranger pair, the same substitution as in (B.16) gives

$$\frac{dV_n(x)}{d\pi} - \frac{v_n(x)}{\pi} = e^{x-x_I}\frac{V_n'(x_I)}{V_r'(x_I)}\left[\frac{dV_r(x)}{d\pi} - \frac{v_r(x)}{\pi}\right]. \tag{B.47}$$

To obtain this identity, subtract the direct flow effect v_θ/π from each linearized equation. The remaining moving-cutoff terms are proportional under (B.14), including their matching conditions. The identity applies both to the rogue-only adjustment just described and to the mutually consistent adjustment of both strangers, with x_I fixed.

Case 1: $x_r \leq x_I$. First retain the rogue-only adjustment. Evaluating (B.47) at x_r, and using (B.40) and (B.45), gives $dV_n(x_r)/d\pi > 0$. Away from x_r, the normative value derivative has a strictly positive unit flow source and positive tail limits. With its positive value at x_r, the maximum principle implies

$$\frac{dV_n(x)}{d\pi} > 0 \quad \text{for every finite } x. \tag{B.48}$$

Appendix B.1 proves that the normative consistency condition, after substituting the consistent rogue response $x_r(x_n)$, crosses zero strictly downward as x_n increases. Its profit derivative, holding x_n and x_I fixed and adjusting the rogue, is strictly positive by (B.48). Implicit differentiation therefore gives

$$\left.\frac{\partial x_n}{\partial \pi}\right|_{x_I} > 0, \qquad \left.\frac{\partial x_r}{\partial \pi}\right|_{x_I} = \left.\frac{\partial x_r}{\partial \pi}\right|_{x_n,x_I} + \frac{dx_r(x_n)}{dx_n}\left.\frac{\partial x_n}{\partial \pi}\right|_{x_I} > 0. \tag{B.49}$$

To sign the insider's value response, first increase profits and adjust only the rogue. At $x_r \leq x_I$, the spatial derivatives of both discounted accusation probabilities in (B.33) are positive: below the lowest cutoff these probabilities are positive multiples of $\exp(\xi_1(\mu_\theta^0)x)$, with the same derivatives at the boundary. Raising x_r creates a positive drift source for each probability; for the rogue it also creates a positive source by delaying her own quitting. Their derivatives with respect to x_r are therefore positive at every finite initial reputation. At fixed cutoffs, these probabilities are independent of π.

Equations (B.34), (B.45), and (B.48), together with $L_n \geq F_n$, now give

$$\left.\frac{dW(x_I)}{d\pi}\right|_{x_n,x_I} > 0. \tag{B.50}$$

Allowing the normative cutoff to adjust adds

$$\left[\frac{\partial W(x_I)}{\partial x_n} + \frac{dx_r(x_n)}{dx_n}\frac{\partial W(x_I)}{\partial x_r}\right]\left.\frac{\partial x_n}{\partial \pi}\right|_{x_I}.$$

The bracket is strictly positive. It is the sum of $1 - dx_r(x_n)/dx_n$ times (B.42) and $dx_r(x_n)/dx_n$ times (B.43). Consequently,

$$\left.\frac{dW(x_I)}{d\pi}\right|_{x_I} > 0, \tag{B.51}$$

where both strangers now adjust consistently.

Case 2: $x_I < x_r < x_n$. In this case, $dV_\theta(x)/d\pi$ includes both mutually consistent

stranger responses, with x and x_I fixed. I first show the stronger comparison

$$\left.\frac{\partial x_n}{\partial \pi}\right|_{x_I} > \left.\frac{\partial x_r}{\partial \pi}\right|_{x_I} > 0. \tag{B.52}$$

By (B.33) and the opportunity-arrival representation,

$$\frac{v_\theta(x)}{\pi} + \frac{V_\theta(x)}{\lambda L_\theta - \pi} = \frac{\lambda L_\theta}{\lambda L_\theta - \pi} \mathbb{E}_x^\theta \left[\int_0^\tau e^{-\rho t} \mathbb{1}\{X_t < x_I\}\, dt \right] > 0. \tag{B.53}$$

Only reputations below x_I contribute to this occupation value. Since neither stranger quits below x_I, the lower-tail Green kernels have constant-coefficient exponential forms. Their factorization and the same exponential substitution as in (B.14) give, for $x \geq x_I$,

$$e^{x_I - x} \frac{\dfrac{v_n(x)}{\pi} + \dfrac{V_n(x)}{\lambda L_n - \pi}}{\dfrac{v_r(x)}{\pi} + \dfrac{V_r(x)}{\lambda L_r - \pi}} = \frac{L_n(\lambda L_r - \pi)}{L_r(\lambda L_n - \pi)} \frac{\xi_1(\mu_n^0)}{\xi_1(\mu_r^0)}. \tag{B.54}$$

More explicitly, below x_I the destination-state factor of the rogue Green kernel is proportional to $\exp(\xi_1(\mu_n^0)(y - x_I))$. The normative kernel introduces the additional factor e^{x-y}. Integrating the two factors over $y < x_I$ gives the ratio of roots in (B.54).

The lower-tail formulas for V_θ also imply

$$\begin{aligned} \frac{V_n'(x_I)}{V_r'(x_I)} &= \frac{\xi_1(\mu_n^0)}{\xi_1(\mu_r^0)} \frac{\pi/\rho - V_n(x_I)}{\pi/\rho - V_r(x_I)} \\ &< \frac{\xi_1(\mu_n^0)}{\xi_1(\mu_r^0)} < \frac{L_n(\lambda L_r - \pi)}{L_r(\lambda L_n - \pi)} \frac{\xi_1(\mu_n^0)}{\xi_1(\mu_r^0)}. \end{aligned} \tag{B.55}$$

The first strict inequality uses $V_n(x_I) > V_r(x_I)$; the second uses $L_r > L_n$ and $\pi > 0$. Combining (B.14), (B.47), and (B.54)–(B.55) therefore gives

$$\frac{\dfrac{dV_n(x)}{d\pi} + \dfrac{V_n(x)}{\lambda L_n - \pi}}{-V_n'(x)} > \frac{\dfrac{dV_r(x)}{d\pi} + \dfrac{V_r(x)}{\lambda L_r - \pi}}{-V_r'(x)}, \qquad x \geq x_I. \tag{B.56}$$

Consider the quotient on the right of (B.56), namely

$$\frac{\dfrac{dV_r(x)}{d\pi} + \dfrac{V_r(x)}{\lambda L_r - \pi}}{-V_r'(x)}. \tag{B.57}$$

At x_r, differentiated rogue indifference makes this quotient equal to $\partial x_r/\partial\pi|_{x_I}$. At x_n, differentiated normative indifference and (B.56) imply

$$\frac{\left.\frac{dV_r(x)}{d\pi}\right|_{x=x_n} + \frac{V_r(x_n)}{\lambda L_r - \pi}}{-V_r'(x_n)} < \left.\frac{\partial x_n}{\partial \pi}\right|_{x_I}. \tag{B.58}$$

Between x_r and x_n, the numerator and denominator of (B.57) satisfy the same homogeneous second-order equation. The numerator has zero flow source there: adding $V_r/(\lambda L_r - \pi)$ to the linearized value cancels the unit source above x_I. It follows from the Wronskian equation that the derivative of the quotient cannot change sign on (x_r, x_n). Above x_n, numerator and denominator are multiples of the same decaying exponential, so the quotient is constant.

At x_n, the derivative of the numerator jumps by $-2\lambda[-V_r'(x_n)]\,\partial x_n/\partial\pi|_{x_I}/\psi^2$, whereas the derivative of the denominator jumps by $-2\lambda[-V_r'(x_n)]/\psi^2$. Its zero right derivative therefore implies that the left derivative of the quotient equals

$$\frac{2\lambda}{\psi^2}\left[\left.\frac{\partial x_n}{\partial \pi}\right|_{x_I} - \frac{\left.\frac{dV_r(x)}{d\pi}\right|_{x=x_n} + \frac{V_r(x_n)}{\lambda L_r - \pi}}{-V_r'(x_n)}\right] > 0.$$

Thus the quotient is strictly increasing between x_r and x_n. Its endpoint values and (B.58) give

$$\left.\frac{\partial x_n}{\partial \pi}\right|_{x_I} > \left.\frac{\partial x_r}{\partial \pi}\right|_{x_I}.$$

The chain rule through $x_r(x_n)$ yields

$$\begin{aligned}&\left(1 - \frac{dx_r(x_n)}{dx_n}\right)\left.\frac{\partial x_r}{\partial \pi}\right|_{x_I} \\ &\quad = \left.\frac{\partial x_r}{\partial \pi}\right|_{x_n, x_I} + \frac{dx_r(x_n)}{dx_n}\left[\left.\frac{\partial x_n}{\partial \pi}\right|_{x_I} - \left.\frac{\partial x_r}{\partial \pi}\right|_{x_I}\right] > 0,\end{aligned}$$

by (B.45) and (B.46). This proves both strict inequalities in (B.52).

The insider's direct profit derivative at fixed cutoffs is $([1-p(x_I)]v_n(x_I)+p(x_I)v_r(x_I))/\pi >$

0. Allowing both strangers to adjust gives

$$\begin{aligned}\left.\frac{dW(x_I)}{d\pi}\right|_{x_I} &= \frac{[1-p(x_I)]v_n(x_I)+p(x_I)v_r(x_I)}{\pi} \\ &\quad + \left[\frac{\partial W(x_I)}{\partial x_r}+\frac{\partial W(x_I)}{\partial x_n}\right]\left.\frac{\partial x_r}{\partial \pi}\right|_{x_I} \\ &\quad + \frac{\partial W(x_I)}{\partial x_n}\left[\left.\frac{\partial x_n}{\partial \pi}\right|_{x_I} - \left.\frac{\partial x_r}{\partial \pi}\right|_{x_I}\right] > 0.\end{aligned} \tag{B.59}$$

Every term is positive by (B.42), (B.43), and (B.52). This establishes (B.51) in the second case as well.

Equilibrium adjustment of the insider cutoff. Under either ordering, increasing π and allowing both strangers to adjust raises the insider's continuation advantage at a fixed candidate x_I, because $\bar{u}(w; x_I)$ does not depend on π. It remains to translate this strict gain into the equilibrium cutoff response.

For fixed π, (B.17) translates the consistent stranger pair one-for-one with x_I. If the insider also uses the candidate cutoff, the conditional continuation payoffs evaluated at that cutoff are unchanged by this translation. At equilibrium, prescribing the candidate insider rule gives the same first derivative as optimization, by the indifference envelope noted above. Hence, evaluated at equilibrium,

$$\frac{d}{dx_I}\big[W(x_I)-\bar{u}(w;x_I)\big] = p'(x_I)\big[W_r(x_I)-W_n(x_I)-G_n-F_n\big] < 0. \tag{B.60}$$

Indeed, $W_n(x_I) > -F_n$ because continuation earns positive flow and discounts any accusation loss. Also $W_r(x_I) < G_n$: every terminal payoff conditional on the rogue type is at most G_n, and $\pi/\rho < G_n$. The opportunity bound makes this upper bound strict.

Implicit differentiation of insider indifference therefore gives

$$\frac{\partial x_I^*}{\partial \pi} = -\frac{\left.\dfrac{dW(x_I)}{d\pi}\right|_{x_I}}{p'(x_I)\big[W_r(x_I)-W_n(x_I)-G_n-F_n\big]} > 0, \tag{B.61}$$

with all quantities evaluated at the equilibrium. Finally, (B.17) and the positive fixed-x_I stranger responses imply

$$\frac{\partial x_\theta^*}{\partial \pi} = \left.\frac{\partial x_\theta}{\partial \pi}\right|_{x_I} + \frac{\partial x_I^*}{\partial \pi} > 0, \qquad \theta \in \{n, r\}. \tag{B.62}$$

If x_I coincides with a stranger cutoff, the moving-boundary contributions are added as in Appendix B.1. The values, their first spatial derivatives, and the coefficients in the linearized matching conditions are continuous through the coincidence. The consistency derivatives remain nonsingular: rogue consistency and reduced normative consistency have strictly negative derivatives, and (B.60) is strictly negative. Thus the one-sided first derivatives agree. Case 1 gives the same strict inequalities at $x_I = x_r$ and at $x_I = x_n$. This proves claim (i).

Claims (ii) and (iii): Accusation payoffs. Fix π and a candidate insider cutoff x_I. Neither G_n nor F_n enters the strangers' flow payoffs, accusation losses, or type-conditional reputation dynamics. Their optimization problems in (B.12) are therefore unchanged at fixed x_I. Uniqueness of the consistent stranger pair and (B.17) imply the cutoff-distance identities in claims (ii) and (iii). It remains to sign the common movement.

Evaluate the insider's continuation value at the candidate x_I against its consistent stranger pair. At the equilibrium, the indifference envelope used above permits derivatives to be computed under the prescribed equilibrium policy. The discounted accusation probability in (B.33) is at most $\lambda/(\rho+\lambda) < 1$, because accusation cannot precede the insider's first opportunity. Since the strangers do not adjust at this fixed x_I, differentiating the payoff identities (B.34) gives

$$\frac{\partial}{\partial G_n}[W(x_I) - \bar{u}(w; x_I)]\bigg|_{x_I} = -p(x_I)\left[1 - \frac{v_r(x_I) - V_r(x_I)}{L_r}\right] < 0, \tag{B.63}$$

$$\frac{\partial}{\partial F_n}[W(x_I) - \bar{u}(w; x_I)]\bigg|_{x_I} = [1 - p(x_I)]\left[1 - \frac{v_n(x_I) - V_n(x_I)}{L_n}\right] > 0. \tag{B.64}$$

The derivative of this advantage with respect to the candidate insider cutoff, allowing the consistent stranger pair to translate, is strictly negative by (B.60). Implicit differentiation gives a strictly negative insider-cutoff response to G_n and a strictly positive response to F_n. The common translation (B.17) gives the same response for each stranger cutoff, proving claims (ii) and (iii).

Claim (iv): Accusation and quitting probabilities. Changing G_n or F_n shifts all three equilibrium cutoffs by the same amount and leaves their distances unchanged. Conditional on the stranger's true type, the reputation drift depends on the current reputation only through its position relative to the stranger cutoffs, as in (B.18). The stopping rules (4) likewise depend only on the position relative to their respective cutoffs,

and the opportunity rates do not change.

Consequently, shifting the initial reputation and all three cutoffs by the same amount leaves the joint law of the termination time and the terminating action unchanged. Equivalently, shifting all the cutoffs while fixing the initial reputation has the same effect as moving the initial reputation by the opposite amount under the original cutoff profile. The backward equations associated with (B.22) and (B.23) give continuously differentiable probability functions, including at the cutoffs. The chain rule therefore gives exactly the two identities in claim (iv), and the same argument applies to F_n. Here primes hold the equilibrium cutoffs fixed, whereas parameter derivatives allow them to adjust. □

C. Omitted Proofs

Proof of Proposition 1. The law of motion is derived for an arbitrary player k, then specialized to player i and her partner j separately.

By Bayes' rule, the posterior odds ratio for player k given the event $\{\tau > t\}$ decomposes as

$$\frac{\mathbb{P}(\theta_k = r \mid \mathcal{Z}_t, \{\tau > t\})}{\mathbb{P}(\theta_k = n \mid \mathcal{Z}_t, \{\tau > t\})} = \frac{\mathbb{P}(\theta_k = r \mid \mathcal{Z}_t)}{\mathbb{P}(\theta_k = n \mid \mathcal{Z}_t)} \cdot \frac{\mathbb{P}(\{\tau > t\} \mid \mathcal{Z}_t, \theta_k = r)}{\mathbb{P}(\{\tau > t\} \mid \mathcal{Z}_t, \theta_k = n)},$$

where the first term on the right-hand side is the signal contribution and the second term is the survival contribution. To compute the signal contribution, recall that Z^k satisfies $dZ_t^k = \mu(\theta_k)\,dt + \sigma\,dB_t^k$. Let $\phi_r(\cdot)$ and $\phi_n(\cdot)$ denote the corresponding Gaussian densities of Z_t^k under types r and n, respectively. Since the drift is constant, Z_t^k is a sufficient statistic for θ_k: by Girsanov's theorem, the likelihood ratio of the path $(Z_s^k)_{s \le t}$ depends on it only through Z_t^k and equals the ratio of these densities,

$$\frac{\phi_r(Z_t^k)}{\phi_n(Z_t^k)} = \exp\left(\frac{\mu(r) - \mu(n)}{\sigma^2} Z_t^k - \frac{(\mu(r) + \mu(n))(\mu(r) - \mu(n))}{2\sigma^2} t\right).$$

Thus, given the prior p_0

$$\frac{\mathbb{P}(\theta_k = r \mid \mathcal{Z}_t)}{\mathbb{P}(\theta_k = n \mid \mathcal{Z}_t)} = \frac{p_0}{1 - p_0} \frac{\phi_r(Z_t^k)}{\phi_n(Z_t^k)}.$$

Taking logs and differentiating yields the signal contribution to dX_t^k:

$$\frac{\mu(r) - \mu(n)}{\sigma^2} dZ_t^k - \frac{(\mu(r) + \mu(n))(\mu(r) - \mu(n))}{2\sigma^2} dt. \tag{C.65}$$

Player i knows her type drift $\mu(\theta_i)$, and substituting it into dZ_t^i in (C.65) yields

$$\frac{\mu(r) - \mu(n)}{\sigma^2} \mu(\theta_i)\,dt - \frac{(\mu(r) + \mu(n))(\mu(r) - \mu(n))}{2\sigma^2}\,dt + \frac{\mu(r) - \mu(n)}{\sigma} dB_t^i.$$

Using $\psi = (\mu(r) - \mu(n))/\sigma$, this simplifies to

$$\frac{\psi^2}{2}\big(\mathbb{1}\{\theta_i = r\} - \mathbb{1}\{\theta_i = n\}\big)\,dt + \psi\,dB_t^i.$$

Player j's type is unknown to player i, so the drift of Z^j is replaced by its posterior mean given the public information, $\mathbb{E}[\mu(\theta_j) \mid \mathcal{Z}_t, \{\tau > t\}] = \mu(r)P_t^j + \mu(n)(1 - P_t^j)$. The *innovation* process associated with Z^j is obtained by subtracting this conditional drift

from the observed signal,

$$d\widetilde{B}_t^j := \frac{1}{\sigma}\left(dZ_t^j - \mathbb{E}[\mu(\theta_j) \mid \mathcal{Z}_t, \{\tau > t\}]\, dt\right).$$

By standard filtering arguments, $\widetilde{B}^j$ is a Brownian motion with respect to the public information. Player i's private information, her type and her opportunities, is independent of θ_j given the public information, so conditioning on it leaves the posterior mean unchanged, and $\widetilde{B}^j$ is a Brownian motion from her perspective as well. Substituting the resulting innovation representation into (C.65) yields

$$\psi^2\left(P_t^j - \frac{1}{2}\right) dt + \psi\, d\widetilde{B}_t^j.$$

Equivalently, the drift can be written in log-odds form as

$$\frac{\psi^2}{2}\tanh\left(\frac{X_t^j}{2}\right) dt + \psi\, d\widetilde{B}_t^j.$$

Next, consider the information conveyed by the fact that the game has survived up to time t. Unlike the signal process, survival is endogenous to players' stopping behavior and therefore strategy-dependent. Let α and β denote the strategies of players i and j, respectively, and let τ_i and τ_j denote their respective stopping times. Since the game ends when either player stops, $\tau = \tau_i \wedge \tau_j$, and hence

$$\{\tau > t\} = \{\tau_i > t\} \cap \{\tau_j > t\}.$$

Consider first the information conveyed about θ_i. Conditional on the realized public state path, player j's stopping behavior depends on her own type, but not directly on θ_i. Player j's survival, $\{\tau_j > t\}$, enters with the same likelihood under $\theta_i = r$ and $\theta_i = n$ and cancels from the likelihood ratio between the two types. Thus, the survival contribution to beliefs about θ_i is determined entirely by player i's own survival.

Conditional on θ_i, player i stops at the first opportunity at which $(X_t^i, X_t^j) \notin \mathcal{C}_{\theta_i}(\alpha)$. Thus, conditional on the public signal history $\mathcal{Z}_t$, the probability that player i has not stopped by time t is

$$\mathbb{P}(\{\tau_i > t\} \mid \mathcal{Z}_t, \theta_i) = \exp\left(-\lambda \int_0^t \mathbb{1}\{(X_s^i, X_s^j) \notin \mathcal{C}_{\theta_i}(\alpha)\} ds\right).$$

It follows that the likelihood ratio associated with player i's survival is

$$\frac{\mathbb{P}(\{\tau_i > t\} \mid \mathcal{Z}_t, \theta_i = r)}{\mathbb{P}(\{\tau_i > t\} \mid \mathcal{Z}_t, \theta_i = n)} = \exp\left(\lambda \int_0^t \left[\mathbb{1}\{(X_s^i, X_s^j) \notin \mathcal{C}_n(\alpha)\} - \mathbb{1}\{(X_s^i, X_s^j) \notin \mathcal{C}_r(\alpha)\}\right] ds\right).$$

Taking logs, survival contributes $+\lambda$ to the drift of the log-odds at states in which only rogue types collaborate, i.e., $\mathcal{I}_r(\alpha)$. Conversely, it contributes $-\lambda$ at states in which only normative types collaborate, i.e., $\mathcal{I}_n(\alpha)$. Hence, survival contributes the drift $\Lambda(x, y; \alpha)$ to dX_t^i. An analogous argument for player j yields the drift $\Lambda(y, x; \beta)$ in dX_t^j. Finally, because the likelihood contributions of the signal and survival multiply under Bayes' rule, their log-likelihood contributions are additive. Combining the two therefore yields the stated log-odds dynamics. □

Proof of Lemma A.1. For the profile (α, β), use the generator $\mathcal{L}_\theta^{\alpha,\beta}$ defined in (A.1), with drifts (A.2)–(A.3). With the type and profile fixed, suppress their arguments and write $\mathcal{L}_\theta$. The drifts are bounded and Borel measurable, and the constant diffusion matrix is nondegenerate because $\psi > 0$. Thus $\mathcal{L}_\theta$ is uniformly elliptic.

Existence and uniqueness of a strong solution follow from Theorem 1 of Veretennikov (1981). Moreover, by Stroock and Varadhan (2007), the martingale problem associated with $\mathcal{L}_\theta$ is well-posed (Theorem 7.2.1), the resulting process is strong Markov (Theorem 6.2.2), and its transition semigroup is strong Feller (Lemma 9.2.2). □

Proof of Proposition 6. The proof first reformulates the problem in opportunity time and shows that the corresponding Bellman operator is a contraction. It then connects this discrete-time recursion to the original continuous-time problem by iterating over successive opportunities. This identifies the unique fixed point of the Bellman operator with the value function and yields the myopic characterization of optimal behavior.

Fix a strategy α, a type θ, and a bounded Borel-measurable candidate continuation value f. Use the opportunity maps Υ_θ and Γ_θ from Appendix A.1, suppressing the fixed strategy argument.

Let $T := \min\{T_1^i, T_1^j\} \sim \text{Exp}(2\lambda)$ denote the first opportunity time of either player. Conditional on $T = t$, the opportunity belongs to each player with probability $1/2$. The value admits a decomposition into the discounted flow payoff accrued up to T and the discounted value at T, which depends on which player receives the opportunity.

Integrating this decomposition against the density of T yields the Bellman operator

$$\Phi_\theta^\alpha[f](x,y) := \int_0^\infty 2\lambda e^{-2\lambda t}\left(\int_0^t e^{-\rho s}\pi\, ds + \frac{e^{-\rho t}}{2}\,\mathbb{E}_{x,y}^\theta\big[\Upsilon_\theta(X_t^i,X_t^j;f)+\Gamma_\theta(X_t^i,X_t^j;f)\big]\right)dt.$$

The operator Φ_θ^α maps $C_b^0(\mathbb{R}^2)$ into itself. Boundedness follows directly from bounded payoffs and continuation values. For continuity, the flow-payoff term $\int_0^\infty 2\lambda e^{-2\lambda t}\int_0^t e^{-\rho s}\pi\, ds\, dt$ is constant in (x,y). The remaining term has the form

$$\int_0^\infty \lambda e^{-(2\lambda+\rho)t}\,\mathbb{E}_{x,y}^\theta\big[\Upsilon_\theta(X_t^i,X_t^j;f)+\Gamma_\theta(X_t^i,X_t^j;f)\big]\,dt. \tag{C.66}$$

The function $\Upsilon_\theta(\cdot;f)+\Gamma_\theta(\cdot;f)$ is bounded and Borel measurable. By the strong Feller property in Lemma A.1, the map $(x,y)\mapsto \mathbb{E}_{x,y}^\theta[\Upsilon_\theta(X_t^i,X_t^j;f)+\Gamma_\theta(X_t^i,X_t^j;f)]$ is continuous for each $t>0$. Since the expectation is uniformly bounded, dominated convergence applies using a constant multiple of $\lambda e^{-(2\lambda+\rho)t}$ as an integrable bound, and therefore implies continuity in (x,y) after integration over t. Hence $\Phi_\theta^\alpha[f]\in C_b^0(\mathbb{R}^2)$.

To show Φ_θ^α is a contraction, take $f,g\in C_b^0(\mathbb{R}^2)$ then,

$$|\Upsilon_\theta(x,y;f)-\Upsilon_\theta(x,y;g)|\le |f(x,y)-g(x,y)|.$$

This follows from $|\max\{a,b\}-\max\{a,c\}|\le|b-c|$. For Γ_θ, the terms involving $u^p(\cdot;\theta)$ do not depend on f or g and cancel. The remaining terms, weighted by $p(y)\mathbb{1}\{(y,x)\in\mathcal{C}_r(\alpha)\}$ and $(1-p(y))\mathbb{1}\{(y,x)\in\mathcal{C}_n(\alpha)\}$, have nonnegative coefficients that sum to at most one. Therefore,

$$|\Gamma_\theta(x,y;f)-\Gamma_\theta(x,y;g)|\le |f(x,y)-g(x,y)|.$$

Using these two bounds in (C.66) yields

$$\begin{aligned}
|\Phi_\theta^\alpha[f](x,y)-\Phi_\theta^\alpha[g](x,y)| &\le \int_0^\infty \lambda e^{-(2\lambda+\rho)t}\cdot 2\mathbb{E}_{x,y}^\theta\big[|f(X_t^i,X_t^j)-g(X_t^i,X_t^j)|\big]\,dt\\
&\le \|f-g\|_\infty \int_0^\infty 2\lambda e^{-(2\lambda+\rho)t}\,dt\\
&= \frac{2\lambda}{2\lambda+\rho}\,\|f-g\|_\infty.
\end{aligned}$$

Since $\rho > 0$, Φ_θ^α is a contraction on $C_b^0(\mathbb{R}^2)$ with modulus $\frac{2\lambda}{2\lambda+\rho} < 1$. By the Banach fixed-point theorem, Φ_θ^α has a unique fixed point $V_\theta(\cdot;\alpha) \in C_b^0(\mathbb{R}^2)$, establishing boundedness and continuity of the value function.

It remains to identify the fixed point with the value function defined in (6) and to derive the myopic characterization. To this end, construct a candidate optimizer φ_θ^*, show that it attains V_θ, and verify that no alternative decision rule yields a higher payoff.

For the remainder of this proof, with θ and α fixed, write $\varphi^* := \varphi_\theta^*(\cdot;\alpha) = \alpha_\theta[V(\cdot;\alpha)]$ for the selected rule from (7).

For any measurable decision rule φ and any $f \in C_b^0(\mathbb{R}^2)$, define

$$\Upsilon_\theta^\varphi(x,y;f) := \mathbb{1}\{\varphi(x,y) = c\}\, f(x,y) + \mathbb{1}\{\varphi(x,y) \neq c\}\, \overline{u}(\varphi(x,y);y),$$

and let $\Phi_\theta^{\alpha,\varphi}$ denote the operator obtained from Φ_θ^α by replacing Υ_θ with Υ_θ^φ. Note that $\Gamma_\theta(x,y;f)$ depends only on α, not on φ, so Φ_θ^α and $\Phi_\theta^{\alpha,\varphi}$ differ only through their Υ-terms. Since $\Upsilon_\theta(x,y;f) = \max\{M_\theta(y), f(x,y)\}$ dominates any single choice among stopping or continuing, while Υ_θ^φ commits to the choice prescribed by φ,

$$\Upsilon_\theta^\varphi(x,y;f) \leq \Upsilon_\theta(x,y;f) \qquad \text{for every } \varphi, \tag{C.67}$$

with equality when $\varphi = \varphi^*$. Consequently $\Phi_\theta^{\alpha,\varphi^*}[V_\theta] = \Phi_\theta^\alpha[V_\theta] = V_\theta$. The same contraction argument used for Φ_θ^α applies verbatim to $\Phi_\theta^{\alpha,\varphi}$, hence V_θ is the unique fixed point of $\Phi_\theta^{\alpha,\varphi^*}$ in $C_b^0(\mathbb{R}^2)$.

The operator $\Phi_\theta^{\alpha,\varphi}$ integrates against the first opportunity time, whereas J_θ runs until termination. Bridging the two requires re-expressing $\Phi_\theta^{\alpha,\varphi}$ along the sequence of opportunity times. Let $T_1 < T_2 < \cdots$ denote the ordered opportunity times of either player, so that T_k is the k-th arrival of a Poisson process of rate 2λ. Given φ and α, let τ be the termination time of the game: at each T_k, whoever has the opportunity consults their decision rule: φ for player i; α_{θ_j} for j. Let Ψ_θ^φ denote the expected terminal payoff accrued at τ. Unpacking the definition of $\Phi_\theta^{\alpha,\varphi}$ yields the one-step representation:

$$\Phi_\theta^{\alpha,\varphi}[f](x,y) = \mathbb{E}_{x,y}^{\theta;\alpha,\varphi}\left[\int_0^{\tau\wedge T_1} e^{-\rho s}\pi\, ds + e^{-\rho\tau}\Psi_\theta^\varphi \mathbb{1}\{\tau \leq T_1\} + e^{-\rho T_1} f(X_{T_1}^i, X_{T_1}^j)\mathbb{1}\{\tau > T_1\}\right].$$

The claim is that iterating k times gives

$$(\Phi_\theta^{\alpha,\varphi})^k[f](x,y) = \mathbb{E}_{x,y}^{\theta;\alpha,\varphi}\Bigg[\int_0^{\tau\wedge T_k} e^{-\rho s}\pi\,ds + e^{-\rho\tau}\Psi_\theta^{\varphi}\mathbb{1}\{\tau \le T_k\} + e^{-\rho T_k} f(X^i_{T_k}, X^j_{T_k})\mathbb{1}\{\tau > T_k\}\Bigg]. \tag{C.68}$$

Equation (C.68) follows by induction from the one-step identity, using the strong Markov property at T_k and the memorylessness of the opportunity processes.

Attainment. Apply (C.68) with $f = V_\theta$ and $\varphi = \varphi^*$. The left-hand side of (C.68) equals V_θ for every k, since V_θ is the fixed point of $\Phi_\theta^{\alpha,\varphi^*}$. For the right-hand side, as $k \to \infty$, $T_k \to \infty$ a.s., so $\tau \wedge T_k \uparrow \tau$ and $\mathbb{1}\{\tau \le T_k\} \uparrow \mathbb{1}\{\tau < \infty\}$ a.s. Moreover,

$$|e^{-\rho T_k} V_\theta(X^i_{T_k}, X^j_{T_k})| \le \|V_\theta\|_\infty e^{-\rho T_k} \to 0 \quad \text{a.s.},$$

and the flow term is uniformly bounded by π/ρ. Dominated convergence then yields

$$V_\theta(x,y;\alpha) = \mathbb{E}_{x,y}^{\theta;\alpha,\varphi^*}\left[\int_0^{\tau} e^{-\rho s}\pi\,ds + e^{-\rho\tau}\Psi_\theta^{\varphi^*}\mathbb{1}\{\tau < \infty\}\right] = J_\theta(x,y;\varphi^*,\alpha).$$

Optimality. For any measurable φ, (C.67) gives $\Phi_\theta^{\alpha,\varphi}[V_\theta] \le \Phi_\theta^{\alpha}[V_\theta] = V_\theta$. Since both $\Upsilon_\theta^{\varphi}$ and Γ_θ are monotone in f, the operator $\Phi_\theta^{\alpha,\varphi}$ is monotone, so iterating yields $(\Phi_\theta^{\alpha,\varphi})^k[V_\theta] \le V_\theta$ for every k. Applying (C.68) to the left-hand side and passing to the limit $k \to \infty$ by dominated convergence as above,

$$J_\theta(x,y;\varphi,\alpha) \le V_\theta(x,y;\alpha).$$

The attainment and optimality results give $V_\theta(x,y;\alpha) = \sup_\varphi J_\theta(x,y;\varphi,\alpha)$, attained by φ^*. By construction, this response is myopically optimal and has the collaboration region in (8). □

Proof of Lemma A.2. The proof proceeds in three steps. First, Sobolev regularity is established by solving the equation on bounded domains and showing, through stochastic representations, that the resulting solutions converge to the value function. Second, Sobolev embedding upgrades this regularity to local $C^{1,\gamma}$-smoothness. Finally, uniqueness is obtained from a Feynman–Kac representation and the contraction property in-

herited from the Bellman operator.

Fix $\theta \in \{n, r\}$ and a myopically optimal conjecture α. Proposition 6 gives $V_\theta \in C_b^0(\mathbb{R}^2)$. Use the uniformly elliptic generator and opportunity maps from Appendix A.1, with fixed strategy arguments suppressed. Applying Fubini's theorem to the Bellman fixed-point representation yields (A.7), with δ as defined there. The corresponding HJB equation is (A.6).

Claim 1: $V_\theta \in W^{2,p}_{\mathrm{loc}}(\mathbb{R}^2)$ for every $p \in (1, \infty)$.

Because the Krylov–Itô formula used below requires an exponent greater than the dimension of the state space, first fix $p > 2$. Let $B_R \subset \mathbb{R}^2$ be an arbitrary open ball and define

$$\tau_B := \inf\{t \geq 0 : (X_t^i, X_t^j) \notin B_R\}.$$

The exit time τ_B is almost surely finite because the diffusion is nondegenerate and B_R is bounded. Applying the strong Markov property at τ_B to (A.7) gives

$$V_\theta(x, y) = \mathbb{E}^\theta_{x,y}\left[\int_0^{\tau_B} e^{-\delta s}\left[\pi + \lambda(\Upsilon_\theta + \Gamma_\theta)(x, y; V_\theta)\right] ds + e^{-\delta \tau_B} V_\theta(X^i_{\tau_B}, X^j_{\tau_B})\right] \tag{C.69}$$

The direct application of the Dirichlet existence theorem with boundary data V_θ is not available because V_θ is currently known only to be continuous. Choose instead a sequence $g_k \in C^\infty(\overline{B_R})$ such that

$$\sup_{\partial B_R} |g_k - V_\theta| \longrightarrow 0.$$

For each k, consider the Dirichlet problem

$$\begin{cases} \mathcal{L}_\theta \nu_k - \delta \nu_k = -\pi - \lambda(\Upsilon_\theta + \Gamma_\theta)(x, y; V_\theta) & \text{in } B_R, \\ \nu_k = g_k & \text{on } \partial B_R. \end{cases} \tag{C.70}$$

The operator $\mathcal{L}_\theta - \delta$ is uniformly elliptic with constant principal coefficients, bounded first-order coefficients, and a strictly negative zeroth-order coefficient. Moreover, the right-hand side of (C.70) is bounded and therefore belongs to $L^p(B_R)$. Since $g_k \in W^{2,p}(B_R)$,Gilbarg and Trudinger (2001, Theorem 9.15) gives a unique strong solution

$$\nu_k \in W^{2,p}(B_R), \qquad \nu_k - g_k \in W^{1,p}_0(B_R).$$

Because $p > 2$, ν_k has a continuous representative on $\overline{B_R}$ whose boundary values agree with g_k. Applying the Krylov–Itô formula to $e^{-\delta(t \wedge \tau_B)} \nu_k(X^i_{t \wedge \tau_B}, X^j_{t \wedge \tau_B})$, using the usual

localization to balls compactly contained in B_R, and then letting $t \to \infty$, gives

$$\nu_k(x,y) = \mathbb{E}^{\theta}_{x,y}\bigg[\int_0^{\tau_B} e^{-\delta s}\,[\pi + \lambda(\Upsilon_\theta + \Gamma_\theta)(x,y;V_\theta)]\,ds + e^{-\delta\tau_B} g_k(X^i_{\tau_B}, X^j_{\tau_B})\bigg]. \tag{C.71}$$

Comparing (C.71) with (C.69) yields

$$\begin{aligned}|\nu_k(x,y) - V_\theta(x,y)| &\le \mathbb{E}^{\theta}_{x,y}\big[e^{-\delta\tau_B}\,\big|g_k(X^i_{\tau_B}, X^j_{\tau_B}) - V_\theta(X^i_{\tau_B}, X^j_{\tau_B})\big|\big]\\ &\le \sup_{\partial B_R} |g_k - V_\theta|.\end{aligned}$$

Consequently,

$$\nu_k \longrightarrow V_\theta \qquad \text{uniformly on } \overline{B_R}. \tag{C.72}$$

Applying the interior $W^{2,p}$ estimate on $B_{R/2} \Subset B_R$ gives

$$\|\nu_k\|_{W^{2,p}(B_{R/2})} \le C\bigg(\|\nu_k\|_{L^p(B_R)} + \big\|\pi + \lambda(\Upsilon_\theta + \Gamma_\theta)(x,y;V_\theta)\big\|_{L^p(B_R)}\bigg),$$

where C is independent of k. The right-hand side is uniformly bounded in k by (C.72) and the boundedness of V_θ, Υ_θ, and Γ_θ. Therefore, ν_k is bounded in $W^{2,p}(B_{R/2})$.

Since $1 < p < \infty$, the space $W^{2,p}(B_{R/2})$ is reflexive. Hence, a subsequence of ν_k converges weakly in $W^{2,p}(B_{R/2})$. Its weak limit must equal V_θ, because the full sequence converges uniformly to V_θ. It follows that $V_\theta \in W^{2,p}(B_{R/2})$.

Passing to the weak limit in (C.70) also shows that (A.6), and hence (A.5), holds almost everywhere on $B_{R/2}$. Since the center and radius of B_R and the exponent $p > 2$ were arbitrary,

$$V_\theta \in W^{2,p}_{\mathrm{loc}}(\mathbb{R}^2) \qquad \text{for every } p > 2.$$

It remains to cover $p \in (1,2]$. The preceding argument, applied with $p = 3$, gives $V_\theta \in W^{2,3}_{\mathrm{loc}}(\mathbb{R}^2)$. On every bounded ball B, finite measure implies $L^3(B) \subset L^p(B)$ for every $p \in (1,2]$. Applying this inclusion to V_θ and to each of its weak derivatives of order at most two gives $W^{2,3}(B) \subset W^{2,p}(B)$, for $p \in (1,2]$. Therefore, $V_\theta \in W^{2,p}_{\mathrm{loc}}(\mathbb{R}^2)$ also for every $p \in (1,2]$. Combining the two ranges proves Claim 1.

Claim 2: $V_\theta \in C^{1,\gamma}_{\mathrm{loc}}(\mathbb{R}^2)$ for every $\gamma \in (0,1)$.

Fix $\gamma \in (0,1)$ and choose $p > 2$ sufficiently large that

$$\gamma < 1 - \frac{2}{p}.$$

By the Sobolev embedding theorem in two dimensions (Gilbarg and Trudinger, 2001, Theorem 7.26), $W^{2,p}_{\text{loc}}(\mathbb{R}^2) \hookrightarrow C^{1,\gamma}_{\text{loc}}(\mathbb{R}^2)$. Claim 1 therefore implies $V_\theta \in C^{1,\gamma}_{\text{loc}}(\mathbb{R}^2)$ for every $\gamma \in (0,1)$.

Claim 3: For every $p > 2$, V_θ is the unique bounded $W^{2,p}_{\text{loc}}(\mathbb{R}^2)$ strong solution of (A.5).

Fix $p > 2$ and suppose that $\widetilde{V}_\theta \in W^{2,p}_{\text{loc}}(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2)$ is another strong solution of (A.5). Since $p > 2$, local Sobolev embedding gives a continuous representative of $\widetilde{V}_\theta$, and hence $\widetilde{V}_\theta \in C^0_b(\mathbb{R}^2)$. Applying the same localized Krylov–Itô argument used above to $\widetilde{V}_\theta$, and then letting the localization radius and time horizon tend to infinity, gives

$$\widetilde{V}_\theta(x,y) = \mathbb{E}^\theta_{x,y}\left[\int_0^\infty e^{-\delta s}\left[\pi + \lambda(\Upsilon_\theta + \Gamma_\theta)(X^i_s, X^j_s; \widetilde{V}_\theta)\right] ds\right].$$

Here the terminal term vanishes because $\widetilde{V}_\theta$ is bounded and $\delta > 0$. Reversing the Fubini calculation leading to (A.7), this representation is equivalent to $\widetilde{V}_\theta = \Phi^\alpha_\theta[\widetilde{V}_\theta]$. Proposition 6 establishes that Φ^α_θ is a contraction on $C^0_b(\mathbb{R}^2)$ and therefore has a unique fixed point, namely V_θ. Since $\widetilde{V}_\theta$ is also a fixed point of Φ^α_θ, it follows that $\widetilde{V}_\theta = V_\theta$, proving uniqueness. □

Proof of Lemma A.3. Fix a myopically optimal conjecture α and a type θ. Suppress α from V_θ and the indifference set $\mathcal{U}_\theta$ defined in (9). The selected collaboration region is given by (8). Since V_θ and M_θ are continuous, this region is open and

$$\partial \mathcal{C}^{BR}_\theta(\alpha) \subseteq \mathcal{U}_\theta.$$

It therefore suffices to prove that $\mathcal{U}_\theta$ has two-dimensional Lebesgue measure zero. No identification of $\mathcal{C}_\theta(\alpha)$ with $\mathcal{C}^{BR}_\theta(\alpha)$ is required.

The definition of M_θ and Table 1 implies that $M_\theta = 0$ for $y < y_\theta$ and that $M_\theta(y)$ is affine in $p(y)$ for $y > y_\theta$. The threshold line $\{y = y_\theta\}$ has two-dimensional Lebesgue measure zero. Thus, it is without loss to work separately on the two open half-planes $\{y < y_\theta\}$ and $\{y > y_\theta\}$, on each of which M_θ is smooth. By Lemma A.2, $V_\theta \in W^{2,p}_{\text{loc}}(\mathbb{R}^2)$ for every $p \in (1,\infty)$. Hence, $V_\theta - M_\theta \in W^{2,p}_{\text{loc}}$ on each of the two half-planes separated by $y = y_\theta$. Suppose that the contact set has positive measure in one of these half-planes. Since $V_\theta - M_\theta = 0$ on that set, Gilbarg and Trudinger (2001, Lemma 7.7), applied first

to $V_\theta - M_\theta$ and then to each of its weak first derivatives, implies

$$\nabla(V_\theta - M_\theta) = 0 \quad \text{and} \quad \text{Hess}(V_\theta - M_\theta) = 0$$

almost everywhere on the contact set, where Hess denotes the matrix of second-order weak derivatives. Consequently, $\mathcal{L}_\theta V_\theta = \mathcal{L}_\theta M_\theta$ at almost every contact point. At such a point, the own-opportunity term $(M_\theta - V_\theta)^+$ vanishes. The HJB equation therefore reduces to the contact identity

$$0 = \pi + \mathcal{L}_\theta M_\theta(y) - \rho M_\theta(y) + \lambda \left[\Gamma_\theta(x, y; M_\theta) - M_\theta(y)\right]. \tag{C.73}$$

The expression on the right-hand side of (C.73) has only finitely many possible functional forms, at any state:

1. each partner-type continuation indicator is either zero or one;
2. whenever the partner stops, the action is either quit or the type-specific action;
3. the survival drift $\Lambda(y, x; \alpha)$ belongs to $\{-\lambda, 0, \lambda\}$.

Conditional on any of these finitely many configurations, the right-hand side of (C.73) is a real-analytic function of y.

The quitting region. Suppose $y < y_\theta$. Then

$$M_\theta(y) = 0, \qquad \mathcal{L}_\theta M_\theta(y) = 0,$$

and (C.73) reduces to

$$0 = \pi + \lambda \Gamma_\theta(x, y; 0). \tag{C.74}$$

For any configuration of the partner's stopping behavior, $\Gamma_\theta(x, y; 0)$ is affine in the posterior $p(y)$. Then (C.74) becomes

$$0 = \pi + \lambda \Big[\mathbb{1}\{(y, x) \notin \mathcal{C}_n\} b_\theta^n + p(y) \Big(\mathbb{1}\{(y, x) \notin \mathcal{C}_r\} b_\theta^r - \mathbb{1}\{(y, x) \notin \mathcal{C}_n\} b_\theta^n \Big) \Big].$$

where, for each partner type $\tilde{\theta}$, the indicators determine whether that type stops, and $b_\theta^{\tilde{\theta}}$ denotes the resulting terminal payoff to a type-θ player. Depending on the terminal action taken,

$$b_r^r \in \{0, G_r\}, \qquad b_r^n \in \{0, -L_r\},$$
$$b_n^r \in \{0, -L_n\}, \qquad b_n^n \in \{0, -L_n\}.$$

Thus, conditional on any fixed configuration (C.74) is an affine function of $p(y)$ and can be identically zero only if

$$\mathbb{1}\{(y,x) \notin \mathcal{C}_r\} b_\theta^r = \mathbb{1}\{(y,x) \notin \mathcal{C}_n\} b_\theta^n = -\frac{\pi}{\lambda}. \tag{C.75}$$

For $\theta = r$, the first term in (C.75) belongs to $\{0, G_r\}$ and the second to $\{0, -L_r\}$, so the equality is impossible. For $\theta = n$, however, the terms belong to $\{0, -L_n\}$. Thus, (C.75) can hold when both types stop through their type-specific terminal actions and $\pi/\lambda = L_n$. The payoff assumptions rules this out. Therefore, under every fixed configuration, the right-hand side of (C.74) is a nonzero affine function of $p(y)$. Since $p(\cdot)$ is strictly increasing, it can vanish at no more than one value of y.

The type-specific region. Suppose $y > y_\theta$. Using

$$p'(y) = p(y)(1-p(y)); \quad p''(y) = p(y)(1-p(y))(1-2p(y)); \quad \tanh(y/2) = 2p(y) - 1,$$

it follows that

$$\frac{\psi^2}{2} M_\theta''(y) + \frac{\psi^2}{2} \tanh(y/2) M_\theta'(y) = 0.$$

Consequently,

$$\mathcal{L}_\theta M_\theta(y) = \Lambda(y, x; \alpha) M_\theta'(y).$$

Since $\Lambda(y, x; \alpha) = \lambda \left[\mathbb{1}\{(y,x) \notin \mathcal{C}_n\} - \mathbb{1}\{(y,x) \notin \mathcal{C}_r\}\right]$, the term $\Lambda(y, x; \alpha) M_\theta'(y)$ cancels the terms quadratic in $p(y)$ that $\Gamma_\theta(x, y; M_\theta)$ generates. With this cancellation, (C.73) for the normative type reduces to

$$\begin{aligned} 0 = {} & \pi + \rho F_n + \lambda \mathbb{1}\{(y,x) \notin \mathcal{C}_n\}(F_n + b_n^n) \\ & + p(y) \Big\{ -\rho(G_n + F_n) + \lambda \big[\mathbb{1}\{(y,x) \notin \mathcal{C}_r\}(b_n^r - G_n) - \mathbb{1}\{(y,x) \notin \mathcal{C}_n\}(F_n + b_n^n) \big] \Big\}. \end{aligned}$$

This expression is affine in $p(y)$, so it vanishes at more than one posterior value only if it vanishes identically, which requires its constant term to be zero. Since $F_n + b_n^n \in \{F_n, F_n - L_n\}$, that term is strictly positive unless normative types whistleblows at (y, x). In that case it equals $\pi + \rho F_n - \lambda(L_n - F_n)$, which is nonzero by the genericity condition in footnote 2. The same steps for the rogue type give an affine expression with constant term $\pi + \rho L_r + \lambda \mathbb{1}\{(y,x) \notin \mathcal{C}_n\}(L_r + b_r^n)$, which is strictly positive because $L_r + b_r^n \in \{L_r, 0\}$. For both types, the expression therefore has at most one zero in $p(y)$, and since $p(\cdot)$ is strictly increasing, (C.73) has at most one solution y under each fixed configuration.

There are only finitely many configurations of the partner's continuation indicators and terminal actions. For each fixed configuration, the preceding arguments show that, in either half-plane, the contact identity can hold at no more than one value of y. Therefore, up to the null set on which either the HJB equation or the derivative identities fail, the contact set is contained in a finite union of horizontal lines, each has two-dimensional Lebesgue measure zero. Since

$$\partial \mathcal{C}_\theta^{BR}(\alpha) \subseteq \mathcal{U}_\theta,$$

the established equality $\mathrm{Leb}_2(\mathcal{U}_\theta) = 0$ implies

$$\mathrm{Leb}_2(\partial \mathcal{C}_\theta^{BR}(\alpha)) = 0.$$

This completes the proof. □

Proof of Lemma A.4. Fix θ and a state with $|V_\theta(x, y; \alpha) - M_\theta(y)| > \varepsilon$. The uniform bound on $V_\theta(\cdot; \alpha) - V_\theta(\cdot; \widetilde{\alpha})$ implies that both values lie strictly on the same side of $M_\theta(y)$. The common selection (7) therefore prescribes collaboration for both if they lie above it, and $m_\theta(y)$ for both if they lie below it. Thus the selected rules agree outside the stated ε-indifference region. □

Proof of Proposition 7. Throughout, the terminal payoffs, ρ, ψ, and $\pi/\lambda \in (0, L_n)$ are fixed. Constants denoted by C are finite, may increase between occurrences, and depend only on these fixed quantities. They are independent of λ, the conjectured strategy, and the initial state. Since $\pi/\lambda < L_n < L_r$, the restrictions $\lambda L_\theta > \pi$ hold throughout this family. For sufficiently small λ, $\rho G_\theta > \pi$ also holds, and the expression excluded in footnote 2 is strictly positive:

$$\pi + \rho F_n - \lambda(L_n - F_n) = \rho F_n + \lambda \left(\frac{\pi}{\lambda} - L_n + F_n \right) > 0.$$

For a bounded continuous pair f, use the selected rule $\alpha[f]$ from (7) and define

$$\mathcal{T}(f)_\theta := V_\theta(\cdot; \alpha[f]).$$

The value on the right is the one already defined in (6). A fixed point gives a symmetric equilibrium by Proposition 6. I construct a complete invariant set containing every equilibrium value and prove that $\mathcal{T}$ is a contraction on this set.

Uniform bounds and discounted occupation. For any myopically optimal conjecture α,

the opportunity maps in (A.7) satisfy

$$\|\Upsilon_\theta(\cdot;V_\theta)\|_\infty + \|\Gamma_\theta(\cdot;V_\theta)\|_\infty \le C + 2\|V_\theta\|_\infty.$$

Consequently, (A.7) gives $(\rho+2\lambda)\|V_\theta\|_\infty \le \pi + \lambda C + 2\lambda\|V_\theta\|_\infty$. Since π/λ is fixed,

$$\|V_\theta(\cdot;\alpha)\|_\infty \le C\lambda. \tag{C.76}$$

No sign restriction is imposed. Applying the uniform interior $W^{2,4}$ estimates used in Lemma A.2 to (A.5), followed by Sobolev embedding, yields

$$\|\nabla V_\theta(\cdot;\alpha)\|_\infty \le C\lambda. \tag{C.77}$$

Indeed, the value and the source in the equation are $O(\lambda)$ on every ball of fixed radius, while the ellipticity and drift bounds in (1)–(2) are uniform for bounded λ.

I also use a discounted occupation estimate. For either own type $\vartheta \in \{n, r\}$, bounded measurable drift and constant nondegenerate diffusion give the uniform Gaussian bound

$$\mathbb{P}^{\vartheta;\alpha}_{x,y}\big((X^i_t, X^j_t) \in A\big) \le \int_A \frac{C}{t}\exp\!\left(-\frac{|(x,y)-(x',y')|^2}{Ct}\right) dx'\,dy', \qquad 0 < t \le 1. \tag{C.78}$$

If every vertical section of D has length at most $h \le 1$, integration first over that section bounds this probability by $C\min\{1, h/\sqrt{t}\}$. For $t \ge 1$, the Markov property gives Ch. Integration over time therefore yields

$$\sup_{x,y}\mathbb{E}^{\vartheta;\alpha}_{x,y}\left[\int_0^\infty e^{-\rho t}\mathbb{1}\{(X^i_t, X^j_t) \in D\}\,dt\right] \le Ch. \tag{C.79}$$

The same estimate holds for horizontal sections, for a finite union of sets with either kind of section bound, and with discount rate $\rho + 2\lambda$. It also holds when survival drift is zero.

A comparison value that permits quitting. Let $\mathcal{L}^0_\theta$ be the generator of (1)–(2) with survival drift omitted. Let V^0_θ be the unique bounded solution of

$$\begin{aligned}(\rho - \mathcal{L}^0_\theta)V^0_\theta(x,y) = \frac{\pi}{\lambda} + M_\theta(y)\\ + p(y)u^p(m_r(x);\theta) + (1-p(y))u^p(m_n(x);\theta).\end{aligned} \tag{C.80}$$

This is the discounted resolvent under the signal-only dynamics. For these dynamics,

the superscript label 0 indicates that survival drift is omitted: $\mathbb{E}_x^{\theta;0}$ is expectation for the own coordinate starting at x, conditional on own type θ, and $\mathbb{E}_y^0$ is expectation for the partner coordinate starting at y, with the partner's type averaged according to $p(y)$. The comparison value V_θ^0 is independent of λ because π/λ is fixed. The regularity argument in Lemma A.2 applies to this linear equation; in particular, V_θ^0 has bounded continuous first derivatives.

Suppose $\max_\theta \|f_\theta\|_\infty \le C_0\lambda$, with C_0 fixed, and write $V_\theta = V_\theta(\cdot;\alpha[f])$. The myopic rule and the positive right derivative of M_ϑ at y_ϑ imply that a partner can collaborate instead of taking a type-specific terminal action only inside

$$y_\vartheta < x < y_\vartheta + C\lambda, \qquad \vartheta \in \{n, r\}.$$

Below y_ϑ, her terminal action is quitting, so replacing it by collaboration changes the opportunity payoff only by $V_\theta = O(\lambda)$.

Subtract (C.80) from (A.5) divided by λ. Under the operator $\rho - \mathcal{L}_\theta^{\alpha[f]}$, the source for $V_\theta/\lambda - V_\theta^0$ is bounded in absolute value by

$$C\lambda + C \sum_{\vartheta\in\{n,r\}} \mathbb{1}\{y_\vartheta < x < y_\vartheta + C\lambda\}. \tag{C.81}$$

Indeed, the own-opportunity discrepancy is $(M_\theta - V_\theta)^+ - M_\theta$, whose magnitude is at most $|V_\theta|$, and the partner-opportunity discrepancy has the bound just described. The remaining terms are $-V_\theta$ and $(\mathcal{L}_\theta^{\alpha[f]} - \mathcal{L}_\theta^0)V_\theta^0$. Both are $O(\lambda)$ by (C.76), $|\Lambda| \le \lambda$, and boundedness of ∇V_θ^0.

The bounded resolvent argument used in (A.7), together with (C.79), gives

$$\left\| \frac{V_\theta}{\lambda} - V_\theta^0 \right\|_\infty \le C\lambda. \tag{C.82}$$

In any ball of fixed radius, each strip in (C.81) has area $O(\lambda)$, so its L^4 norm is $O(\lambda^{1/4})$. Uniform interior $W^{2,4}$ estimates and Sobolev embedding then give

$$\left\| \nabla\left(\frac{V_\theta}{\lambda} - V_\theta^0 \right) \right\|_\infty \le C\lambda^{1/4}. \tag{C.83}$$

Choose C large enough and then λ sufficiently small. Let $\mathcal{K}_\lambda$ consist of bounded

continuous pairs satisfying

$$\|f_\theta - \lambda V^0_\theta\|_\infty \le C\lambda^2, \tag{C.84}$$

$$\mathrm{Lip}(f_\theta - \lambda V^0_\theta) \le C\lambda^{5/4}. \tag{C.85}$$

The pair λV^0 belongs to this set, and the bounds are preserved by uniform limits, so it is nonempty and complete in the sup norm. Its elements satisfy a common bound $\|f_\theta\|_\infty \le C_0\lambda$. Equations (C.82)–(C.83) imply $\mathcal{T}(\mathcal{K}_\lambda) \subseteq \mathcal{K}_\lambda$.

Geometry of the quitting indifference sets. Under the signal-only dynamics the coordinates are independent, and $p(X^j_t)$ is a martingale. The partner-payoff contribution to V^0_θ is therefore a weighted sum of one-dimensional discounted occupations of $[y_n, \infty)$ and $[y_r, \infty)$ by X^i. Each occupation has strictly positive derivative in the starting reputation x. The discounted expectation of $M_\theta(X^j_t)$ has strictly positive derivative in y at every finite y. The payoffs in Table 1 consequently give

$$\partial_x V^0_n(x, y) < 0, \qquad \partial_y V^0_r(x, y) > 0. \tag{C.86}$$

For the first inequality, both normative partner-payoff coefficients are negative. For the second, the rogue partner-payoff contribution increases with $p(y)$, as does the own-opportunity contribution.

The tails make these crossing properties uniform near zero. As $x \to -\infty$, the partner-payoff contribution vanishes uniformly in y, leaving a positive limiting value at least $(\pi/\lambda)/\rho$. As $y \to -\infty$, uniformly in x,

$$V^0_\theta(x, y) \longrightarrow \frac{\pi/\lambda}{\rho} - L_\theta\, \mathbb{E}^{\theta;0}_x \left[\int_0^\infty e^{-\rho t} \mathbb{1}\{X^i_t \ge y_n\}\, dt \right]. \tag{C.87}$$

The right-hand side decreases from $(\pi/\lambda)/\rho > 0$ to $((\pi/\lambda) - L_\theta)/\rho < 0$, with strictly negative derivative at every finite x. The negative upper limit uses precisely $\pi/\lambda < L_n < L_r$.

For the normative type, as $x \to \infty$,

$$V^0_n(x, y) \longrightarrow \frac{(\pi/\lambda) - L_n}{\rho} + \mathbb{E}^0_y \left[\int_0^\infty e^{-\rho t} M_n(X^j_t)\, dt \right]. \tag{C.88}$$

Its derivative in y is strictly positive on every compact interval. The relevant first derivatives also converge in these tail statements, by the one-dimensional occupation

representations.

Here is an explicit finite partition of the near-zero sets on $y \le y_\theta$. Choose $Y < \min(y_n, y_r)$ sufficiently negative. Equation (C.87) places all near-zero points with $y < Y$ in a fixed bounded interval $a < x < b$; there, $\partial_x V_\theta^0$ is bounded above by a negative constant. For the normative type, enlarge $[a, b]$ so that the remaining near-zero points either have $a \le x \le b$, where $\partial_x V_n^0$ has a uniform negative bound, or have $x > b$ and $Y \le y \le y_n$, where (C.88) gives a uniform positive bound on $\partial_y V_n^0$. For the rogue type, on $Y \le y \le y_r$ the derivative of the discounted expectation of M_r bounds $\partial_y V_r^0$ away from zero, uniformly in x. Outside these finitely many rectangular pieces, $|V_\theta^0|$ is bounded away from zero on $y \le y_\theta$.

By (C.85), these coordinate crossing bounds persist for f_θ/λ when λ is small. Thus $\{|f_\theta| \le \varepsilon\}$ near zero, restricted to $y < y_\theta$, is a finite union of pieces whose horizontal or vertical sections have length at most $C\varepsilon/\lambda$.

Disagreement between nearby strategies. Take $f, g \in \mathcal{K}_\lambda$ and put $\varepsilon := \max_\theta \|f_\theta - g_\theta\|_\infty$. By (C.84), $\varepsilon \le 2C\lambda^2$. If $\alpha_\theta[f] \ne \alpha_\theta[g]$, their comparisons with M_θ have opposite signs, so $|f_\theta - M_\theta| \le \varepsilon$. This is the sign argument in Lemma A.4. Split the disagreement set into D_θ^q on $y < y_\theta$ and D_θ^+ on $y > y_\theta$; the threshold line is null and does not affect occupation or payoffs.

On D_θ^q the preceding argument gives section lengths at most $C\varepsilon/\lambda$ in finitely many pieces. On D_θ^+, $M_\theta(y) \le C\lambda$, so $y_\theta < y < y_\theta + C\lambda$. In this strip M_θ' is bounded below by a positive constant, whereas $\mathrm{Lip}(f_\theta) \le C\lambda$. Thus every vertical section of D_θ^+ has length at most $C\varepsilon$. By (C.79), for either own type $\vartheta \in \{n, r\}$,

$$\sup_{x,y} \mathbb{E}_{x,y}^{\vartheta;\alpha} \left[\int_0^\infty e^{-(\rho+2\lambda)t} \mathbb{1}_{D_\theta^q}(X_t^i, X_t^j)\, dt \right] \le C\frac{\varepsilon}{\lambda}, \tag{C.89}$$

$$\sup_{x,y} \mathbb{E}_{x,y}^{\vartheta;\alpha} \left[\int_0^\infty e^{-(\rho+2\lambda)t} \mathbb{1}_{D_\theta^+}(X_t^i, X_t^j)\, dt \right] \le C\varepsilon. \tag{C.90}$$

The same bounds hold for the transposed sets, which exchange the players' coordinates, and are uniform over the conjectured strategy generating the diffusion.

Contraction. Write $\alpha = \alpha[f]$, $\widetilde{\alpha} = \alpha[g]$, $V_\theta = \mathcal{T}(f)_\theta$, and $\widetilde{V}_\theta = \mathcal{T}(g)_\theta$. Subtracting their

equations (A.6), with the generator induced by α on the left, gives

$$\begin{aligned}
&(\rho + 2\lambda - \mathcal{L}_\theta^\alpha)(V_\theta - \widetilde{V}_\theta) \\
&= \lambda\big[\Upsilon_\theta(V_\theta) - \Upsilon_\theta(\widetilde{V}_\theta)\big] \\
&\quad + \lambda\big[\Gamma_\theta^\alpha(V_\theta) - \Gamma_\theta^\alpha(\widetilde{V}_\theta)\big] \\
&\quad + \lambda\big[\Gamma_\theta^\alpha(\widetilde{V}_\theta) - \Gamma_\theta^{\widetilde{\alpha}}(\widetilde{V}_\theta)\big] \\
&\quad + (\mathcal{L}_\theta^\alpha - \mathcal{L}_\theta^{\widetilde{\alpha}})\widetilde{V}_\theta. \qquad \text{(C.91)}
\end{aligned}$$

The strategy superscripts only indicate the dependence already present in Γ_θ and $\mathcal{L}_\theta$. The bounded resolvent argument in Lemma A.2 applies to this difference equation. Both opportunity maps are one-Lipschitz in the continuation value, as shown in the proof of Proposition 6. The first two terms contribute at most

$$\frac{2\lambda}{\rho + 2\lambda}\|V_\theta - \widetilde{V}_\theta\|_\infty.$$

On a transposed quitting disagreement set, the partner chooses between quitting, which pays zero, and collaboration, which leaves $\widetilde{V}_\theta = O(\lambda)$. On a transposed type-specific disagreement set, the payoff change is bounded by a constant. Hence

$$\left|\Gamma_\theta^\alpha(\widetilde{V}_\theta) - \Gamma_\theta^{\widetilde{\alpha}}(\widetilde{V}_\theta)\right| \le C\lambda \sum_\vartheta \mathbb{1}_{(D_\vartheta^q)^\top} + C \sum_\vartheta \mathbb{1}_{(D_\vartheta^+)^\top}.$$

After multiplication by λ, equations (C.89)–(C.90) bound the discounted contribution by $C\lambda\varepsilon$.

The generators differ only through survival drift, by (1)–(2). Its change is bounded by λ times disagreement indicators in the relevant coordinate order. Together with (C.77), this gives

$$\left|(\mathcal{L}_\theta^\alpha - \mathcal{L}_\theta^{\widetilde{\alpha}})\widetilde{V}_\theta\right| \le C\lambda^2 \sum_\vartheta \left(\mathbb{1}_{D_\vartheta^q} + \mathbb{1}_{D_\vartheta^+} + \mathbb{1}_{(D_\vartheta^q)^\top} + \mathbb{1}_{(D_\vartheta^+)^\top}\right).$$

The same occupation estimates bound this contribution by $C\lambda\varepsilon$. Combining the bounds and absorbing the direct continuation-value terms yields

$$\|\mathcal{T}(f) - \mathcal{T}(g)\|_\infty \le C\lambda\|f - g\|_\infty. \qquad \text{(C.92)}$$

For sufficiently small λ, $\mathcal{T}$ is a contraction on the complete invariant set $\mathcal{K}_\lambda$. Banach's theorem gives a unique fixed point, and Proposition 6 identifies its induced strategy as

a symmetric Markov perfect equilibrium.

First, modifications on Lebesgue-null sets do not affect equilibrium outcomes, as the following argument shows. Suppose two strategy profiles differ only on such sets, and let $N \subseteq \mathbb{R}^2$ contain these sets and their coordinate transposes. The transition-density property in Lemma A.1 implies, under either profile, for every own type θ, initial state (x, y), and finite t,

$$\mathbb{E}^{\theta}_{x,y}\left[\int_0^t \mathbb{1}\{(X^i_s, X^j_s) \in N\}\, ds\right] = 0.$$

The associated survival drifts therefore differ only on a set of zero occupation time. Pathwise uniqueness implies that the reputation processes can be identified under the two profiles. Moreover, the compensation formula for each player's Poisson opportunity process implies that no positive opportunity time falls in N, almost surely. Consequently, these modifications preserve reputation dynamics, stopping and terminal-action laws, and deviation payoffs.

Now let α be any symmetric stationary Markov perfect equilibrium, with value pair V. Its values satisfy (C.76)-(C.77). The verification argument for Proposition 6 implies that each equilibrium decision rule chooses an optimal opportunity action almost everywhere. Any nonmyopic terminal choices are therefore confined to a null set. By the preceding argument, those choices can be replaced by m_θ, yielding a pointwise myopically optimal representative without changing values or equilibrium outcomes

Apply Lemma A.3 to this representative. The indifference set $\mathcal{U}_\theta$ in (9) is null for each type. Set $\beta := \alpha[V]$, using (7). Outside the collaboration–stopping indifference set and the terminal-action tie line $y = y_\theta$, the optimal opportunity action is unique. Both sets are null. Hence α and β agree almost everywhere. The preceding null-set argument shows that β generates the same values and equilibrium outcomes as α. In particular,

$$V = \mathcal{T}(V).$$

Applying the approximation estimates (C.82)-(C.83) with $f = V$ places V in $\mathcal{K}_\lambda$. Since $\mathcal{T}$ is a contraction on $\mathcal{K}_\lambda$, every symmetric stationary equilibrium has the unique fixed-point value already constructed. That value determines a unique strategy under the specified pointwise selection. Any other equilibrium strategy agrees with this selected strategy almost everywhere and generates the same reputation, stopping, terminal-action, and payoff laws. This establishes uniqueness within the stated equilibrium class, up to outcome-irrelevant modifications on null sets

□

Proof of Proposition 8. Fix a symmetric Markov equilibrium and suppress the equilibrium argument in the values. Use δ from Appendix A.1, set $\eta := (1 + 8\delta/\psi^2)^{-1/2}$, and write $p = p(y)$ when the argument is suppressed. For both types, define the fixed-belief comparison values and their information corrections together:

$$B_{\theta,0}(y) := \begin{cases} \dfrac{\pi - \lambda L_n + \lambda M_n(y)}{\delta}, & \theta = n, \\ \dfrac{\pi - \lambda L_r(1 - p(y)) + \lambda M_r(y)}{\delta}, & \theta = r, \end{cases} \tag{C.93}$$

$$B_\theta(y) := B_{\theta,0}(y) + \frac{\lambda\eta G_\theta p(y)}{\delta} \exp\left[-\frac{y - y_\theta}{2} - \frac{|y - y_\theta|}{2\eta}\right].$$

For each type, the correction cancels the first-derivative jump of $B_{\theta,0}$ at y_θ. Direct differentiation away from that threshold, followed by continuity of the right-hand side, shows that B_θ is bounded and C^2 and satisfies

$$\delta B_\theta - \frac{\psi^2}{2}\big(B''_\theta + \tanh(y/2)B'_\theta\big) = \delta B_{\theta,0}(y). \tag{C.94}$$

Part (i)**.** For the normative comparison, the auxiliary partnership ends at the first opportunity of either player. At her own opportunity, the player takes her myopically optimal terminal action; at her partner's opportunity, she receives $-L_n$. The function B_n accounts for information arriving before that opportunity, and will provide a lower bound on V_n. Now consider $B_n(y) - M_n(y)$. The correction in (C.93) attains its maximum at y_n and since $M_n(y) \geq 0$,

$$B_n(y) - M_n(y) \leq B_n(y_n) = \frac{\lambda}{\delta}\big(\bar{L}_n^\infty - L_n\big), \qquad \bar{L}_n^\infty = \frac{\pi}{\lambda} + \eta\frac{G_n F_n}{G_n + F_n}. \tag{C.95}$$

Trust. Suppose $L_n < \bar{L}_n^\infty$. The correction satisfies

$$\frac{B_n - B_{n,0}}{1 - p} \leq \frac{\lambda\eta F_n}{\delta} \quad (y \leq y_n), \qquad \frac{B_n - B_{n,0}}{p} \leq \frac{\lambda\eta G_n}{\delta} \quad (y \geq y_n).$$

Differentiating (C.93) and using these bounds gives

$$\begin{aligned} (1 - p)(B_n + L_n) - B'_n &\geq \frac{(1 - p)[\pi + (\rho + \lambda)L_n - \lambda F_n]}{\delta} > 0, \\ p(B_n + L_n) + B'_n &\geq \frac{p[\pi + (\rho + \lambda)L_n]}{\delta} > 0. \end{aligned} \tag{C.96}$$

The first strict inequality uses $\pi + (\rho + \lambda)L_n - \lambda F_n \geq \pi + \rho L_n > 0$. Continuity supplies

both bounds at y_n.

Equation (C.94) treats both partner types as taking their harmful terminal actions. By (A.5) and (A.3), replacing a normative partner's action by collaboration changes the Bellman residual at B_n by $\lambda[(1-p)(B_n+L_n)-B'_n]$; replacing a rogue partner's action by collaboration changes it by $\lambda[p(B_n+L_n)+B'_n]$. Quitting replaces the loss $-L_n$ with zero, and own optimization adds $\lambda(B_n-M_n)^+$. All these changes are nonnegative by (C.96), so B_n is a subsolution of (A.5). Lemma A.2 and bounded discounted comparison give $V_n(x,y) \geq B_n(y)$: both opportunity maps are nondecreasing and one-Lipschitz in the continuation value, so their linearization leaves a discount coefficient at least $\rho > 0$. Evaluating at y_n and applying (C.95) yields

$$V_n(x,y_n) \geq \frac{\lambda}{\delta}\big(\bar{L}_n^\infty - L_n\big) > 0 = M_n(y_n) \qquad \text{for every } x \in \mathbb{R}.$$

Proposition 6 therefore gives trust on $\mathbb{R}$, with the same partner reputation y_n.

Fear. Suppose $L_n > \bar{L}_n^\infty$. By (A.8)–(A.9), whenever $x > b$, a normative partner whistleblows and a rogue partner schemes, regardless of y. Thus $\Gamma_n = -L_n$ and $\Lambda(y,x;\alpha) = 0$ there. Subtracting (C.94) from (A.5), using (A.3), gives

$$(\delta - \mathcal{L}_n)(V_n - B_n) = \lambda(V_n - M_n)^+, \qquad x > b. \tag{C.97}$$

Let $\tau_b = \inf\{t \geq 0 : X_t^i \leq b\}$. The bounded drift in (1) makes the probability of reaching b over any fixed finite horizon vanish uniformly in y as $x \to \infty$. Splitting the discounted expectation at that horizon therefore gives, for every $q > 0$,

$$\sup_y \mathbb{E}_{x,y}^n\big[e^{-q\tau_b}\mathbb{1}\{\tau_b < \infty\}\big] \longrightarrow 0 \qquad (x \to \infty). \tag{C.98}$$

By (C.95), $B_n - M_n$ is uniformly strictly negative, so $(V_n - M_n)^+ \leq (V_n - B_n)^+$. Positive-part comparison in (C.97), stopped at τ_b, then gives a finite constant C with

$$(V_n - B_n)^+(x,y) \leq C\mathbb{E}_{x,y}^n\big[e^{-(\rho+\lambda)\tau_b}\mathbb{1}\{\tau_b < \infty\}\big]. \tag{C.99}$$

Combining (C.99), (C.98), and (C.95) gives

$$V_n(x,y) - M_n(y) \leq -\frac{\lambda}{\delta}\big(L_n - \bar{L}_n^\infty\big) + o(1) < 0$$

for all sufficiently high x, uniformly in y. Proposition 6 therefore gives fear on an upper half-line, as claimed.

Part (ii)**.** Use the rogue comparison B_r in (C.93), which treats normative partners as whistleblowers and rogue partners as quitters. Its comparison equation is (C.94) with $\theta = r$.

Trust. The correction satisfies

$$\frac{B_r - B_{r,0}}{1-p} \le \frac{\lambda\eta L_r}{\delta} \quad (y \le y_r), \qquad \frac{B_r - B_{r,0}}{p} \le \frac{\lambda\eta G_r}{\delta} \quad (y \ge y_r).$$

Differentiating (C.93) gives

$$\begin{aligned} (1-p)(B_r + L_r) - B_r' &\ge \frac{(1-p)(\pi + \rho L_r)}{\delta} > 0, \\ pB_r + B_r' &\ge \frac{p\pi}{\delta} > 0. \end{aligned} \tag{C.100}$$

Continuity supplies both inequalities at y_r. By (A.5) and (A.3), the Bellman residual at B_r is

$$\begin{aligned} R_{B_r}(x,y) = \lambda(B_r - M_r)^+ & \\ + \lambda \mathbb{1}\{\alpha_n(y,x) = c\}\big[(1-p)(B_r + L_r) - B_r'\big] & \\ + \lambda \mathbb{1}\{\alpha_r(y,x) = c\}\big[pB_r + B_r'\big] & \\ + \lambda(1-p)L_r \mathbb{1}\{\alpha_n(y,x) = q\} & \\ + \lambda p G_r \mathbb{1}\{\alpha_r(y,x) = s\}. & \end{aligned} \tag{C.101}$$

Here the residual is the right-hand side of (A.5) minus ρB_r, evaluated at B_r. Every term is nonnegative by (C.100). Thus B_r is a subsolution, and Lemma A.2 and bounded discounted comparison as in part (i) give $V_r \ge B_r$. The comparison is strict. Linearizing the two opportunity maps gives

$$(\delta' - \mathcal{L}_r)(V_r - B_r) = R_{B_r}, \qquad \rho \le \delta'(x,y) \le \delta.$$

Whenever $x > y_r$, a rogue partner either collaborates or schemes. Her contribution to (C.101) is therefore either $\lambda[pB_r + B_r']$ or $\lambda p G_r$, both strictly positive. Hence R_{B_r} is bounded away from zero on compact rectangles contained in $\{x > y_r\}$. The nondegenerate reputation diffusion has positive discounted occupation of such a rectangle from every initial state; the bounded discounted resolvent representation therefore gives $V_r > B_r$ everywhere. Since $p(y_r) = L_r/(G_r + L_r)$ and $M_r(y_r) = 0$,

$$B_r(y_r) = \frac{\pi - \lambda(1-\eta)G_r L_r/(G_r + L_r)}{\delta}.$$

Consequently, whenever $\pi \geq \lambda(1-\eta)G_r L_r/(G_r+L_r)$,

$$V_r(x, y_r) > B_r(y_r) \geq 0 = M_r(y_r) \qquad \text{for every } x \in \mathbb{R}.$$

Proposition 6 gives trust on $\mathbb{R}$, with the same partner reputation y_r for every own reputation. Since $G_r L_r/(G_r+L_r) < G_r$, taking $\bar{\pi} = \lambda(1-\eta)G_r$ proves the trust claim whenever $\pi \geq \bar{\pi}$.

Fear. Suppose $\pi < \bar{\pi}$ and fix all primitives except L_r. Below, C denotes a sufficiently large finite constant, independent of L_r and of the selected equilibrium. Direct differentiation of (C.93) shows that $B_r - M_r$ is maximized at y_r, so

$$\sup_y \{B_r(y) - M_r(y)\} = \frac{\pi - \lambda(1-\eta)G_r L_r/(G_r+L_r)}{\delta} \longrightarrow \frac{\pi - \bar{\pi}}{\delta} < 0.$$

For $y \leq y_r$, the correction is at most $\lambda\eta G_r/\delta$, whereas $L_r(1-p(y)) \leq G_r$ for $y \geq y_r$. Consequently, for sufficiently large L_r,

$$\frac{B_r(y) - M_r(y)}{1 + L_r(1-p(y))} \leq -\frac{1}{C} \qquad \text{for every } y.$$

The normative threshold b_n in (A.9) can be chosen independently of L_r, since M_n and $\overline{V}_n$ do not depend on L_r. Above this threshold a normative partner whistleblows regardless of her own reputation. Since $y_r = \log(L_r/G_r) \to \infty$ as $L_r \to \infty$, the interval $y_r/4 < x < 3y_r/4$ eventually lies above this threshold. Thus normative partners whistleblow throughout this interval for all sufficiently large L_r. Rogue partners there can only collaborate or quit. For a rogue partner with $y \geq y_r/4$, the bounded drift in (1) makes the probability of remaining above that fixed reputation for the next one unit of time at least $1/2$. Conditional on a normative counterpart, that counterpart receives the first opportunity before this time with probability $(1-e^{-2\lambda})/2$. Since realized rogue payoffs are at most G_r, this gives

$$V_r(y, x) \leq G_r - \frac{e^{-\rho}(1-e^{-2\lambda})}{4} L_r(1-p(x)) < 0 = M_r(x)$$

uniformly for $y \geq y_r/4$ and $y_r/4 < x < 3y_r/4$, once L_r is large: $L_r(1-p(x)) \geq G_r e^{y_r}/(1+e^{3y_r/4}) \to \infty$. Thus these rogue partners quit. On this strip, (C.101) reduces to

$$R_{B_r} = \lambda \mathbb{1}\{\alpha_r(y, x) = c\}(pB_r + B_r') \leq C\mathbb{1}\{y < y_r/4\},$$

where direct differentiation also gives $0 < pB_r + B_r' \leq (\pi + \lambda G_r)/\delta$. Moreover, (A.3)

together with the martingale property of the posterior imply

$$\mathcal{L}_r[1 + L_r(1-p)] = -\lambda L_r p' \mathbb{1}\{\alpha_r(y,x) = c\} \leq 0.$$

It remains to control $V_r - B_r$. Divide the linearized equation above by $1 + L_r(1-p)$ and apply the product rule. Since this weight has nonpositive generator, the resulting equation has discount at least ρ. Its drifts remain bounded, and the own-reputation drift is unchanged because there is no mixed diffusion term. The normalized right-hand side is bounded by $C/[1 + L_r(1 - p(y_r/4))]$, while the uniform value bound gives normalized boundary values at most C.

Apply the bounded discounted representation up to exit from the strip. For $y_r/3 \leq x \leq 2y_r/3$, each vertical boundary is at least $y_r/12$ away. The drift bound in (1) therefore gives

$$\begin{aligned}\frac{V_r(x,y) - B_r(y)}{1 + L_r(1-p(y))} &\leq \frac{C}{\rho[1 + L_r(1 - p(y_r/4))]} \\ &\quad + 2C \exp\left(-\frac{\rho y_r}{12(\rho + \lambda + \psi^2)}\right) \longrightarrow 0,\end{aligned}$$

uniformly in y on this inner interval. The first term bounds the discounted residual; the second bounds the contribution from the two vertical boundaries.

Combining this estimate with the negative bound on $(B_r - M_r)/[1 + L_r(1-p)]$ gives $V_r(x,y) < M_r(y)$ there for sufficiently large L_r. Proposition 6 establishes fear on $(x_1, x_2) = (y_r/3, 2y_r/3)$.

□

Proof of Proposition 9. Fix the selected equilibrium strategy α. For the argument, extend the reputation diffusion beyond τ using the same equilibrium coefficients. This auxiliary extension agrees with the actual reputation process before τ. Conditional on the actual type pair, it satisfies

$$dX_t^k = \left[\frac{\psi^2}{2}\left(\mathbb{1}\{\theta_k = r\} - \mathbb{1}\{\theta_k = n\}\right) + \Lambda(X_t^k, X_t^\ell; \alpha)\right] dt + \psi\, dB_t^k, \qquad k, \ell \in \{i,j\}, \quad k \neq \ell,$$

where B^i and B^j are independent Brownian motions, independent of the opportunity processes. The extension depends only on these Brownian paths and the fixed strategy α.

A uniform stopping region at high partner reputations.

Use the bound and threshold b in (A.8)–(A.9). Whenever a player's partner has reputation above b, her next opportunity induces whistleblowing if she is normative and

scheming if she is rogue, regardless of her own reputation.

Finite termination whenever at least one player is rogue.

Define the stopping indicator

$$d_\theta(x,y) := \mathbb{1}\{\alpha_\theta(x,y) \neq c\},$$

and, for each actual player k with counterpart ℓ, define

$$A_k(t) := \int_0^t d_{\theta_k}(X_u^k, X_u^\ell)\, du, \qquad A_k(\infty) := \lim_{t\to\infty} A_k(t).$$

Conditional on the entire auxiliary reputation path, the independent Poisson opportunity processes give

$$\mathbb{P}_{x,y}^{\theta_i,\theta_j}\left(\tau = \infty \,\middle|\, (X_t^i, X_t^j)_{t\geq 0}\right) = \exp\left\{-\lambda\big[A_i(\infty) + A_j(\infty)\big]\right\}.$$

In particular, almost surely on $\{\tau = \infty\}$,

$$A_i(\infty) < \infty \qquad \text{and} \qquad A_j(\infty) < \infty.$$

Suppose that player i is rogue. Using

$$\Lambda(x,y;\alpha) = \lambda\big[d_n(x,y) - d_r(x,y)\big],$$

her reputation satisfies

$$\begin{aligned} X_t^i &= x + \frac{\psi^2}{2}t + \psi B_t^i + \lambda \int_0^t d_n(X_u^i, X_u^j)\, du - \lambda A_i(t) \\ &\geq x + \frac{\psi^2}{2}t + \psi B_t^i - \lambda A_i(t). \end{aligned}$$

On $\{\tau = \infty\}$, the final term is bounded below by $-\lambda A_i(\infty)$. Since $B_t^i/t \to 0$ almost surely, it follows that

$$X_t^i \longrightarrow +\infty \qquad \text{on } \{\tau = \infty\},$$

apart from a null event.

Eventually, therefore, $X_t^i > b$ at every subsequent time. The uniform stopping region

established above then implies

$$d_{\theta_j}(X_t^j, X_t^i) = 1$$

at every sufficiently large time. Hence $A_j(\infty) = \infty$, contradicting the necessary condition for $\tau = \infty$. Thus,

$$\mathbb{P}_{x,y}^{\theta_i,\theta_j}(\tau < \infty) = 1 \qquad \text{whenever } \theta_i = r \text{ or } \theta_j = r.$$

For a mixed pair, every terminal action dissolves the partnership. Therefore $\tau^P = \tau$, proving part (i). The same argument also establishes the almost-sure finiteness of τ asserted in part (iii).

Positive survival probability for two normative players.

Choose

$$b_0 < \min\{y_n, y_r\}.$$

Whenever a player's own reputation is below b_0, either partner type, if stopping, chooses quitting rather than her type-specific terminal action. Such an exit pays the player zero.

Consider a normative player who quits at her first opportunity T_1^i. This strategy receives flow profits at least until the first opportunity of either player. A negative terminal payoff can occur only if her own reputation reaches b_0 before T_1^i. Since her reputation drift is bounded above by $\lambda - \psi^2/2$,

$$V_n(x,y) \geq \frac{\pi}{\rho + 2\lambda} - L_n \, \mathbb{P}\left(\sup_{0 \leq u \leq T_1^i} \left[\left(\lambda - \frac{\psi^2}{2} \right) u + \psi B_u^i \right] \geq b_0 - x \right), \qquad x < b_0.$$

This bound is uniform in y. The random maximum is finite almost surely because $T_1^i < \infty$ almost surely. Its tail probability therefore tends to zero as $x \to -\infty$. Consequently, there exists $\ell < b_0$ such that

$$V_n(x,y) > 0 \qquad \text{for every } x < \ell \text{ and every } y.$$

If both reputations are below ℓ, then $M_n(x) = M_n(y) = 0$, so both normative players strictly prefer collaboration. Throughout this region,

$$\Lambda(x, y; \alpha) = \lambda \left[\mathbb{1}\{\alpha_r(x,y) = c\} - 1 \right] \leq 0,$$

and similarly for the transposed state. Conditional on two normative types, each reputation thus has drift at most $-\psi^2/2$ until the pair leaves the region. Starting from

$x, y < \ell$, compare the reputations with

$$x - \frac{\psi^2}{2}t + \psi B_t^i, \qquad y - \frac{\psi^2}{2}t + \psi B_t^j.$$

If neither comparison process reaches ℓ, the actual reputations remain in the collaboration region forever. A Brownian motion with drift $-\psi^2/2$ and volatility ψ, starting at $x < \ell$, reaches ℓ with probability $e^{x-\ell}$. Independence therefore gives

$$\mathbb{P}_{x,y}^{n,n}(\tau = \infty) \geq (1 - e^{x-\ell})(1 - e^{y-\ell}) > 0.$$

Now consider any finite initial profile. Fix $t_0 > 0$. The global drift bound gives

$$X_{t_0}^i \leq x + \left(\lambda - \frac{\psi^2}{2}\right) t_0 + \psi B_{t_0}^i, \qquad X_{t_0}^j \leq y + \left(\lambda - \frac{\psi^2}{2}\right) t_0 + \psi B_{t_0}^j.$$

There is positive probability that both Brownian endpoints are sufficiently negative to place both reputations below $\ell - 1$. Independently, no opportunity occurs before t_0 with probability $e^{-2\lambda t_0} > 0$. Conditional on this event, permanent collaboration thereafter has probability at least $(1 - e^{-1})^2$.

Thus every finite initial profile has strictly positive probability of $\tau = \infty$. For two normative players, every terminal action dissolves the partnership, so $\tau^P = \tau$. This proves part (ii).

Positive survival probability for two rogues.

It remains to show that two rogues reach a successful scheme with positive probability. Fix any finite initial profile and a deterministic $t_0 > 0$. Conditional on two rogue types, the global lower drift bound gives

$$X_{t_0}^i \geq x + \left(\frac{\psi^2}{2} - \lambda\right) t_0 + \psi B_{t_0}^i, \qquad X_{t_0}^j \geq y + \left(\frac{\psi^2}{2} - \lambda\right) t_0 + \psi B_{t_0}^j.$$

There is therefore positive probability that both reputations exceed $b + 1$ at t_0. Independently, neither player receives an opportunity before t_0 with probability $e^{-2\lambda t_0} > 0$.

Choose $h > 0$ sufficiently small that $(\psi^2/2 + \lambda)h < 1/2$. There is positive probability that, for each $k \in \{i, j\}$,

$$\inf_{0 \leq u \leq h} \psi(B_{t_0+u}^k - B_{t_0}^k) > -\frac{1}{2}.$$

On this event, both reputations remain above b throughout $[t_0, t_0 + h]$, because the absolute drift is bounded by $\psi^2/2 + \lambda$.

Independently, the probability of at least one opportunity during this interval is

$1 - e^{-2\lambda h} > 0$. At the first such opportunity, the acting rogue's partner has reputation above b, so the prescribed action is a scheme. Since both players are rogue, this scheme is successful and the pair enters the absorbing accomplice relationship. All events in this construction have jointly positive probability. Consequently,

$$\mathbb{P}^{r,r}_{x,y}(\tau < \infty,\ a^\tau = s) > 0.$$

Together with the already established equality $\mathbb{P}^{r,r}_{x,y}(\tau < \infty) = 1$, the definition of τ^P yields

$$\mathbb{P}^{r,r}_{x,y}(\tau^P = \infty) = \mathbb{P}^{r,r}_{x,y}(\tau < \infty,\ a^\tau = s) > 0.$$

This proves part (iii). □

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